INSTRUCTION MANUAL

FC-2900 SERIES
Mass Flow Controllers
and Flowmeters



Mass Flow Controllers

FC-2900, FC-2900O, FC-2901, FC-2901O, FC-2902, FC-2902O
FC-2910, FC-2910O, FC-2911, FC-2911O, FC-2912, FC-2912O
FC-2900B, FC-2900BO, FC-2910B, FC-2910BO
FC-2900M, FC-2900MO, FC-2901M, FC-2901MO, FC-2902M, FC-2902MO

Mass Flowmeters

FM-3900, FM-3901, FM-3902
FM-3910, FM-3911, FM-3912
FM-3900M, FM-3901M, FM-3902M

JUNE 1997

ADDENDUM

Instruction Manual for Millipore FC-2900 Series

Mass Flow Controllers and Flowmeters

Replaces content on introduction page (i) and

pages 45–48 of version 908609-001, Rev. P

Note to Our Customers

Early in 1997, Millipore Corporation acquired Tylan General, and Span Instruments to further strengthen and expand its product and technology offerings to microelectronics customers. Tylan General was a leader in flow technologies since 1965, and started developing products for the semiconductor applications in 1985. Because Millipore is able to focus on the gas filtration, purification, process control and delivery needs of the microelectronics industry, customers can count on sales and service support from specialists who understand their applications thoroughly.

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When returning mass flow products for service, please be sure to specify the type of service required, the product's failure mode, return shipment information, a purchase order to cover non-warranty charges and the name and telephone number of a person to contact if questions should arise.

In addition, you must complete an appropriate ***Certificate of Non-Hazardous Containment*** and attach it to the outside of the shipping container of the product being returned. Disclosure of all chemicals exposed to the equipment is required by federal common carrier transportation laws. This will helps us to protect our personnel and environment, and also helps us to determine if special handling is required to process your repair. For example, ***California Hazardous Communication Regulation (general Industry Safety Orders Sec. 5194 Title 8, California Code CCR Title 22)*** defines the regulations that must be met for products returned to California, U.S.A for service. ***Certificates of Non-Hazardous Contamination*** vary from state to state, so we recommend that you contact your nearest service center to determine what certificates are required. They can provide you with the specific documents needed to comply with state and federal laws.

Omission of the information or required Non-Hazardous Contamination form may result in a delay in processing your order or return of the material. This information will help us to avoid unnecessary delays in processing your request for service.

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GENERAL DESCRIPTION

Tylan General Model FC-2900 series Flow Controllers accurately and reliably measure and control the mass flow rate of gases. They have been specifically designed to allow operation on any gas having a known molar specific heat (pC_p).

The FC-2900 series has a user-adjustable range from 10 sccm to 30 slm full-scale flow rate. This is accomplished in the flowmeter section by replacing the bypass. In the controller section, many changes in flow range can be accomplished by an external adjustment of the control valve alone. Very large range changes may involve resizing of the valve orifice.

The instrument operates on ±15 VDC power, provides a 0 to 5.0 VDC indicated output signal linearly proportional to the flow rate, is RFI protected and is both mechanically and electrically interchangeable with most Tylan General mass flow controllers. The FC-2901/FC-2911 is equipped with a 15-pin "D" connector and the FC-2902/FC-2912 is equipped with a 9-pin "D" connector. Either can have 0-5 VDC or 4-20mA input/output signals. For simplicity, references in this manual are to a card edge style P.C. Board, ±15 VDC power source and 0-5 VDC input/output signals. The FC-2900B/2910B operates from a ±12 VDC battery source. Other inherent design features include:

- Balanced Power Load
- Enhanced stability of both zero and span
- Auto Zero Function (except FC-2900B/2910B and flowmeters)
- Metal seals with ultra-high leak integrity of 1×10^{-10} atm-cc/sec. He (FC-2900M/2901M/2902M/3900M/3901M/3902M)
- Fast response with minimal overshoot
- Battery Powered (FC-2900B/2910B)
- Normally closed or normally open solenoid valve with both mechanical and electrical "purge" capability.

The FC-2900 series of mass flow instruments include:

Mass Flow Controllers

FC-2900(O)*	10 sccm to 10 slm with card edge connector
FC-2901(O)*	10 sccm to 10 slm with 15-pin "D" connector
FC-2902(O)*	10 sccm to 10 slm with 9-pin "D" connector
FC-2910(O)*	100 sccm to 30 slm with card edge connector
FC-2911(O)*	100 sccm to 30 slm with 15-pin "D" Connector
FC-2912(O)*	100 sccm to 30 slm with 9-pin "D" connector
FC-2900B(O)*	10 sccm to 10 slm, Battery Powered
FC-2910B(O)*	100 sccm to 30 slm, Battery Powered
FC-2900M(O)* (EP)**	10 sccm to 30 slm with Metal Seals and card edge connector
FC-2901M(O)* (EP)**	10 sccm to 30 slm with Metal Seals and 15-pin "D" connector
FC-2902M(O)* (EP)**	10 sccm to 30 slm with Metal Seals and 9-pin "D" connector.

*"O" designates normally open valve option.

Mass Flowmeters

FM-3900	10 sccm to 10 slm with card edge connector
FM-3901	10 sccm to 10 slm with 15-pin "D" connector
FM-3902	10 sccm to 10 slm with 9-pin "D" connector
FM-3910	100 sccm to 30 slm with card edge connector
FM-3911	100 sccm to 30 slm with 15-pin "D" connector
FM-3912	100 sccm to 30 slm with 9-pin "D" connector
FM-3900M(EP)**	10 sccm to 30 slm with Metal Seals and card edge connector
FM-3901M(EP)**	10 sccm to 30 slm with Metal Seals and 15-pin "D" connector
FM-3902M(EP)**	10 sccm to 30 slm with Metal Seals and 9-pin "D" connector.

"J" option	4-20 mA input/output signals (available with instruments that have a "D" connector).

NOTE: This manual describes the operation and adjustment procedures for flow controllers. All sections pertaining to the flowmetering portion of the controller also pertain to flowmeters, while sections pertaining to the valve do not. It should also be noted that the autozero function is not available on any of the flowmeter models.

PRINCIPLE OF OPERATION

The flow controller is a self-contained, closed-loop control system which measures the mass rate of gaseous flow through the instrument, compares this with an externally commanded flow rate, and adjusts the valve to control the flow to the commanded level. The flow controller consists of four basic elements which accomplish this function.

- The flow sensor
- The bypass, or flow-splitter, which determines the full-scale flow range of the measuring section
- The control valve
- The electronics which condition the flow signal and drive the control valve.

**"EP" designates Electro-polished option.

Flow Sensor

The flow sensor consists of two self-heated resistance thermometers wound around the outside diameter of a thin-walled capillary tube. These coils, each having a resistance of 330 ±13 ohms at 24°C, are connected in a bridge circuit and supplied with a regulated current. The heat generated by the power dissipated in the coils raises the tube temperature approximately 70°C above ambient. At no flow, this heat is symmetrically distributed along the tube. With gas flowing in the sensor tube, heat is carried downstream. The resulting shift in temperature makes the upstream sensor cooler than the downstream sensor. This temperature difference (and its corresponding electrical resistance difference) is directly proportional to the mass flow rate of the gas through the tube. The bridge output, being a direct function of the resistance difference, is amplified and further linearized by the electronics to give a 0 to 5.0 VDC indication of flow rate. Increasing the flow rate well above the full-scale range of the instrument will eventually cool the entire sensor tube and the output signal will reverse and asymptotically approach zero.

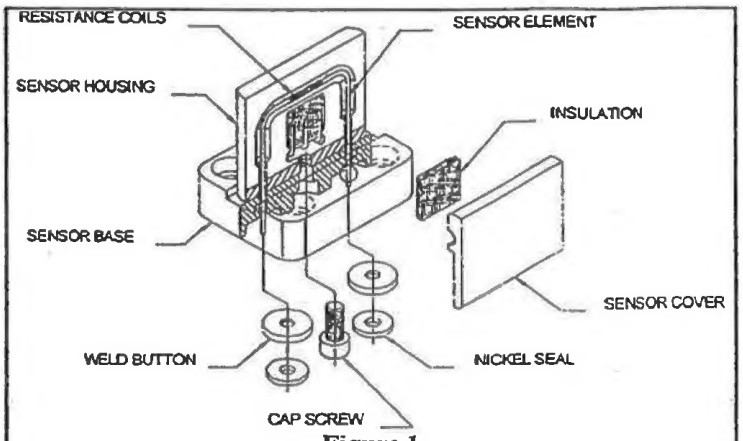


Figure 1
Flow Sensor (Metal Sealed Instruments)

MNL-004A

The capillary tube is dimensioned to have minimal mass (for fast response) and an extremely large length-to-diameter ratio to ensure laminar flow over the full operating range.

The housing which encases the sensor tube is precisely configured to minimize both external and internal natural convection currents from one coil to the other, thus allowing the instrument to be mounted in any position with no zero adjustment required to re-establish the original calibration. This patent-pending configuration minimizes the mass of the sensor resulting in a time constant that is one-third that of other sensors of similar design. Overall flow controller response can therefore be dynamically controlled to eliminate overshoot.

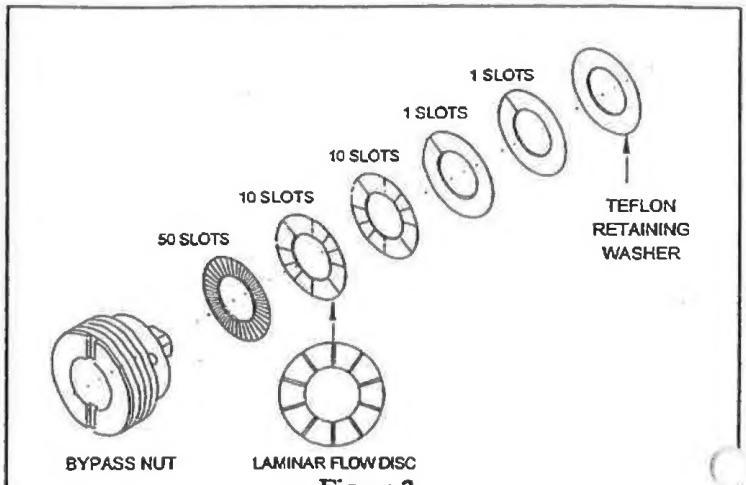


Figure 2
Bypass (Exploded View)

MNL-005A

Bypass (Flow-splitter)

The bypass, which is located in the primary flow path in the base assembly, produces a linear pressure drop versus flow rate between the inlet and outlet of the sensor tube, which in turn produces a 0 to 100% sensor flow for 0 to 100% flow of the instrument. In order to ensure a constant ratio between sensor flow and total flow (independent of pressure, temperature and gas properties), the bypass arrangement has been designed to maintain the flow well within the laminar region of fluid flow over the entire range of the instrument.

The bypass consists of a stack of slotted discs held in place by a bypass nut. Gas flows through the nut and is directed to the outer diameter of the discs where it flows radially through the slots and exits through the opening between the inner diame of the discs and the triangular extension of the nut.

The slots have a sufficient length to diameter ratio to maintain laminar flow over the operating range of the sensor, thereby providing the constant and independent flow split.

By varying the slot depth, the number of slots per disc, and the number of discs per assembly, the bypass can be adjusted to provide the required full scale pressure drop of the sensor coincident with the desired full scale range of the flowmeter. Bypass assemblies are factory-built to specific ranges with the discs held in place by a removable Teflon washer. Each assembly is packaged separately and identified by part number and/or range.

Valve (Mass Flow Controllers only)

The control valve is a normally closed or normally open solenoid valve which is extremely fast, easily adjusted, and has 2% or better shutoff capability. Except for the dynamic plunger, all

wetted metal surfaces are of type 316L stainless steel. The plunger is constructed of type 446 stainless steel.

The range of the valve is determined by the size of the control orifice, which is screwed into the controller base. These precision valve seats are type 316L stainless steel and provide excellent control resolution by engagement with the Teflon face of the valve poppet. The valve seats are easily replaceable for cleaning and instrument range changing.

The valve seat is designed to provide a relatively constant control range (stroke) over the entire flow range. Different ranges within the full range of a particular valve seat are accommodated by the amount of voltage applied to the valve. This arrangement provides consistent dynamic response and static stability with only a minimal need for electronic "tuning."

An adjustment on top of the valve is provided to allow for setting the relative position between the valve poppet and valve seat following re-assembly after cleaning or seat replacement *(see Adjustment Procedures)*. This adjusting nut may also be used to mechanically open the valve for purging purposes in the event of a power failure or an instrument malfunction, which may necessitate removing the instrument from the gas system.

Electronics

The electronics consist of a circuit board with a hybrid autozero circuit soldered in place. Surface mount technology is utilized in the FC-2900 series.

Referring to the Electrical Hook-Up Diagram, J0 is a card edge, 9-pin "D" or 15-pin "D" connector that interfaces with the external control system. A positive 5 volt regulator supplies power to the Auto-Zero circuit and also provides the reference voltage required for the "Valve OFF" function. The negative 5 volt regulator provides the reference for the sensor bridge constant current source and the linearization circuit.

The Current Source delivers a constant 12.7 milliamps to the sensor bridge which "floats" between the plus and minus supplies. This arrangement gives extremely high common-mode rejection and allows insensitive operation over a wide range of supply voltage levels. The Sensor Bridge circuit contains the zero and span temperature compensation networks.

The Flowmeter Amplifier circuit amplifies the sensor output signal, the gain of which is adjusted by RP2 at 50% full scale flow rate. This circuit also incorporates a speed-up filter which provides for matching the sensor response to the actual flow rate response during a transient flow condition. The AC gain versus frequency is adjusted by RP3. This adjustment is fine-tuned by observing the output waveform in response to step changes in setpoint and actual flow rate.

The Linearity circuit provides a variable gain versus input voltage allowing correction for any non-linearity of the sensor output signal. The adjustment is made by RP4 at 100% full-scale flow rate.

The Auto-Zero circuit provides periodic correction of zero offset due to long term sensor drift, temperature and pressure variations, or other environmental causes. This feature ensures long term calibration accuracy and eliminates the need for periodic manual zero adjustment. The range of correction is limited to ±2.5% of full scale to prevent erroneous correction of an undetectable leak or defective sensor.

The auto-zero function is activated either by commanding ≤1% of full scale flow rate which automatically closes the control valve or connecting pin L to common. After 80 to 100 seconds, the output signal is compared with (and driven to) zero within 2.5 seconds. This correction is maintained until the next update. The current or latest correction can be permanently stored in memory by an external command such that in the event of power loss, the output will return to zero when power is re-applied. The auto-zero functions may be inhibited either externally by connecting pin J to common *(See Electrical Hook-Up Diagram)* or internally, by removing jumper JP2. See Appendix C for more information.

The Valve Control circuit compares the flowmeter output signal with the command voltage (setpoint) and varies the voltage to the valve to throttle the flow to the commanded level via closed loop control. The ramp circuit limits the rate of change of the setpoint allowing the sensor output to "keep-up", thus minimizing overshoot during step changes in commanded flow rate. If a slower response is desired, jumper JP1 can be removed to give a 10-15% per second rate of change to the desired flow rate.

While the integral and derivative functions of the PID control circuit are relatively fixed, the proportional band (gain) of the circuit is adjusted by the ratio of R52 to R50. The values of these resistors are selected during dynamic response testing to optimize the transient response of the control loop, and vary depending on the range, gas, and intended operating conditions of the instrument.

The control logic can be overridden externally to either open or close the valve in accordance with specific process requirements. *(See Electrical Hook-Up Diagram)*.

NOTE: While connecting pin D to common will partially open the normally closed valve and partially close the normally open valve, connecting pin D to +15 VDC will drive the normally closed valve to the full "purge" position and close the normally open valve.

The Balanced Power Load circuit has been designed to limit the current in the common line to a maximum of ±5 milliamps. This limits the control error to less than 0.2% per 100 feet of interconnect cable between the flow controller and its setpoint/readout control system. This is a 20:1 improvement over most conventional designs.

Cover

The cover is made of steel and is grounded to the flow base and the chassis ground connection. This, in conjunction with the roll-off and filter capacitors in the electronics provides excellent EMI protection.

INSTALLATION

For maximum performance and service life, the instrument should be installed in a clean, dry atmosphere, relatively free of shock and vibration. Sufficient room for access to the electronics and plumbing should be provided to facilitate maintenance and removal for cleaning. Fitting caps should not be removed from the instrument until installation.

Gas Connections

Since the control valve in the instrument is not intended to provide positive shut-off, it is recommended that a separate shut-off valve be installed in series (upstream to eliminate a flow surge during turn-on, downstream to prevent back migration of contaminants into the control valve, or both).

ELASTOMER SEALED INSTRUMENTS: Three types of fittings are available including VCR™, VCO™ and Swagelok™. Tubing should be pre-cleaned and polished to eliminate particulate contamination and ensure leak-tight operation. After installation of the instrument and prior to its use, the plumbing system should be thoroughly leak-tested and purged.

FC-2900M METAL SEALED: Two types of ultra-high vacuum fittings are available including VCR™ and tube weld. Due to the degree of electropolishing and ultraclean preparation, the FC-2900M should be installed in a prepared ultrapure gas delivery system. After installation of the instrument and prior to its use, the gas delivery system should be thoroughly leak tested and purged. If positive shut-off is required it is recommended that a pneumatic positive shut-off valve be installed in series with the mass flow controller.

VCR, Swagelok and VCO are Trademarks of Crawford Fitting Co.

Electrical Connections

Refer to the appropriate electrical hookup diagram.

POWER: Any ±15 VDC power supply meeting the requirements as designated in the specifications may be used to energize the instrument.

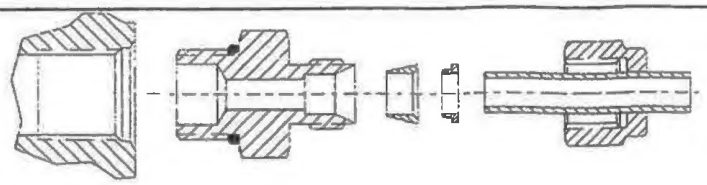


Figure 3
Swagelok Compression
Gas Fitting Configuration

MNL-0001A

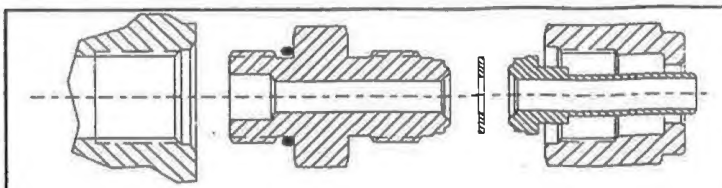


Figure 4
VCR Vacuum
Gas Fitting Configuration

MNL-0001A

CONTROL SIGNAL: Any 0 to 5.0 VDC command voltage having a source impedance of 2500 ohms or less may be used. The input impedance of the flow controller is 0.5 megohm (minimum). 4-20 mA current option is available on the FC-2901 and FC-2911.

OUTPUT INDICATION: Any 0 to 5.0 VDC meter with at least 1000 ohms/volt can be used to provide visual indication of the mass flow rate. Recorders, voltage dividers (for conversion to engineering units), and other instrumentation may be added, provided the total load impedance is no less than 2000 ohms. The source impedance of the flowmeter output signal is less than 0.5 ohm.

VALVE OFF: A TTL low signal will override the control signal, close the valve and trigger the autozero function. TTL high (or open connection) allows normal control operation. (Does not apply to 9-pin "D" PCB).

VALVE TEST (OVERRIDE): The valve voltage may be monitored during normal operation using a high input impedance DVM. Connecting this pin to common or the +15VDC supply will override the control signal and open the valve. (A maximum current of 75 milliamps is required). (For 9-pin PCB, see Figure 8 options "W" or "Y" for valve voltage monitoring).

ZENER TEST: The internal negative reference voltage (-5VDC) may be monitored for troubleshooting purposes using a high impedance DVM. (Does not apply to 9-pin PCB and 15-pin PCB).

AUTO-ZERO INHIBIT: The auto-zero function may be inhibited (deactivated) by applying a TTL low signal. This pin may also be used as an output, as it goes high during the 2.5 second adjustment cycle. (Card edge PCB only).

AUTO-ZERO STORE: Momentary application of a TTL low signal stores the latest zero correction in memory. This is automatically recalled on power up. This feature is not available on MFMs or MFCs with a "D" connector. (Does not apply to 9-pin or 15-pin "D" PCB because AZ Store function is automatic).

AUTO-ZERO TRIGGER: The auto-zero function may be triggered by applying a TTL low signal to pin L. This will override the setpoint signal and close the valve. (AZ Triggering is internally automatic with 9-pin "D" and 15-pin "D" PCBs.)

NOTE: Shorting to common may be substituted for TTL low, and open connection substituted for TTL high.

Autozero is removed on the FC-2900B/2910B to accommodate ±12 VDC battery power feature.

Environmental Requirements

Refer to the specifications.

TEMPERATURE: The instrument may be operated at any temperature between 0°C and 50°C, provided the gas and ambient temperatures are maintained equally. Since the indicated flow rate has a temperature coefficient of ±0.05% per °C, the user may want to calibrate the instrument at the actual operating temperature to maximize the measurement accuracy.

PRESSURE: Flow controllers may be operated at any gas pressure up to 1135 kPa (150 psig). Since the indicated flow rate varies in direct proportion to specific heat (Cp), which varies differently with pressure and temperature depending upon the molecular structure of the gas, it is recommended that the instrument be calibrated at the actual operating pressure.

NOTE: The pressure coefficient of ±0.001% per kPa (±0.007% per psi) generally applies to monatomic and diatomic gases only.

The flow controllers, which can be pressurized to 10,500 kPa (1500 psig) without damage, have a maximum operating pressure of 1135 kPa (150 psig) and the differential pressure (inlet to outlet) must be maintained within the specified range. Lower differential pressure ranges are available on special order.

The flow controllers may be operated into a vacuum as long as the inlet pressure is 105 kPa (15 psia) or greater. Lower differential pressure ranges are available on special order.

Calibration

Each instrument is factory calibrated for the specific flow range and gas indicated on the nameplate. Standard factory calibration is within ±1.0% and is referenced to standard temperature and pressure.* The calibration for other gases can be approximated to ±5% using the Conversion Factor Charts. Factory calibration utilizes the test gases shown in Appendix A, Gas Flow Conversion Factors. Calibration checks with other gases can show discrepancies of up to ±5%. To obtain calibration accuracy of 1% after range change or for other gases, precision calibration equipment will be required. As detailed in the Adjustment and Calibration Procedures section, range changes can be made to within ±15% by removing the inlet fitting and replacing the bypass. Fine tuning to the desired accuracy level can then be accomplished by adjusting the potentiometers on the printed circuit board in conjunction with a reference flow standard.

**NOTE: In accordance with Semiconductor Equipment and Materials Institute Standard E12-91, Standard Pressure and Temperature are defined as 101.325 kPa (760mm Hg) and 0°C respectively.*

CAUTION: An elastomer sealed instrument originally calibrated for a non-reactive, non-corrosive gas should not be converted for use with a reactive or corrosive gas unless all the seals are replaced with the suitable compound. Additionally, any instrument which has been in reactive or corrosive gas service should be thoroughly purged and cleaned prior to conversion to another gas. Instruments are factory assembled using Viton for non-reactives, Neoprene for Ammonia, or Kalrez for all other reactive and corrosive gases unless specified otherwise. The FC-2900M is assembled with metal seals compatible with all gases. Refer to spare parts O-ring Kits.

Mounting

Two 8-32 UNC tapped holes are provided for mounting. The instrument is essentially insensitive to mounting position.

OPERATING INSTRUCTIONS

Installation and Start-up

The interconnect cable should be tested separately for continuity, pin-to-pin shorts, and correct pin assignments per the electrical hook-up diagram. Attach the mating connector to the flow controller and secure it to the cover with two 4-40 x 5/8 inch long screws. Do not over-torque.

After installation in the system and prior to operation, any reactive gas system should be thoroughly leak checked and then purged with dry nitrogen to eliminate the presence of air and moisture. If the mass flow controller is equipped with a normally-closed control valve, it cannot be purged unless the valve is commanded to open. This can be done electrically in two ways. With ±15 volt power applied, apply either a full-scale (5 VDC) setpoint or apply +15VDC to the valve test point (pin D).

A "cycle" purging technique is more effective in removing atmospheric contaminants than a simple continuous purge gas flow. To cycle purge, alternate the flow of purge gas with a pumpdown of the gas system to vacuum for several cycles. If vacuum is not available, reducing the pressure to 101.325 kPa (zero psig) for several cycles should be adequate. Cycle purging helps remove contaminants from small, blind cavities in the system which constitute a virtual leak source. A more detailed procedure can be found in Appendix B.

Apply ±15 VDC power and allow a 30 minute warm-up time before pressurizing the system with process gas. If the indicated output does not settle to zero to within 1% full scale (or the level of zero desired), re-zero the instrument (see *Adjustment Procedures*) before proceeding. Once the zero has been verified, pressure may be applied. After establishing that no hazardous condition will be created by the venting of process gas, flow controller operation may then be verified over the complete operating range. If the flow output voltage is within ten millivolts of the setpoint, the flow controller is controlling properly. Check for this condition at the highest and lowest flows anticipated at both the highest and lowest input-output pressure differentials anticipated.

Safety Features and Precautions

Every effort has been made in the design of the instrument to provide safe, trouble-free operation. Key features include reverse power supply polarity protection, output over-voltage and short circuit protection, low component temperatures, and conformance to intrinsic-safety design criteria. Additionally, a built-in "threshold" circuit ensures that the valve is commanded closed at zero command independent of any existing output offset, and the valve automatically closes in case of a power failure (normally closed). The normally open valve automatically opens with power failure.

Two other features are provided for the operator's convenience during start-up, abnormal, or fault conditions. First, the valve command circuit can be overridden externally to either drive the valve fully open or fully closed. For normally closed valve, connecting pin D (for card edge PCB) or pin 12 (for 15-pin "D") to 0 VDC or +15 VDC will drive the valve open, while connecting pin L (for card edge) or pin 15 (for 15-pin PCB) to 0 VDC will drive the valve closed. For normally open valve, the opposite effect occurs.

The input to close the valve provided on pin L is active low and conforms to TTL Logic levels. Second, should there be a need to purge the system during a power failure or instrument malfunction prior to removing the instrument from the system for repair, the valve may be mechanically opened by first removing the instrument cover and then loosening the lock nut and turning the adjusting nut on top of the valve one turn maximum counterclockwise. Re-adjustment of the valve after purging or repair is described in the *Adjustment Procedures*.

WARNING: PROPER LEAK CHECK EQUIPMENT AND SYSTEM PLUMBING MUST BE AVAILABLE TO LEAK CHECK THE CONTROLLER AFTER ANY MECHANICAL ADJUSTMENT HAS BEEN PERFORMED ON THE VALVE.

The following precautions should be taken to prevent damage, minimize safety hazards, and maximize performance.

1. Use pre-cleaned tubing, free of particulate contamination. Ultraclean tubing is recommended for use with the FC-2900M.
2. Thoroughly leak test the entire system prior to operation (recommended level for elastomer seals is $1x10^{-9}$ atm cc/sec of helium or less, $1x10^{-10}$ with metal seals).
3. Use only clean, moisture-free (dry) gases.
4. Purge only with dry nitrogen (or other inert gas) before and after breaking into the system. Connecting pin D (for card edge PCB) or pin 12 (for 15-pin "D" PCB) to 0 VDC or +15VDC will readily allow for this by driving the valve open. (opposite for normally open valve).
5. DO NOT purge reactive gas systems with inert gas between runs unless it is required by the process. Even "dry" gases contain some amount of moisture which will result in contamination build-up.

WARNING: NEVER INSERT OR UNPLUG THE CONNECTING CABLE OR THE CONTROL VALVE LEADS WITH POWER ON IF THERE IS A POSSIBILITY THAT THE AMBIENT ATMOSPHERE MIGHT BE EXPLOSIVE.

6. ALWAYS command no flow or ground pin L (for card edge PCB) or pin 15 (for 15-Pin "D" PCB) when the gas supply is shut off. The command signal should be interlocked with a series shutoff valve (as shown in Figure 7) to prevent unnecessary over-heating of the control valve during shutoff, and excessive flow surges after turn-on.
7. Avoid installation of the instrument in close proximity to high sources of RF noise and/or mechanical vibration. If this is unavoidable, proven methods

instrumentation filtering, cable shielding, and/or shock mounting should be utilized.

8. See *Caution Note* on page 5 before converting any elastomer sealed instrument for operation on a gas other than what it was originally calibrated for or previously used on.

Operating Modes

In general, the instruments may be operated at any pressure and temperature that falls within the limits stated in the specification. Optimum performance, however, can be achieved and will prevail if the operating pressure and temperature are pre-determined and controlled to a narrow range. Calibrating and "fine tuning" the instruments at the actual operating pressure, temperature and flow rates can significantly improve the performance characteristics.

The instruments are factory calibrated at an ambient temperature of 23°C (±3°C) and the inlet pressure set to the midpoint of the operating pressure range and corrected to standard conditions. Flow controllers are adjusted to pass full rated flow at the minimum inlet pressure, with 101.325 kPa (0 psig) outlet pressure and shut down to less than 2% of full scale at the maximum inlet pressure. Response time is verified at both extremes.

If the control valve is normally closed, it is not necessary to provide an external time-delayed "soft-start" signal provided the command signal is maintained at zero during the "off" mode and is then applied to the instrument at the same time (or after) pressure is applied. Alternately, "Valve Off" pin connected momentarily to common prior to applying gas pressure also prevents excessive flow transients at turn-on. A slower ramp change may also be selected for vacuum systems which have small volumes and large flow rates or in other applications where a slower ramped change in flow rate is desirable.

Flow controllers may be operated at inlet pressures of up to 1135 kPa (150 psig), provided the differential pressure is maintained within the 175 kPa to 380 kPa (10 to 40 psig) range and the outlet pressure of the flow controller is regulated to a relatively constant level. Flow controllers can withstand pressure surges of up to 10,500 kPa (1500 psig) without damage, but are not designed to operate in a controlling mode above 1135 kPa (150 psig) inlet pressure.

Routine and Preventative Maintenance

It is recommended that a routine maintenance schedule be established on each instrument in order to maximize its useful service life. This would include, as a minimum, cleaning and recalibration. It is recommended that calibration be done at least once a year. The optimum (or necessary) service period is dependent on usage, environmental conditions, gas corrosiveness, etc., and must be established by the user based upon historical experience in that particular application.

The recommended steps for routine and preventative maintenance are as follows:

1. Purge the instrument with dry nitrogen for a minimum of 30 minutes prior to removal from the system *(See Appendix B for detailed purge procedure)*.
2. Verify the calibration of the flow metering section by comparison with a suitable reference standard or calibrator.
3. Operate the instrument in the control mode over its entire operating range (2 to 100% full scale) at the minimum and maximum inlet pressures of 175 kPa and 380 kPa (10 and 40 psig). Check response, stability and control resolution.
4. Remove the inlet and outlet fittings of elastomer sealed instruments and visually examine for signs of contamination. Examine the seals for any evidence of cracking, hardening, swelling or contaminant accumulation. Do not attempt to disassemble the FC-2900M unless replacement seals are available.
5. Based upon the results of steps 1 through 4 above, either reinstall in the system (followed by a nitrogen purge) or proceed to the applicable *Cleaning, Adjustment and/or Calibration Procedure* described in this manual.

MAINTENANCE, ADJUSTMENT AND RANGE CHANGE

NOTE: The following instructions and procedures are intended for a laboratory bench top environment. Proper equipment and precautions should be used for any other system or environment.

Cleaning *(Elastomer Sealed Instruments)*

Should the instrument show symptoms of internal flow path contamination, it should be disassembled and cleaned. Elastomer seals on the fittings, sensor and valve should be examined for hardening, swelling, inflexibility, and/or contamination accumulation and should be replaced if any of these symptoms prevail relative to new and unused seals. The recommended sequence of solvents used for cleaning (either for flushing the assembled instrument or for soaking individual components in an ultrasonic cleaner) is as follows.

For instruments used with all non-reactive, non-corrosive gases:

a. Distilled water (2 to 3 minutes)
b. Alcohol (2 to 3 minutes)
c. Blow dry with dry nitrogen or air.

For instruments used with all highly reactive and corrosive gases:

a. A solution of 2% HF, 8% HNO_3, and 90% de-ionized H_2O (3 to 5 minutes)
b. Distilled water (2 to 3 minutes)
c. Alcohol (2 to 3 minutes)
d. Blow dry with dry nitrogen or air.

Individual flow-path elements should be cleaned as follows:

Fittings
Remove the fittings from the base and clean individually.

Sensor
Remove the P.C. Board from the assembly, then remove the two screws which secure the sensor assembly to the base. Run a .18 mm (.007 inch) diameter wire (stainless steel or piano) through the full length of the sensor tube 3 or 4 times. Replace the sensor on the base, torque to 15 in-lb., flush with solvent during the *Bypass Cleaning Procedure* as described below.

Bypass
(See Figure 2, Bypass-Exploded View)
If it is desired to change the range of a flow meter or controller beyond the electronics adjustment capabilities, it will be necessary to replace the bypass assembly to produce the nominal output of 5 ±0.6VDC at full scale flow. In addition, the valve may have to be adjusted and/or the orifice replaced.

The bypass is a preadjusted assembly located in the base, which can be removed and reinstalled with the use of a bypass torque wrench *(see Mass Flow Maintenance Kits)* or a screwdriver.

To remove the bypass, hold the flow controller in a vertical position (inlet down) and carefully remove the assembly. Since the discs are loose on the nut, be careful not to lose or damage them. The nut and discs should be cleaned in an ultrasonic cleaner and reassembled in the same order as originally installed (slots upstream, facing the shoulder of the nut).

New bypass assemblies may be installed with or without the Teflon retaining washer. Reinstall the bypass and torque to 25 in-lb.

NOTE: Installing the nut without the discs provides a means for thorough solvent flushing of the flow sensor. After purging with dry nitrogen, remove the nut and reinstall the discs.

After replacement or removal for cleaning of the bypass assembly, the flowmeter section should be recalibrated per the procedures outlined in the *Calibration* section, since a calibration shift of as much as 10% may occur during this operation depending on the level of contamination.

Valve (See Figure 5)
For minor contamination, the valve can be cleaned in place by mechanically opening the valve, flushing with solvent, purging with dry nitrogen and then resetting the valve during test as described in the *Valve Adjustment Procedure*. To mechanically open the valve, loosen the lock nut and turn the adjusting nut on top of the valve 2 to 3 turns counter-clockwise with a 5/16 inch open-end box wrench.

For severe contamination, the valve assembly and valve seat can be removed from the base and cleaned separately.This is done as follows:

1. Unplug the valve from the P.C. Board.

CAUTION! Always make sure power is off before disconnecting or reconnecting the valve to the P.C. Board.

Remove (or loosen) the P.C. Board from the base assembly if necessary to provide clearance for the removal of the valve assembly.
2. Remove the two screws which secure the valve flange to the base and pull the valve assembly out completely.
3. If further disassembly is necessary (due to the level of contamination), remove the plunger assembly (plunger, poppet, and spring) by holding the inner sleeve assembly at the top of the retaining nut with a 5/16 inch ring-wrench and removing the valve retaining nut with a ½ inch open-end wrench.
4. Thoroughly clean the plunger assembly and the inside of the inner sleeve assembly. Examine the poppet sealing surface and the valve O-ring and replace if necessary.

NOTE: Be sure to re-lubricate the O-ring with Halocarbon grease for Viton O-rings or Krytox grease for Kalrez O-rings prior to re-installation into the base.

5. Reinstall the plunger assembly making certain that the spider spring locates concentrically in the counter-bore of the inner sleeve assembly. Tighten the retaining nut with a torque of 20 in-lb. and verify that the spring is positively held in place with no axial or rotational movement.
6. Remove the orifice from the base and clean separately. Be careful not to nick or damage sealing surface. Replace o-ring if required and re-grease before re-installing.
7. Replace the valve assembly by inserting it straight down into the base. Locate the valve coil leads towards the P.C. Board to provide maximum clearance for the valve adjustment wrench. Secure the valve in place by alternate tightening of the valve flange screws with 20 inch-pounds of torque.
8. Adjust the valve during test as described in the *Valve Adjustment Procedure*.

After cleaning or replacement of the sensor, bypass or valve assemblies, the instrument must be adjusted and recalibrated against a suitable reference standard or calibrator to re-establish the original calibration and performance characteristics.

Cleaning *(Metal Sealed Instruments)*

Should the instrument show symptoms of internal flow path contamination, it may be disassembled and cleaned. Contamination of the ultraclean surface may cause irreparable damage depending on the severity of contamination. Metal seals should be replaced once the factory seal is broken. Before reassembling, inspect metal sealing surfaces carefully for nicks. Use of damaged metal seals will likely result in excess leakage after re-assembly.

Disassembly of the FC-2900M is accomplished as follows:

1. Remove the two screws which secure the cover to the base using a Phillips screwdriver.
2. Unplug the valve from the P.C. Board.

CAUTION: Always make sure power is off before disconnecting or reconnecting the valve to the P.C. Board.

3. Using an Allen wrench, remove the four screws which secure the valve to the base and pull the valve assembly out completely. Discard the nickel seal.
4. If further disassembly of the valve is necessary (due to the level of contamination), remove the plunger assembly by removing the square headed retaining nut from the inner sleeve.
5. Remove the orifice from the base and retain. Discard the Teflon seal and replace with a new seal when reassembling.
6. Remove standoffs which support the cover.
7. Remove the P.C. Board from the assembly and, using an Allen wrench, remove the four screws which secure the sensor assembly to the base. Discard the Teflon centering rings and nickel seals.
8. Using an Allen wrench, remove four screws which secure the inlet fitting and outlet fitting end plates to the base. Discard the nickel seals.
9. To remove the bypass, hold the flow controller in a vertical position (inlet down) and carefully remove the assembly. Since the discs are loose on the nut, be careful not to lose or damage them. Note the order of the installed discs for re-assembly.

Cleaning of all gas wetted parts is accomplished as follows:

1. Run a heated ultrasonic cleaner with D.I. water for a minimum of 5 minutes. Hand clean each part with clean room approved wipes. Reclean using fresh D.I. water.

NOTE: Do not submerge electronics, sensor or valve coil. Use of harsh solvents, chemicals or HF will likely cause irreparable damage to surface finishes.

2. Dry parts in a vacuum oven for 30 minutes. Maintain and observe clean room practices during the cleaning process in preparation for re-assembly.
3. Clean the sensor by running a .18 mm (.007 inch) diameter wire (stainless steel or piano) through the full length of the sensor tube 3 or 4 times. If possible, force DI water through the sensor then dry with ultra-pure nitrogen.

Re-assembly should be done in a Class 10 clean room environment which includes use of tools and fixtures.

1. Inspect all surfaces under high power magnification for surface degradation and replace suspect parts.

NOTE: Carefully inspect all new nickel seal surfaces for nicks before installing.

2. Install outlet fitting plate using a new nickel seal. Note the orientation of the fitting. Torque Allen screws using a cross pattern to 25 in-lb.

NOTE: Hand tighten Allen screws before torquing to be certain that each component is properly seated. Torque in 5 in-lb. increments until required torque is achieved.

3. The nut and discs should be reassembled in the same order as originally installed (slots upstream, facing the shoulder of the nut). Reinstall the bypass and torque to 15 in-lb. using the bypass adapter on the torque wrench.
4. Install the inlet fitting plate using a new nickel seal. Note the orientation of the fitting. Torque Allen screws using a cross pattern to 25 in-lb.
5. Install sensor using new nickel seals. Torque Allen screws to 15 in-lb. Reinstall the P.C. Board to the sensor with the screw and spacer.
6. Install the orifice into the base with a new Teflon seal. Torque to 25 in-lb.
7. Install the valve using a new nickel seal. Torque the Allen screws to 20 in-lb. using a cross pattern to tighten. It will also be necessary to readjust the valve (refer to the Valve Adjustment Procedure).
8. Secure cover support standoffs to the base.
9. Leak check to 1×10^{-10} atm cc/sec Helium.
10. Install the cover following recalibration.

After cleaning or replacement of the sensor, bypass or valve assemblies, the instrument must be adjusted and recalibrated

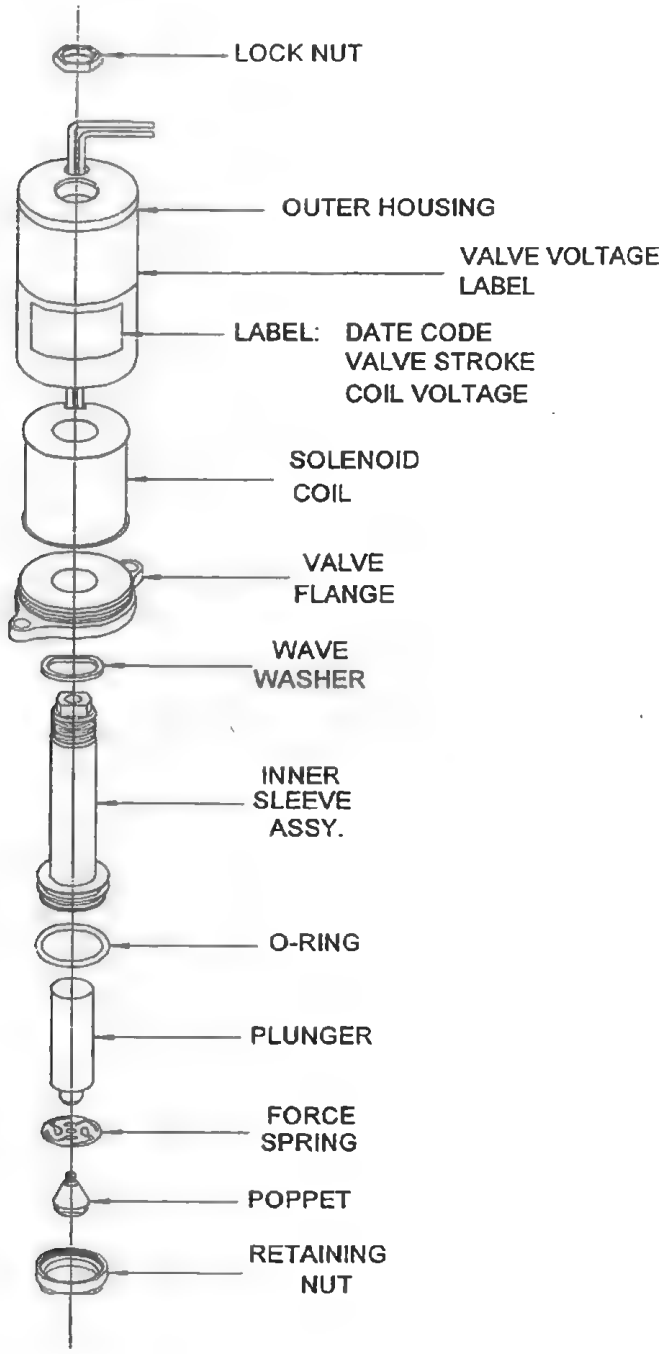


Figure 5
Solenoid Valve (Exploded View)
904149-XXX
(Elastomer Sealed Instruments)

MNL-0025A

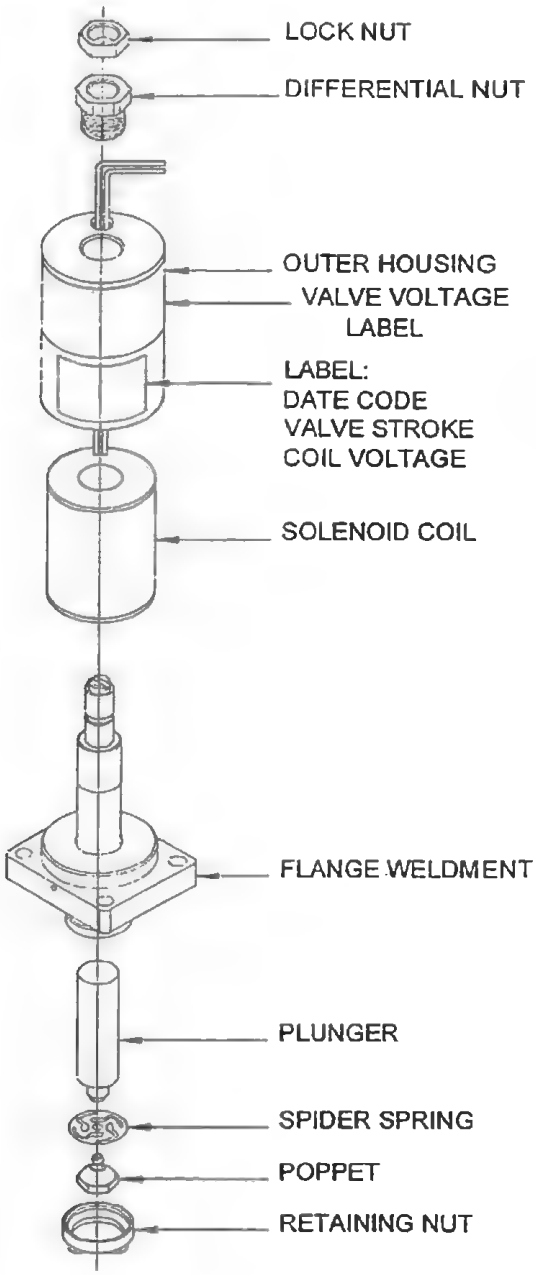


Figure 6
Solenoid Valve (Exploded View)
908778-XXX
(Metal Sealed Instruments)

MNL-0026A

against a suitable reference standard or calibrator to re-establish the original calibration and performance characteristics.

Adjustment and Calibration Procedures

In order to maintain the original factory calibration and performance characteristics of the instrument, the following adjustments may be necessary during routine maintenance and servicing and will definitely be required after disassembly for cleaning and/or parts replacement.

Calibration - Indicated Versus Actual Flow

Each controller is factory calibrated for the specific flow and gases indicated on the nameplate. Standard factory calibration is within ±1.0%. The calibration for other gases can be approximated to ±5% by using the conversion factors shown in Appendix A.

Factory adjustments should not be altered unless precision gas flow measuring equipment is available for calibration. Rotameters do not have sufficient accuracy for flow measurement calibration unless they have been specifically calibrated and the proper corrections are made for temperature and pressure. For access to the electronics, remove the two cover screws and carefully lift the cover from the assembly. If electrical adjustments are made with the cover removed and the PC board exposed, be very careful not to break wires or create short circuits while the instrument is open.

Recalibration may be accomplished as follows: (Refer to Figures 7 through 12).

1. Thoroughly flush and dry the instrument to remove contaminants (see Cleaning Procedures).
2. Connect a source of gas to the inlet and a suitable flow standard to the outlet. If a volumetric calibrator is used, be sure to apply the proper density corrections to maintain the mass flow calibration.
3. Connect the power and indicator wires *(see Figure 7 Electrical Hook-up Diagram)* and allow a 30 minute warm-up period.
4. Either mechanically open the valve by turning the adjusting nut on top of the valve 2 to 3 turns counter-clockwise, or electrically drive the valve open by connecting pin D to pin 4 (+15 VDC). This will override the control loop, allowing flowmeter operation. Command >10% flow to deactivate autozero during calibration.
5. Remove the outlet gas line and cap off the instrument to assure zero flow through the sensor. Adjust ZERO potentiometer RP1 to make the indicator read zero (±5mV).
6. Reconnect the outlet gas line and adjust the flow to 50% full-scale value. Set the output to read 2.50 VDC (±5mV) at 50% full-scale flow rate using the GAIN potentiometer RP2.
7. Recheck zero as described in step 5.
8. Linearity adjustments are not normally required. After achieving calibration at zero and 50% maximum flow, the full scale calibration may be checked by setting the flow to full-scale. If the indicator does not read 5.00 ±0.025 VDC, adjust the output to 5.00 VDC using the LINEARITY potentiometer RP4. Although this adjustment is essentially independent, steps 5 and 6 should be repeated until all three points are within the desired calibration.
9. The flowmeter section of the flow controllers can be calibrated with the instrument operating in the controller mode and using the setpoint control to dial in the desired flow rate. Adjustments should be made as in steps 5 through 8. When adjusting the gain and linearity potentiometers, the actual flow will change rather than the output voltage since the controller acts to control the output voltage to the commanded setpoint voltage.
10. If the instrument cannot be brought into calibration within the adjustment range of RP2 or RP4, the flowmeter amplifier gain and/or bypass will have to be adjusted. This procedure is described in the *Range Change* section of this manual.
11. When using a test gas other than the intended usage gas, a correction factor equaling the ratio of the conversion factors of the two gases must be applied. See Appendix A, *Gas Flow Conversion Factors*, for further explanation.

Valve Adjustment (Mass Flow Controllers only)

WARNING: PROPER LEAK CHECK EQUIPMENT AND SYSTEM PLUMBING MUST BE AVAILABLE TO LEAK CHECK THE CONTROLLER AFTER ANY MECHANICAL ADJUSTMENT HAS BEEN PERFORMED ON THE VALVE.

Valve adjustment is accomplished through the following procedure:

1. Plumb the inlet of the instrument to a regulated supply of the appropriate gas. Connect a reference flowmeter in series, or monitor the flow as measured by the flow metering section of the instrument itself.
2. Remove the cover to permit access to the valve assembly. Plug the valve into the P.C. Board prior to applying power to the instrument. Starting with the valve mechanically open and the input command signal at zero, slowly apply inlet pressure to the controller to 380 kPa (40 psig). Mechanically close the valve by SLOWLY turning the adjusting nut clockwise until the flow is reduced to less than 2% of full scale. Cycle the valve from 0% to 100% then from 100% to 0% full scale to ensure consistent valve closure of less than 2% of full scale.

CAUTION: The control valve is not designed for positive shut-off. Do not attempt to adjust closure to less than 1% as the poppet may be damaged.

3. Set the inlet pressure to the minimum, 175 kPa (10 psig) for up to 5 SLM full scale, and 210 kPa (15 psig) for 10 to 30 SLM full scale, and command 100% flow rate. Verify full-scale output.
4. While monitoring the voltage applied to the valve (pin "D" to pin "F" for card edge PCB or pin 12 to pin 6 for 15-pin "D" PCB), mechanically "fine-tune" the valve adjustment to achieve as close to the optimal 13 VDC control voltage as possible. Repeat this process until the valve controls from <2% to 100% full scale at both extremes of pressure.
5. After the valve is properly adjusted and tuned dynamically (see below), tighten the lock nut on the top of the valve with 10 to 20 inch-pounds of torque. This will lock the adjusted position in place.
6. Leak check the controller.

Dynamic Response Adjustment

Each flow controller is dynamically adjusted prior to shipment using a flow measuring device, in series, which is at least 5 times faster than the instrument under test (for experimental details, refer to SEMI Standard E17-91). This verifies that the tuned indicated response is indicative of the actual flow response and does not mask or filter-out overshoot or oscillatory flow conditions. It is recommended that the adjustments not be altered unless suitable equipment is available to adequately display the transient results. If it is found necessary to readjust the response characteristics following cleaning, re-assembly, valve adjustment, or range change, the following steps may serve as a guideline.

1. Start with the previously set adjustment, or alternately center potentiometer RP3 and set R50 to 200K.

NOTE: A resistance decade box is useful for finding the optimum value of R50.

2. Supply the instrument with the intended usage gas or a suitable substitute gas at the intended operating conditions including inlet pressure.
3. Generate step changes of the setpoint signal. For example, 50% to 100% and vice versa, and watch the output signal using a storage oscilloscope or a fast chart recorder.
4. Adjust RP3 to produce as near a square wave as possible. If a fast actual flow measuring device is used in series to view actual response, adjust RP3 to match the indicated response to the actual response.
5. Once RP3 has been optimized, reduce R50 to minimize the overshoot without causing oscillations or instability. R52 may also be varied if necessary to achieve optimal results. A high ratio of R52/R50 will provide the fastest response and minimal overshoot, but is more inclined to result in oscillation and instability. R50 should be minimized to prevent long delays at low setpoints.
6. Once the response has been optimized, the response characteristics should be checked at various step changes in setpoint over the expected operating pressure range. It is advisable that the value of R50 be increased by 50% as a last step to preclude the possibility of instability developing after installation.

Bypass Adjustment

Since the bypass is pre-assembled to a specific range, it is as such, non-adjustable. Minor adjustments can be made, however, by removing or adding discs to bring the calibration within range of the adjustment potentiometers. (Removing discs reduces the range proportionately, while adding discs increases the range). Bypass adjustment is not recommended by anyone other than a qualified service technician who has been factory trained in the serviceability of the instrument.

Range Change

Changing the range of the instrument is neither recommended nor warranted by anyone other than Tylan General authorized Service Centers or Maintenance Depots. The following information is provided only as a service to the OEM or user to assist them in an emergency situation. To change the range of the instrument it will be necessary to replace and readjust one or all of the following elements:

Valve: Five valve seats with different orifice sizes are available to accommodate full scale ranges of 10 sccm through 30 slm. See Table 11 for the flow ranges of each orifice. All five orifices are available in the Tylan General Spare Parts Kit.

When changing ranges, the valve seat may have to be changed accordingly. This can be accomplished by removing the valve assembly and replacing the valve seat as described in the Cleaning section. Readjustment of the valve is recommended (and in most cases is necessary) for optimal performance following any range change of 2:1 or more. Refer to the *Valve Adjustment Procedure* previously described.

Bypass:
Remove the bypass assembly as described in the Cleaning section and replace with an assembly which has been pre-assembled to the desired range. If the range is too high to bring within the range of the gain pot (RP2), either remove discs or decrease R21 (1.8K: minimum). If the range is too low to bring the calibration within the adjustment range of RP2, increase R21 (3.0K: maximum). Recalibrate per the Calibration Procedure previously described.

Electronics: The following components need to be adjusted or re-selected during the calibration procedure or dynamic response optimization previously described:

Potentiometer RP1	Zero Adjust
Potentiometer RP2	Gain Adjust
Potentiometer RP4	Linearity Adjust
Potentiometer RP3	"DYN" Response Adjust
Resistor R50 and/or R52	Proportional Band Adjust

NOTE: Complete recalibration and dynamic response adjustment will also be required any time the sensor assembly or P.C. Board assemblies are replaced.

TROUBLESHOOTING PROCEDURES

Initial Test

1. Check setup and procedure against connection instructions given in the *Installation* section. Permanent damage to the instrument may result if purging procedures are not followed, or if line power is accidentally applied to the signal leads.
2. Test line cord for compliance with pin assignments and continuity from all wires to correct pins (see Figure 7 *Electrical Hook-up Diagram*). Use a hipot tester to check for pin-to-pin shorts; during the test, flex the cable coming out of the connector to find intermittent shorts.
3. Check insulation resistance from pin 2 to base. It should exceed 50 megohms at 50 VDC. Pin 1 to case should measure less than 1 ohm.
4. Connect a source of gas (same as that for which the instrument is calibrated or equivalent test gas) to the inlet fitting. Apply power and allow a 30 min. warm-up period. With the appropriate inlet pressure applied, the output signal of the controller should follow the command setting (i.e., 50% of full scale at 2.50 VDC commanded voltage). Actual flow (as measured by a suitable flow standard in series with the controller) should follow the command setting and agree with the indicated flow of the controller. If the instrument is malfunctioning or cannot be recalibrated as previously described in the *Adjustment and Calibration Procedure*, check the appropriate symptom in the Troubleshooting Chart.

Tylan General has maintenance kits available which contain custom-designed tools that simplify the maintenance and repair of all Tylan General's mass flow equipment.

Table 1. FC-2900 Series Mass Flow Instrument Maintenance Kit, Part No. 908605-001

Description	Part Number
Torque wrench, 10-inch-lb.(PC bd stndoff)	651780-001
Torque wrench, 15-inch-lb.(sensor)	200241-017
Torque wrench, 25-inch-lb.(bypass)	651780-002
Adapter, torque wrench (bypass)	907328-001
Adapter, torque wrench (bypass)	907328-002
Adapter, torque wrench (valve, sensor)	906278-001
Adapter, orifice (1/4")	906284-001
Adapter, use with orifice	200065-001
Pot adjustment tool	651784-001
Extended board	3780-280-0
Wire, mandrel	3505-122-0001A
Tool kit box	200557-001

Table 2. FC-2900 Series Additional Tools

Description	Part Number
Screwdriver, 1/8" Phillips	200551-001
Wrench, open-end, 5/16" x ¼"	200552-006
Wrench, open-end, 7/16" x 3/8"	200552-007
Wrench, open-end, 11/16" x 13/16"	200552-001
Wrench, open-end, 7/8" x ¾"	200552-005
Wrench, adjustable, 1"	200553-001
Diagonal cutter	200554-001
Pliers, needle-nose	200555-001
X-acto knife	NO-1

Table 3. Other Recommended Tools & Equipment

1.	Alcohol Dispenser
2.	Ultrasonic Cleaner
3.	Plastic Beakers (for use with cleaning solution)
4.	25-10M Halocarbon Grease
5.	Soldering Equipment (for S.I.T. resistors)
6.	Suitable Power Supply and Readout with Interconnect Cable (Tylan General RO-28 recommended)
7.	Tweezers
8.	Artist's brush #4
9.	Krytox Grease

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Table 4. General Troubleshooting

SYMPTOM	POSSIBLE CAUSE	REMEDY
No output	Faulty meter	Read output at pins 3 and 2 directly with alternate meter.
	No actual flow	Check pressures, valve positions, line or filter blockage.
	Sensor clogged	See Cleaning Procedures.
	Valve closed	See Valve section. Measure valve DC resistance (should be 410 ±20Ω). If coil is open, replace coil or entire valve assembly.
	Electronic Failure	See Electronics section.
	No input power	Check for line power between appropriate pins at mating connector.
	Faulty power supply	Check input/output voltages (±15 VDC, +5.00 VDC)
Maximum signal (approx. 200% of full scale): a. Indication correct, flow is high	Valve failed open	Check valve voltage as measured across valve wires. Valve should close when voltage decreases below 4.00 VDC. Higher voltage indicates lack of closing command or electronic failure; repair or replace electronics.
	Faulty power supply or command signal	Check input/output voltages (±15 VDC, +5.00 VDC)
b. Indication erroneous	Open resistance element on sensor	Replace sensor.
	Electronics failure	See Electronics section.

Table 5. Electronics Troubleshooting

SYMPTOM	POSSIBLE CAUSE	REMEDY
General Failure or miscalculation	Power supply voltage off of nominal	Check ±15 VDC. Check P.C. Board interconnect J2.
Flow indication saturated (-0.7 or +12.0 VDC) regardless of flow	Bridge or sensor failure	Check sensor resistances, these should be 330 ±13Ω green to red and green to brown with power off. Voltage across BRN to RED should be 8.0 to 10.0 VDC and pin 6 to circuit common should be -5.0±0.3 VDC.
	Component failure	Check zero pot, gain pot, other components and solder joints.
Valve drive open or driven off (0VDC).	Valve drive transistor Q3 open or Q2 shorted	Check Q2, Q3, and other components; replace as required.
Valve drive saturated(>20VDC)	Q3 short, Op-amp (U1) failed.	Replace if necessary, or replace P.C. Board assembly 909002-001.
All circuits functional, but out of calibration	Contamination, or as a result of cleaning or repairing	Adjust.
Controls, but output voltage does not agree with control input voltage	Mismatched resistors	Check that R34=R37 and R35=R36 (±0.1%)
	Large input voltage offset in Op-amp(U1 of 909002-001)	Check U1. Replace as necessary, or replace P.C. Board 909002-001.

Table 6. Valve Troubleshooting

SYMPTOM	POSSIBLE CAUSE	REMEDY
Signal offset at zero flow	Electronics not adjusted	See Adjustment Procedures.
Valve fails to close	Contamination	See Cleaning Procedures.
	Electronics failure	See Electronics Section
	Mechanical damage from over-pressure or other cause	Adjust or replace valve.
	Operation on wrong gas(e.g., unit adjusted for argon may not shut off completely on hydrogen)	Test on proper gas or mechanically readjust valve.
	Valve failed open (normally open only)	Refer to General Troubleshooting Table on the previous page. Use the same remedy as the "Valve failed open" under Possible Cause.
Valve fails to open	Contamination	See Cleaning Procedures
	Electrically commanded closed or pot shorted	Check command signal (pins A and B) and pot. Check for Electronic failure.
Valve controls at flow rates, but not at minimum	Contamination	See Cleaning Procedures.
	Erosion or corrosion	Replace valve.
	Improper adjustment or inadequate drive	See Adjustment Procedures.
Valve Oscillates or hunts	Erratic pressure regulator	Replace regulator.
	Improper system dynamics due to excessive inlet pressure	Reduce upstream pressure regulator setting.
	Improper dynamics in electronics	Check for failed resistor or capacitor. See Dynamic Response Adjustment Procedure.

CAUTION: Always command zero flow when the gas supply is shut off. Failure to do so will result in excessive flow transients after the gas supply is turned "on" (See Electrical Hook-up Diagram.)

NOTE 1: Pin L may be connected to common prior to turning on the gas supply to achieve "soft start" performance.

NOTE 2: A threshold circuit in the controller ensures zero valve voltage whenever the commanded signal falls below 1.0%

NOTE 3: It is not recommended that a failed electronic component be replaced, as component failures are usually caused by another problem in the device or in the system. It is recommended instead, that the flow controller be returned to the factory or other authorized Service Center for failure analysis and repair.

Sensor Replacement

If it is determined that either of the sensor elements is open or shorted to case, it will be necessary to replace the sensor assembly. This can be done by unsoldering the three sensor leads from the P.C. Board and removing the three screws which attach the sensor assembly to the base and the P.C. Board. The sensor assembly can then be removed and replaced. Before installing the new sensor element, resistance should be checked and measured as follows:

Upstream..... RU=330±13 ohms at 24°C
(Brown to Green)
Downstream..... RD=330±13 ohms at 24°C
(Red to Green)
Isolation ..... >100 megohms
(All leads to sensor housing)

Solenoid Valve Replacement

If it is determined that the coil of the solenoid valve is open, shorted to case or outside the DC resistance range of 390 to 420 ohms, the valve assembly will need to be replaced. Simply remove the locknut and valve housing to gain access to the coil. The valve must by readjusted following the coil replacement.

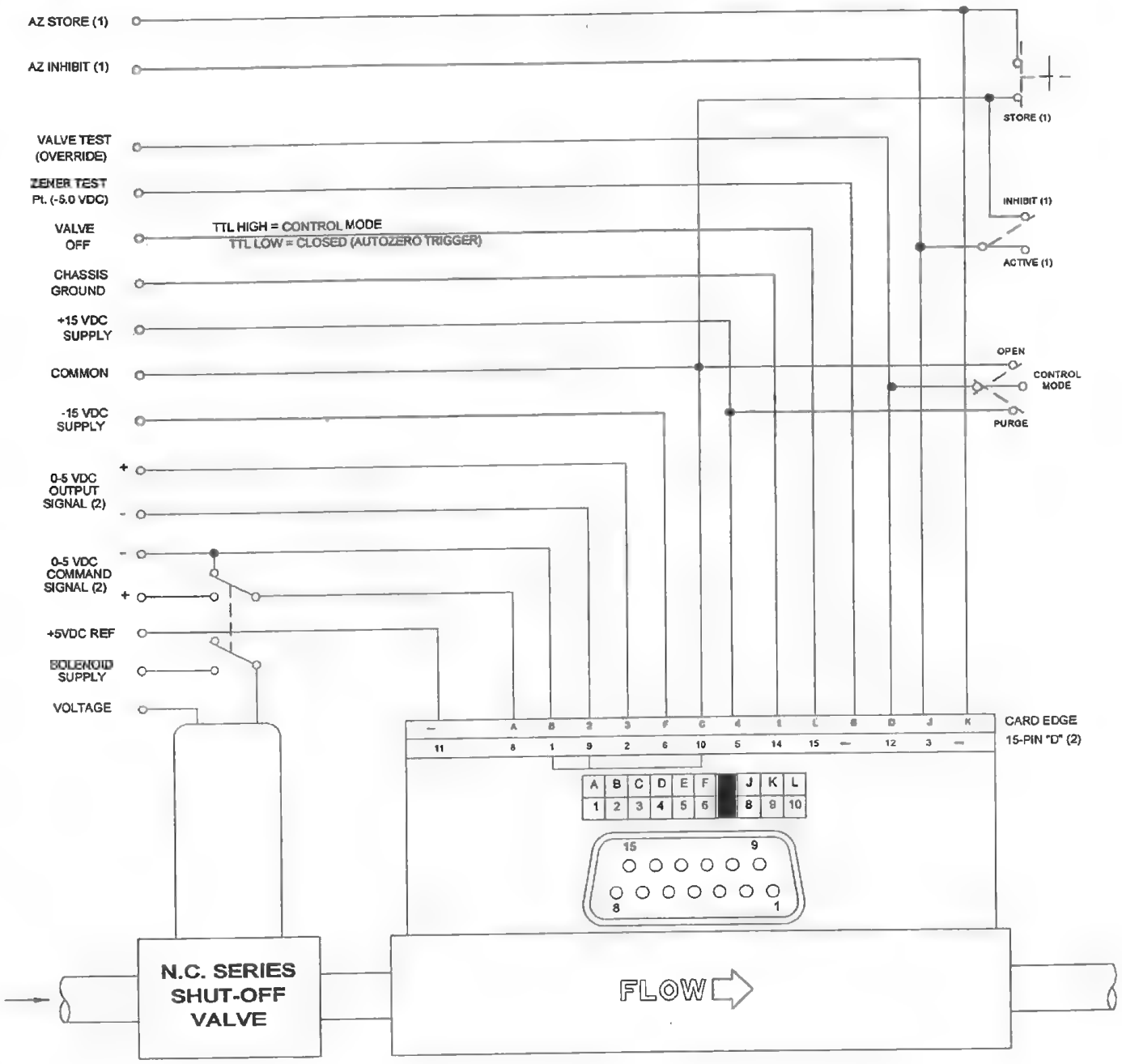
Table 7. FC-2900 Series Mass Flow Controllers Specifications

Specifications	FC-2900(O),FC-2901(O),FC-2902(O) FC-2910(O),FC-2911(O),FC-2912(O)	FC-2900M(O), FC-2901M(O),FC-2902M(O)	FC-2900B(O) FC-2910B(O)
PERFORMANCE			
Full Scale N2 equiv:	FC-2900,2901,2902 = 10 sccm to 10 slm FC-2910,2911,2912 = 100 sccm to 30 slm	10 sccm to 30 slm	FC-2900B = 10 sccm to 10 slm FC-2910B = 100 sccm to 30 slm
Turndown Ratio	50:1	50:1	50:1
Accuracy	±1.0%	±1.0%	±1.0%
Repeatability	±0.2%	±0.2%	±0.2%
Linearity	±0.5%	±0.5%	±0.5%
Step Response Time (0 - 100%)	500 ms (typical)	500 ms (typical)	500 ms (typical)
Settling Time	< 1.5 seconds	< 1.5 seconds	< 1.5 seconds
OPERATING			
Maximum Operating Pressure	1150 kPa (150 psig)	1150 kPa (150 psig)	1150 kPa (150 psig)
Proof Pressure	10,500 kPa (1500 psig)	10,500 kPa (1500 psig)	10,500 kPa (1500 psig)
Differential Pressure:			
10 sccm to 9 slm	70 to 280 kPa (10 to 40 psid)	70 to 280 kPa (10 to 40 psid)	70 to 280 kPa (10 to 40 psid)
10 to 30 slm	105 to 280 kPa (15 to 40 psid)	105 to 280 kPa (15 to 40 psid)	105 to 280 kPa (15 to 40 psid)
(Lower differential pressures available, consult factory)			
Pressure Coefficient	0.001% / kPa (0.007% / psi)	0.001% / kPa (0.007% / psi)	0.001% / kPa (0.007% / psi)
Temperature Range	0 - 50 °C	0 - 50 °C	0 - 50 °C
Temperature Coefficient (Zero and Span)	±0.05%/°C	±0.05%/°C	±0.05%/°C
Warm-up Time	30 minutes	30 minutes	30 minutes
Attitude Sensitivity (Without Autozero)	< 0.25% @ 90° attitude	< 0.25% @ 90° attitude	< 0.25% @ 90° attitude
ENVIRONMENTAL			
Pressure	> 100 Torr (2 psia)	> 100 Torr (2 psia)	> 100 Torr (2 psia)
Humidity	0-95% rel. humidity, non-condensing	0-95% rel humidity, non-condensing	0-95% rel. humidity, non-condensing
ELECTRICAL			
Power Supply Sensitivity	< 0.01%/volt	< 0.01%/volt	< 0.01%/volt
Power Consumption	3.3 watts @ ±15 VDC	3.3 watts @ ±15 VDC	1.8 watts @ ±12 VDC
Common Current	±5mA max.	±5mA max.	±10mA max.
Output Impedance	0.25 ohms	0.25 ohms	0.25 ohms
Supply Voltage	±12 VDC to ±18 VDC	±12 VDC to ±18 VDC	±11 VDC to ±15 VDC
Input/Output Signal	0 to 5 VDC	0 to 5 VDC	0 to 5 VDC
Supply Current	±110mA(125mA max. @ ±18 VDC)	±110mA(125mA max. @ ±18 VDC)	±75mA(95mA max. @ ±15 VDC)
Input Impedance	500K ohms (min.)	500K ohms (min.)	500K ohms (min.)
Minimum Load Impedance	2000 ohms	2000 ohms	2000 ohms
Input/Output Signal (Optional)	4-20 mA (FC-2901/2911/2902/2912)	4-20 mA (FC-2901M/2902M only)	Not available
Supply Current	±130mA (150mA @ + 18VDC)	±130mA (150mA @ ± 18VDC)	
Input Impedance	250 ohms	250 ohms	
Maximum Load Impedance	250 ohms	250 ohms	
Connection Type:			
Card Edge:	FC-2900/FC-2910	FC-2900M	FC-2900B/FC-2910B
15-pin "D":	FC-2901/FC-2911	FC-2901M	Not available
9-pin "D":	FC-2902/FC-2912	FC-2902M	Not available
Autozero:			
Delay Time	80 ±20 sec.	80 ±20 sec.	Not applicable
Correction Time	< 2.5 sec.	< 2.5 sec	Not applicable
Resolution	0.1% full scale	0.1% full scale	Not applicable
Range	±2.5% full scale	±2.5% full scale	Not applicable
MECHANICAL			
Valve (> 1 million cycles)	N.C. & N.O. Solenoid	N.C. & N.O. Solenoid	N.C. & N.O. Solenoid
Materials	316 L ss, 446 ss, Viton™, Kalrez™, or Neoprene™	316 L ss, 446 ss, 321 ss, Nickel 200	316 L ss, 446 ss, Viton™ Kalrez™, or Neoprene™
Leak Integrity	1×10^{-9} atm-cc/sec. He	1×10^{-10} atm cc/sec. He	1×10^{-9} atm-cc/sec. He
Valve Through-Leak	<2% full scale for N.C. <2% full scale for N.O. 10 SCCM to 9.99 SLM <5% full scale for N.O. 10 to 30 SLM	< 2% full scale for N.C. <2% full scale for N.O. 10 SCCM to 9.99 SLM <5% full scale for N.O. 10 to 30 SLM	< 2% full scale for N.C. <2% full scale for N.O. 10 SCCM to 9.99 SLM <5% full scale for N.O. 10 to 30 SLM
Fittings	1/4", 3/8" Swagelok™ 1/4", 3/8" VCR™	1/4" VCR™, 1/4" Butt Weld	1/4", 3/8" Swagelok™ 1/4", 3/8" VCR™
Dimensions	See Outline Diagram	See Outline Diagram	See Outline Diagram
Surface Finish		Non-EP = 0.5 um (20 µinch) Ra EP = 0.25 um (10 µinch) Ra	

Table 8. Mass Flowmeters Specifications

Specifications	FM-3900, FM-3902, FM-3910, FM-3901, FM-3911, FM-3912	FM-3900M, FM-3901M, FM-3902M
PERFORMANCE		
Full Scale N2 equivalent	10 sccm to 10 slm (FM-3900, 3901, 3902) 100 sccm to 30 slm (FM-3910, 3911, 3912)	10 sccm to 30 slm
Accuracy	±1.0%	±1.0%
Repeatability	±0.2%	±0.2%
Linearity	±0.5%	±0.5%
Step Response Time (0 - 100%)	1 second (typical)	1 second (typical)
Settling Time	< 1.5 seconds	< 1.5 seconds
OPERATING		
Maximum Operating Pressure	1150 kPa (150 psig)	1150 kPa (150 psig)
Proof Pressure	10,500 kPa (1500 psig)	10,500 kPa (1500 psig)
Differential Pressure	35 kPa (5 psid max.)	35 kPa (5 psid max.)
Pressure Coefficient	0.001% / kPa (0.007% / psi)	0.001% / kPa (0.007% / psi)
Temperature Range	0 - 50°C	0 - 50°C
Temperature Coefficient (Zero and Span)	±0.05%/°C	±0.05%/°C
Warm-up Time	30 minutes	30 minutes
Attitude Sensitivity	< 0.25% @ 90° attitude	< 0.25% @ 90° attitude
ENVIRONMENTAL		
Pressure	> 100 Torr (2 psia)	> 100 Torr (2 psia)
Humidity	0-95% rel. humidity, non-condensing	0-95% rel. humidity, non-condensing
ELECTRICAL		
Power Supply Sensitivity	< 0.01%/volt	< 0.01%/volt
Power Consumption	1.2 watts @ ±15 VDC	1.2 watts @ ±15 VDC
Common Current	±5mA max.	±5mA max.
Output Impedance	0.25 ohms	0.25 ohms
Supply Voltage	±12 VDC to ±18 VDC	±12 VDC to ±18 VDC
Output Signal	0 to 5 VDC	0 to 5 VDC
Supply Current	±35mA(40mA max. @ ±18 VDC)	±35mA(40mA max. @ ±18 VDC)
Input Impedance	500K ohms (min.)	500K ohms (min.)
Minimum Load Impedance	2000 ohms	2000 ohms
Output Signal (optional)	4-20 mA (FM-3901/3902/3911/3912 only)	4-20 mA (FM-3901M & FM-3902M only)
Supply Current	±55mA (65mA @ ± 18VDC)	±55mA (65mA @ ± 18VDC)
Input Impedance	Not applicable	Not applicable
Maximum Load Impedance	250 ohms	250 ohms
Connection Type:		
Card Edge	FM-3900/FM-3910	FM-3900M
15-pin "D"	FM-3901/FM-3911	FM-3901M
9-pin "D"	FM-3902/FM-3912	FM-3902M
Autozero:	Not available	Not available
MECHANICAL		
Materials	316 L ss, 446 ss, Viton™, Kalrez™, or Neoprene™	316 L ss, 446 ss, 321 ss, Nickel 200
Leak Integrity	1 x 10^{-9} atm-cc/sec He	1 x 10^{-10} atm cc/sec He
Fittings	1/4" , 3/8" Swagelok™, 1/4", 3/8" VCR™	1/4" VCR™, 1/4" Butt weld
Weight	0.9 kg (2 lbs.)	0.9 kg (2 lbs.)
Dimensions	See Outline Diagram	See Outline Diagram
Surface Finish		Non-EP = 0.5 um (20 µinch) Ra EP = 0.25 um (10 µinch) Ra

VCR & Swagelok are Trademarks of Crawford Fitting Co.
Viton, Neoprene and Kalrez are Trademarks of E.I. Dupont de Nemours Co.

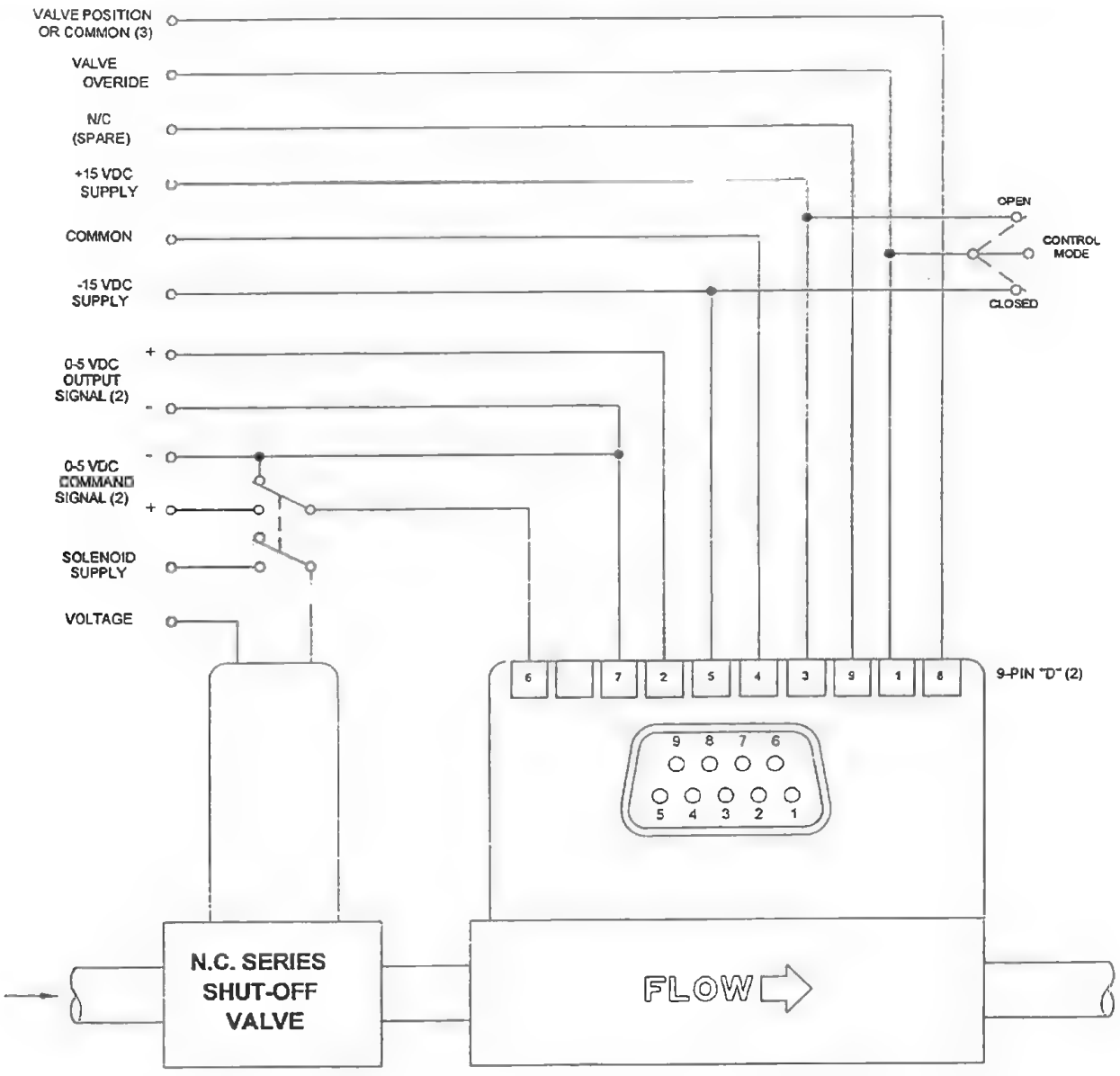


Notes: (1) Autozero circuit is not available on FC-2900B/FC-2910B and flowmeters.
AZ Store is internally automatic on FC-2901/FC-2911/FC-2901M.

(2) The following "-J option" is optional by request on MFCs or FMs with 15-pin or 9-pin "D" connectors.
Pin 7 is 4 to 20 mA Input (MFCs only).
Pin 4 is 4 to 20 mA Output.

Figure 7
Electrical Hook-Up Diagram
FC-2900 Series (Except FC-2902/2902M/2912)

MNL-012A



(1) AZ Store is internally automatic.

(2) The following "-J option" is available by request:
Pin 6 is 4 to 20 mA Input (MFCs only).
Pin 2 is 4 to 20 mA Output.

(3) For valve voltage "W" option, the valve voltage is -15 to +15 VDC.
For valve voltage "Y" option, the valve voltage is 0 to +7 VDC.
For valve voltage "T' option, Pin 8 is Signal Common.

Figure 8
Electrical Hook-Up Diagram
FC-2902, FC-2912, FC-2902M

MNL-011

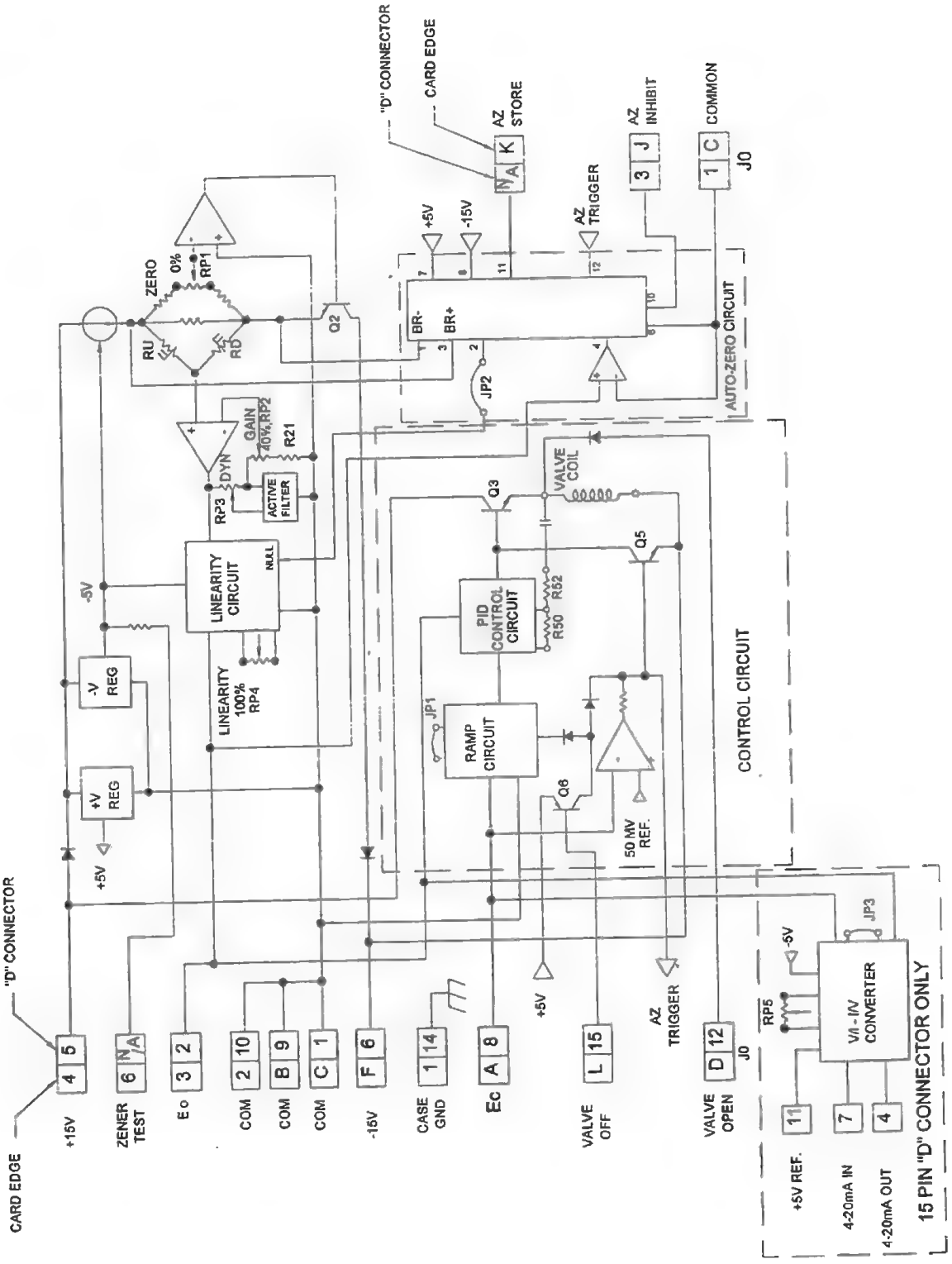


Figure 9
Functional Diagram
FC-2900 Series (Except FC-2902, FC-2902M, FC-2912)

MNL-002A

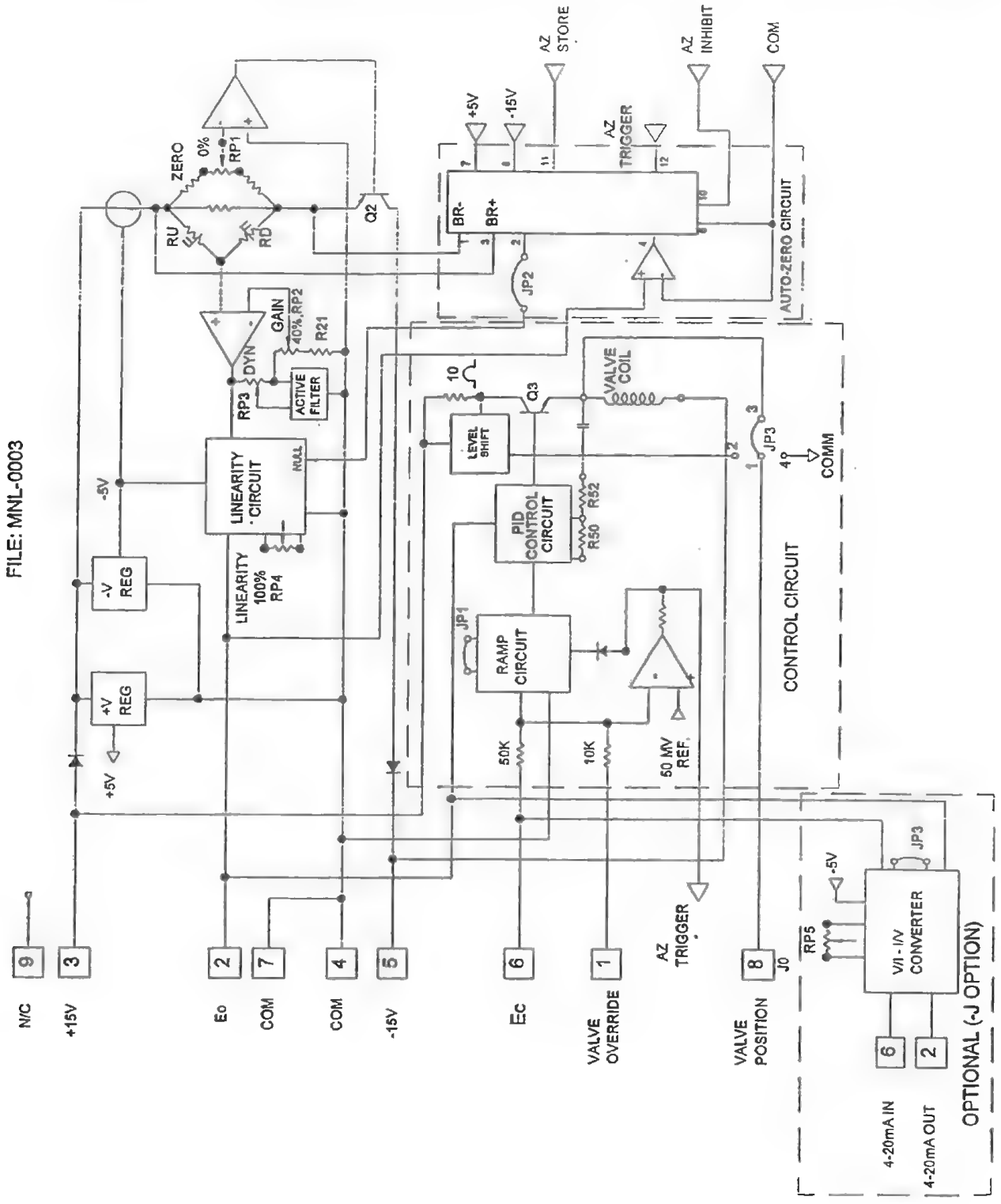
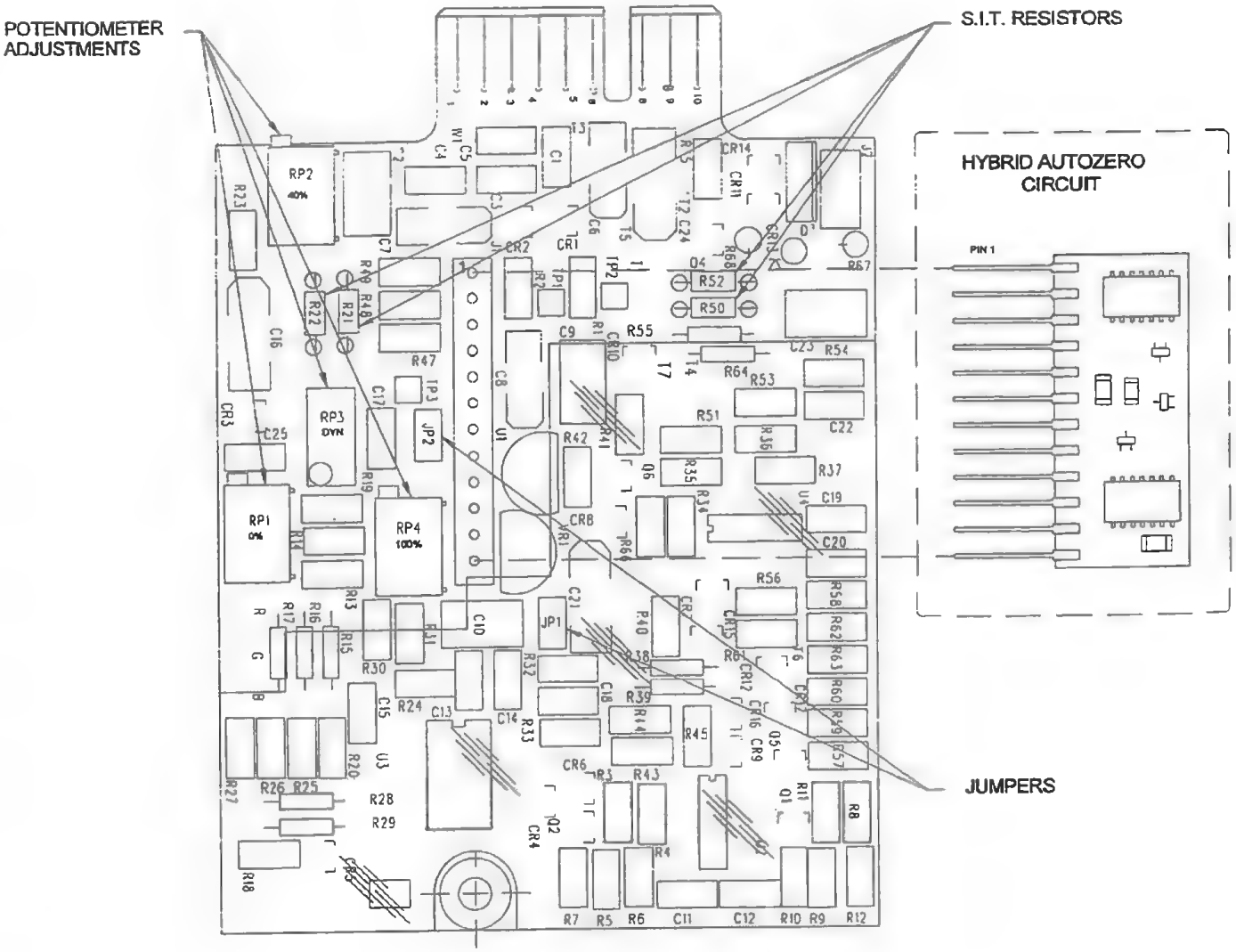


Figure 10
Functional Diagram
FC-2902, FC-2912, FC-2902M

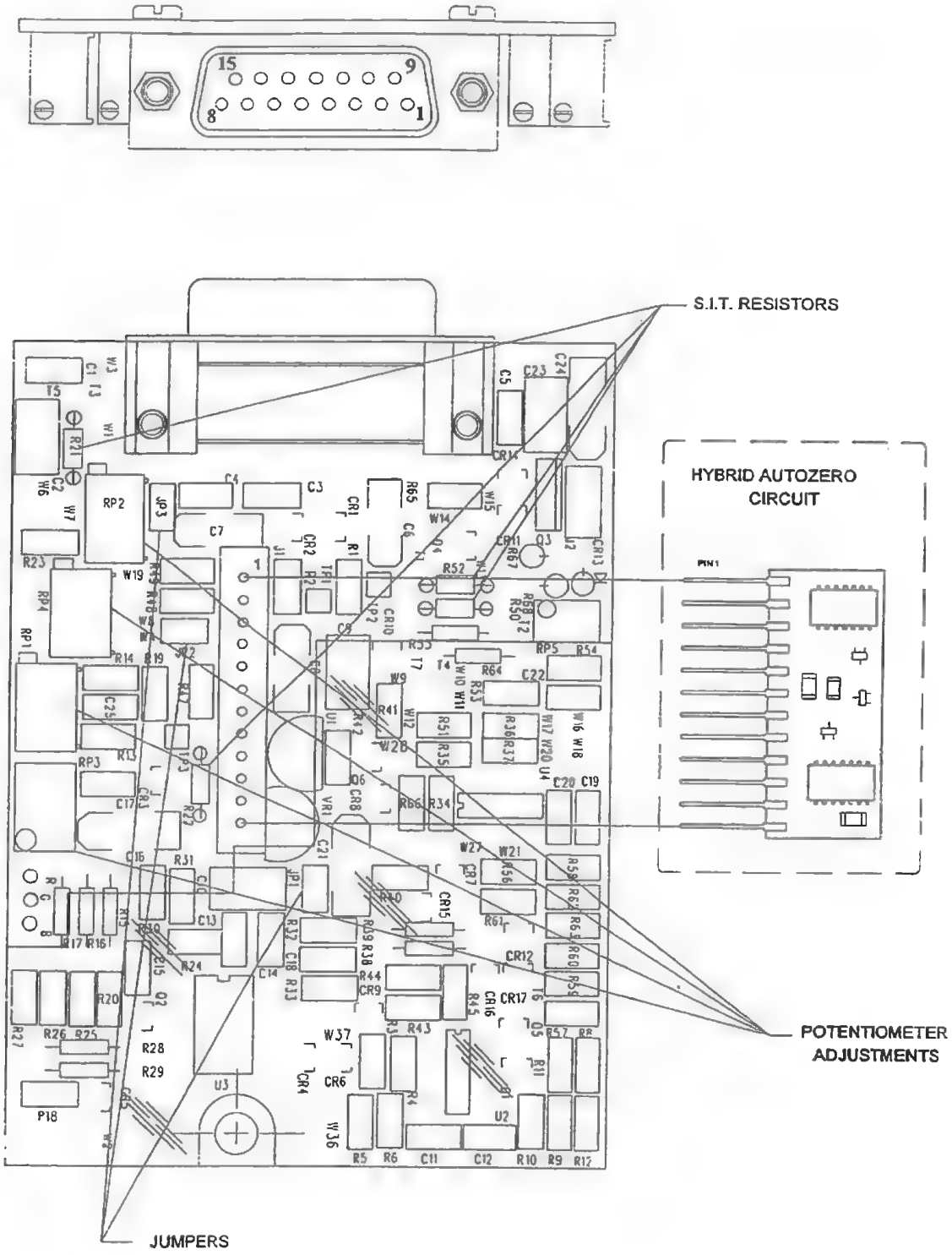
MNL-003A



Note: Use part #908844-001 (has Hybrid Autozero Circuit) for FC-2900(O), FC-2910(O), FC-2900M(O). Use part #908950-001 (does not have Hybrid Autozero Circuit) for FC-2900B(O), FC-2910B(O) and Flowmeters.

Figure 11
P. C. Board Layout
908854-001 (Card Edge Connector)

MNL-007A



Note: Use part #908854-001 (has Hybrid Autozero Circuit) for FC-2901(O), FC-2911(O), FC-2901M(O). Use part #908960-001 (does not have Hybrid Autozero Circuit) for FM-3901, FM-3911, and FM-3901M.

Figure 12
P. C. Board Layout
908854-001 (15-Pin "D" Connector)

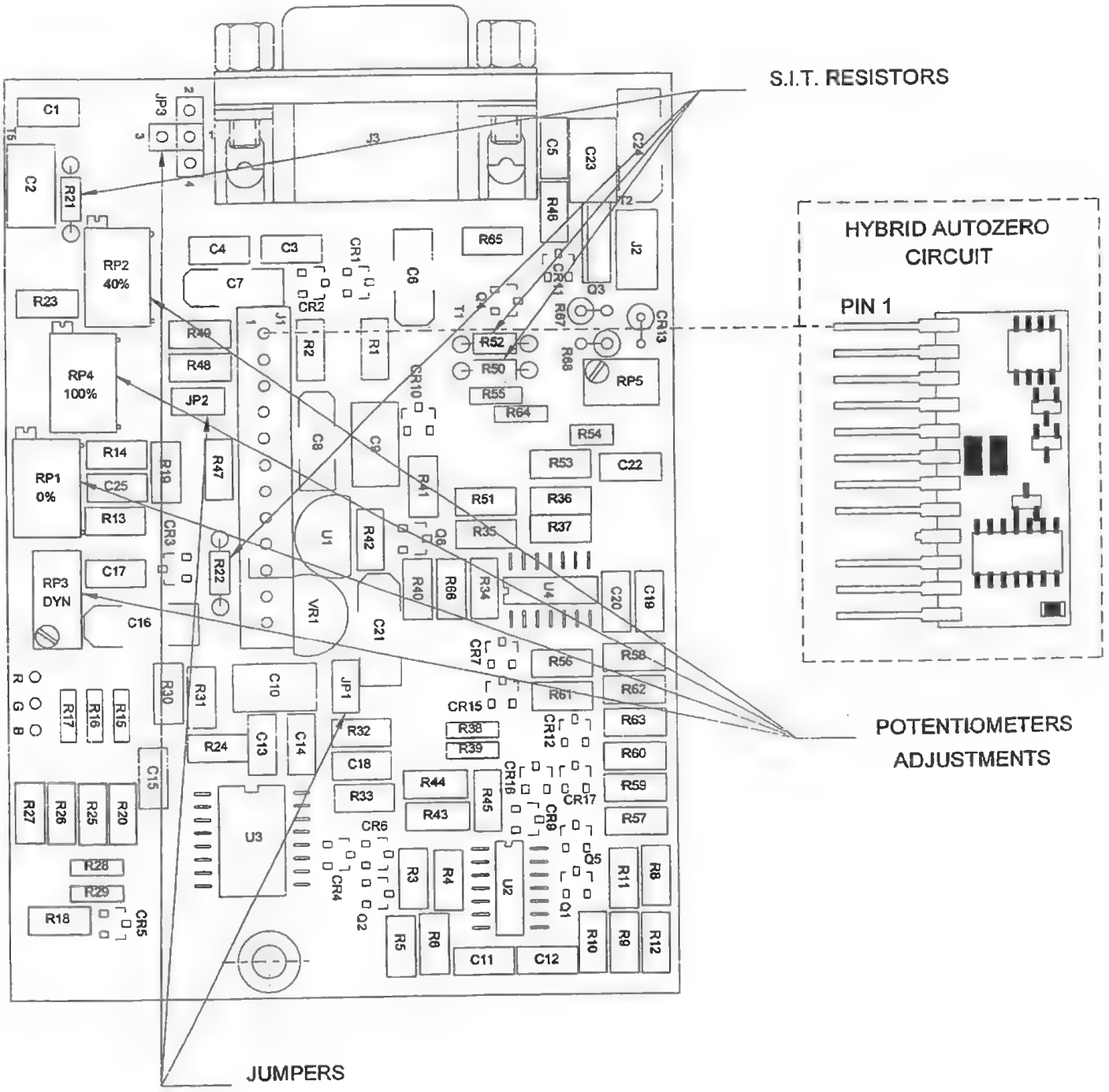
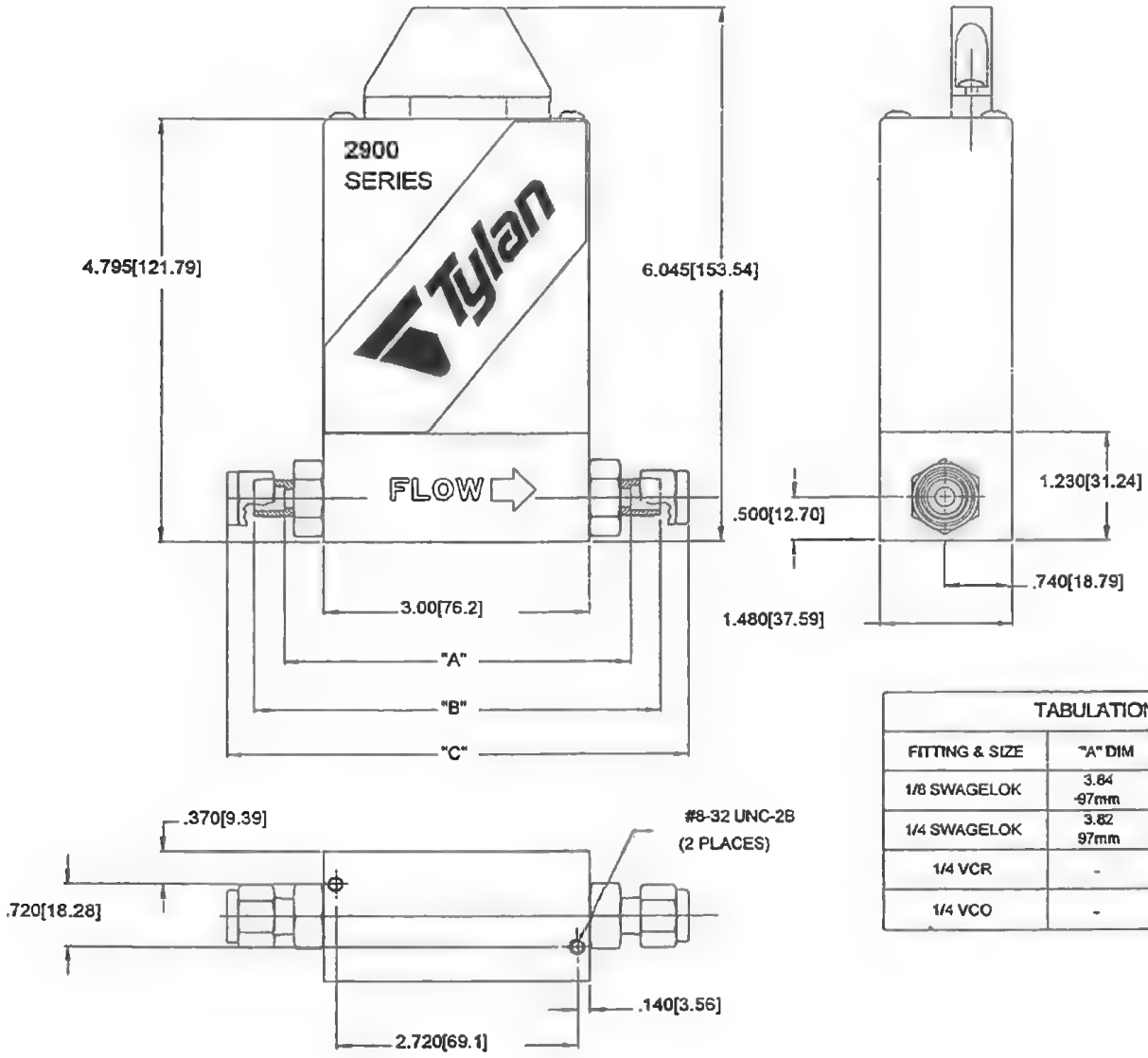


Figure 13
P. C. Board Layout
909330-001 (9-Pin "D" Connector)

MNL-009A



TABULATION BLOCK			
FITTING & SIZE	"A" DIM	"B" DIM	"C" DIM
1/8 SWAGELOK	3.84 97mm	4.32 110mm	4.84 123mm
1/4 SWAGELOK	3.82 97mm	4.44 113mm	5.02 128mm
1/4 VCR	-	4.88 124mm	-
1/4 VCO	-	4.56 116mm	-

Figure 14
Outline Diagram
FC-2900 Series (Elastomer Sealed)

MNL-041

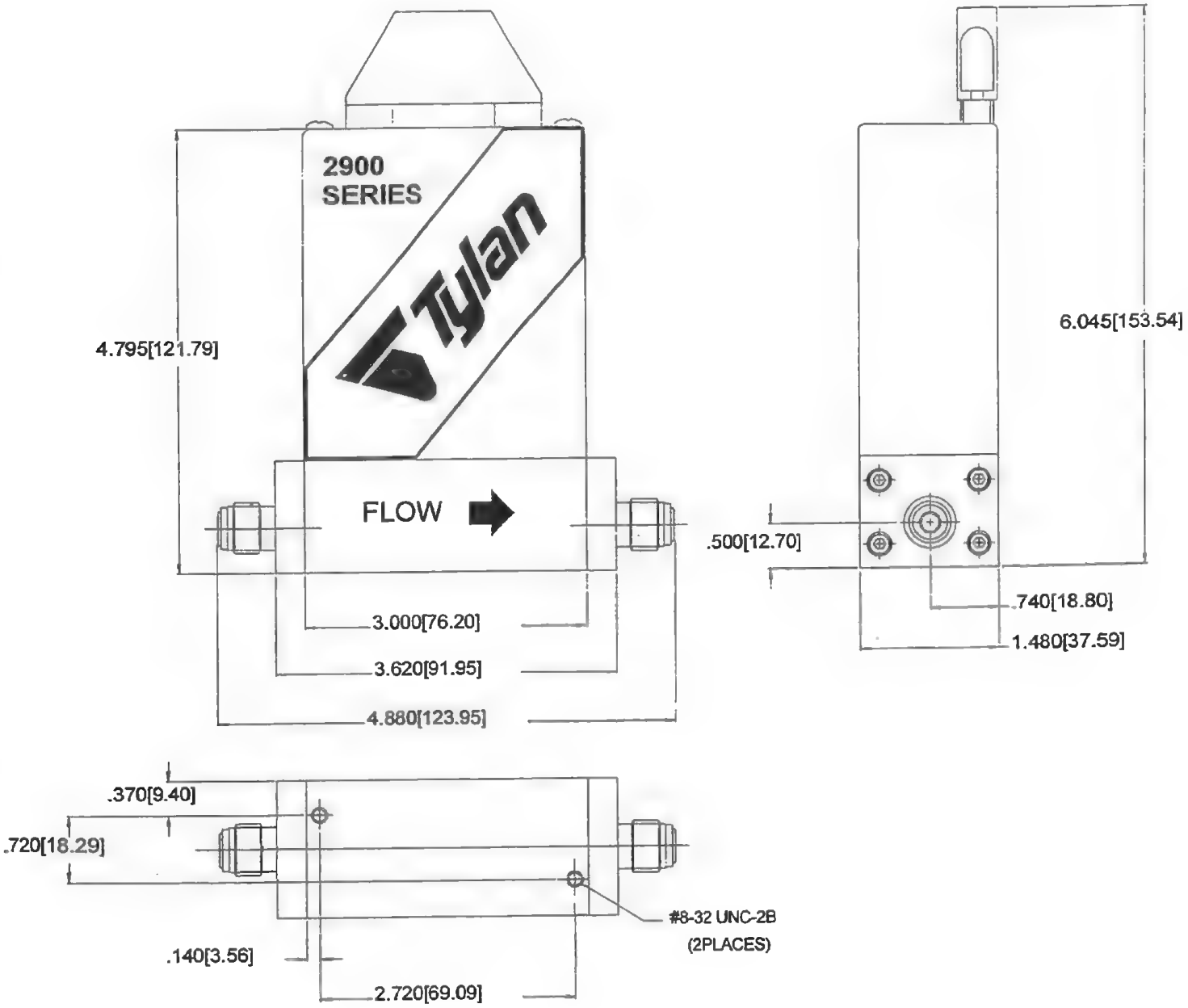


Figure 15
Outline Diagram
FC-2900M Series (Metal Sealed)

MNL-042A

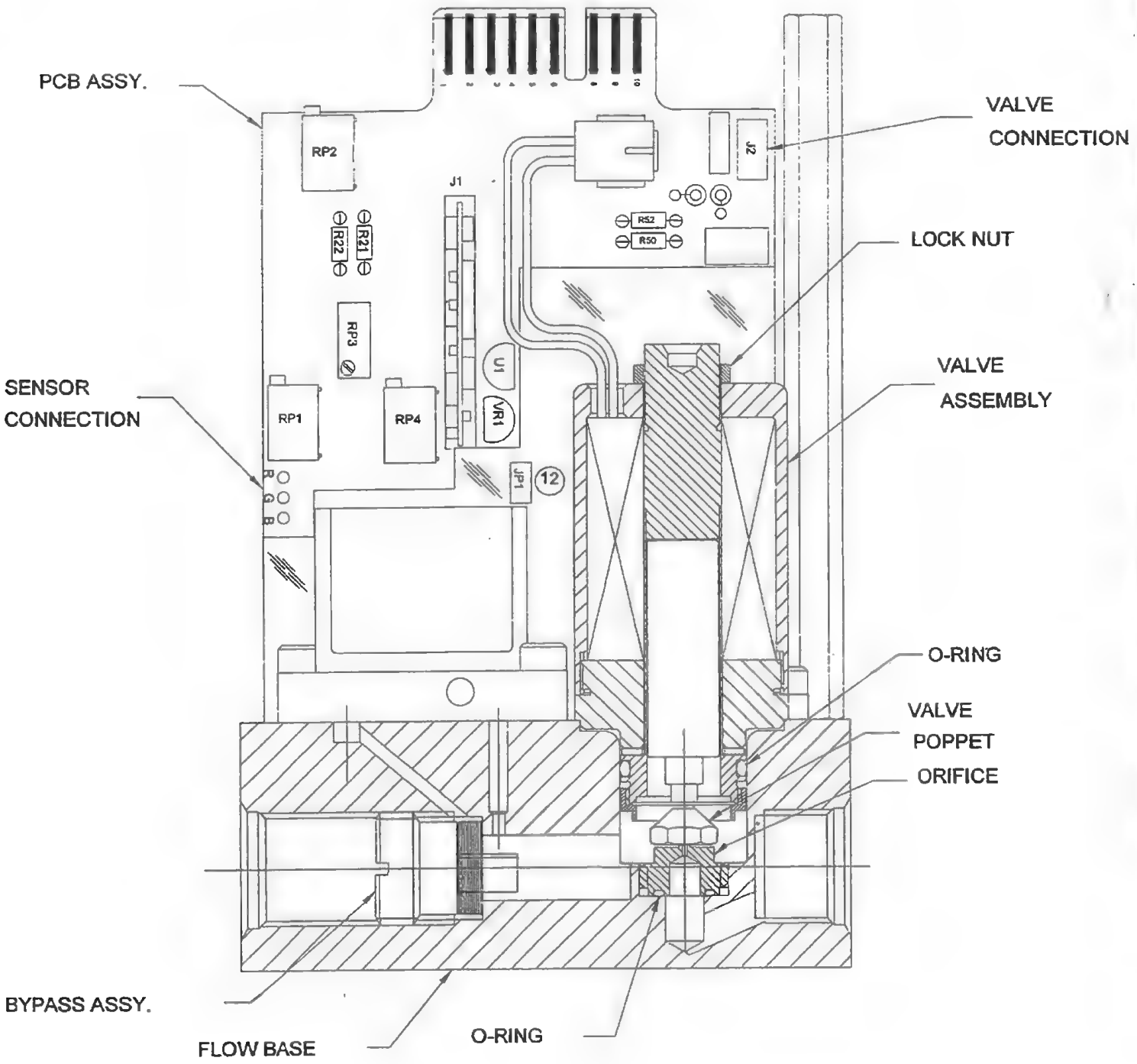


Figure 16
Cross Sectional View
FC-2900 Series (Elastomer Sealed)

MNL

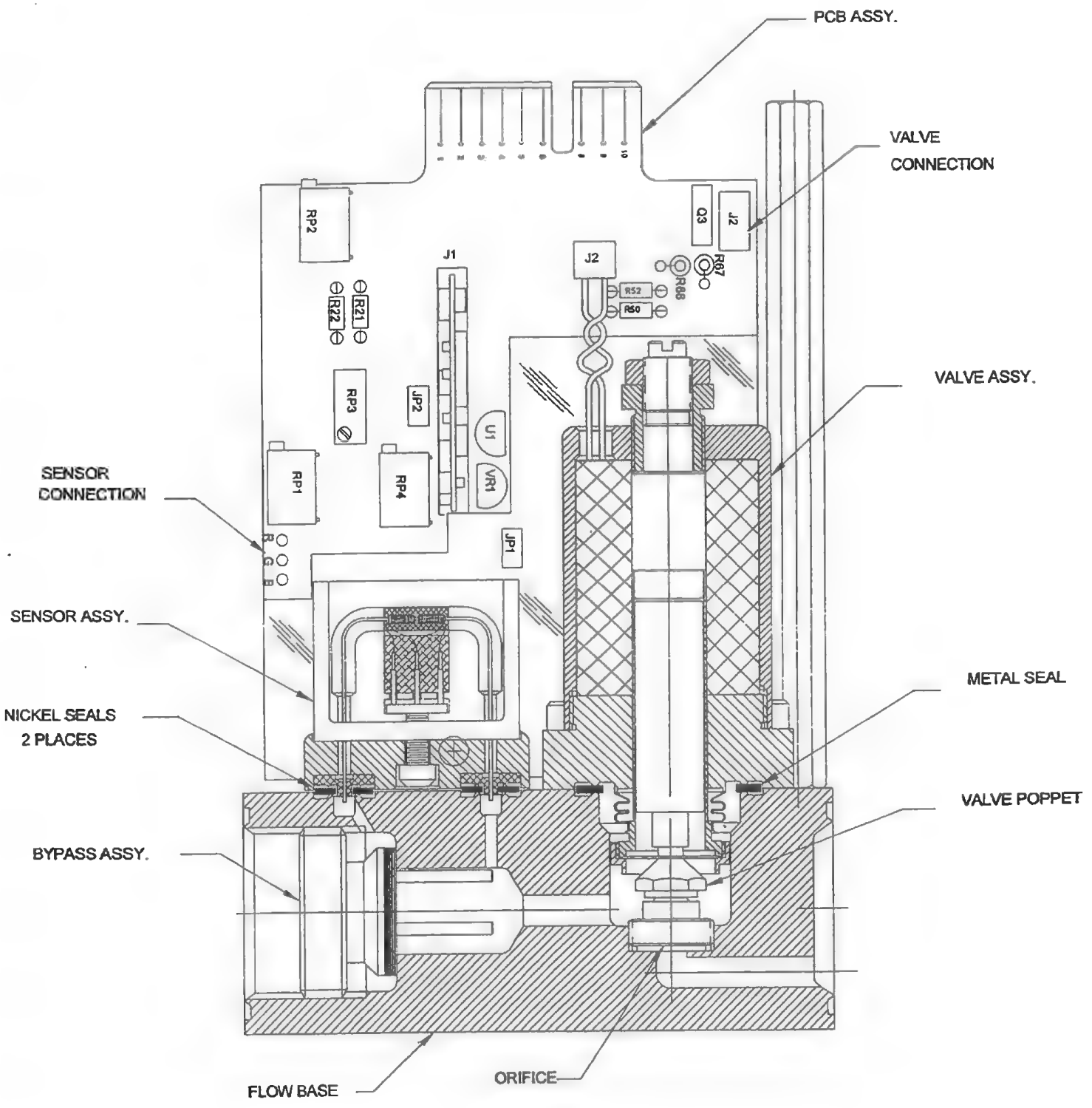


Figure 17
Cross Sectional View
FC-2900M Series (Metal Sealed)

MNL-077

Table 9. - Parts List (Elastomer Sealed Instruments)

DESCRIPTION	Part Number
Cover (FC-2900(O), 2910(O), (FM-3900, 3910)	908459-001
Cover (FC-2901(O), 2911(O)), (FM-3901, 3911)	908459-003
Cover (FC-2902(O), 2912(O), (FM-3902, 3912)	909377-001
Assembly, Circuit Board (FC-2900, 2910), (FM-3900, 3910)	908844-001
Assembly, Circuit Board FC-2900O, FC-2910O	909547-001
Assembly, Circuit Board (FC-2901, 2911), (FM-3901, 3911)	908854-001
Assembly, Circuit Board (FC-2902, 2912), (FM 3902, 3912)	909364-001
Assembly, Circuit Board (FC-2901(O), 2911(O))	909547-002
Assembly, Circuit Board (FC-2902(O), 2912(O))	909547-003
Bypass Assembly (FC-2900(O), 2901(O), 2902(O), (FM-3900, 3901, 3902)	3505-114-XXX
Bypass Assembly [FC-2910(O), 2911(O), 2912(O)], (FM-3910, 3911, 3912)	3518-210-XXX

Fittings, Outlet (1 per assy)	**FC-2900(O),2901(O), 2902(O) FM-3900, 3901, 3902**	**FC-2910(O),2911(O), 2912(O) FM-3910, 3911, 3912**
1/8-inch Swagelok (316 SS)	SS-200-1-6ST	N/A
1/4-inch Swagelok (316 SS)	SS-400-1-6ST	SS-400-1-6ST
1/4-inch VCR (316 SS)	909769-001	909769-001
1/4-inch VCO (316 SS)	909943-001	909943-001
3/8-inch Swagelok (316 SS)	N/A	SS-600-1-6ST
3/8-inch VCR (316 SS)	N/A	908518-001
Fittings, Inlet (1 per assy)		
1/8-inch Swagelok (316 SS)	SS-200-1-6ST	N/A
1/4-inch Swagelok (316 SS)	SS-400-1-6ST	908468-002
1/4-inch VCR (316 SS)	909769-001	908468-006
1/4-inch VCO (316 SS)	909943-001	908468-004
3/8-inch Swagelok (316 SS)	N/A	908468-003
3/8-inch VCR (316 SS)	N/A	908468-007

O-rings, Fittings (1 per fitting)	**Viton**	**Neoprene**	**Kalrez**
FC-2900 (Inlet/Outlet)	200472-015	200475-015	200471-015
FC-2900/2910 VCO face seals	200472-016	200475-016	200471-016
FC-2910 (Inlet)	200472-004	200475-004	200471-004
FC-2910 (Outlet)	200472-015	200475-015	200471-015

	Viton	**Neoprene**	**Kalrez**
Sensor Assembly	902343-008	902343-010	902343-012
Seals only (2 required)	908030-001	908030-002	908030-005

Mass Flow Controller Only

	Viton	**Neoprene**	**Kalrez**
Orifice O-Ring only	200472-002	200475-002	200471-002
Valve Assembly (normally closed with seal),			
10-1000 SCCM	904149-023	904149-025	904149-024
1001-9999 SCCM	904149-011	904149-015	904149-013
10-30 SLM	904149-020	904149-022	904149-021
(normally open with seal))			
10-1000 SCCM	904149-031	904149-033	904149-032
1001-9999 SCCM	904149-034	904149-036	904149-035
10-30 SLM	904149-037	904149-039	904149-038
Seal only (1 required)	200472-003	200475-003	200471-003

	Part Number
Valve Orifice Kit (contains all 5 orifices)	905491-001
10 sccm - 99 sccm	904006-006
100 sccm - 499 sccm	904006-001
0.5 slm - 2.9 slm	904006-002
3 slm - 9.9 slm	904006-005
10 slm - 30 slm	904006-007

Table 9 (Continued)

O-Ring Kit (Contains all O-rings except sensor)	
FC-2900(O), FC-2901(O), FC-2902(O), FM-3900, FM3901, FM3902	
Reactive VCR or Swagelok	909548-001
Reactive VCO	909548-004
Non-Reactive VCR or Swagelok	909548-002
Non-Reactive VCO	909548-005
NH3 VCR or Swagelok	909548-003
NH3 VCO	909548-006
FC-2910(O), FC-2911(O), FC-2912(O), FM03910, FM 3911, FM-3912	
Reactive VCR or Swagelok	909548-007
Reactive VCO	909548-010
Non-Reactive VCR or Swagelok	909548-008
Non-Reactive VCO	909548-011
NH3 VCR or Swagelok	909548-009
NH3 VCO	909548-012

Table 10. - Parts List (Metal Sealed Instruments)

DESCRIPTION	Part Number
Cover (FC-2900M(O), FM-3900M)	908459-001
Cover (FC-2901M(O), FM-3901M)	908459-003
Cover (FC-2902M(O), FM-3902M)	908459-003 with adapter plate 909417-002
Assembly, Circuit Board (FC-2900M, FM-3900M)	908844-001
Assembly, Circuit Board (FC-2900MO)	909547-001
Assembly, Circuit Board (FC-2901M, FM-3901M)	908854-001
Assembly, Circuit Board (FC-2901MO)	909547-002
Assembly, Circuit Board (FC-2902M, FM-3902M)	909364-001
Assembly, Circuit Board (FC-2902MO)	909547-003
Bypass Assembly (non EP)	3518-210-XXX
Bypass Assembly (electropolished)	908805-XXX
Fittings, Inlet/Outlet (2 per assembly required)	
1/4-inch VCR (non E.P.)	908864-001
3/8-inch VCR (non E.P.)	908947-001
1/4-inch Weld tube (non E.P.)	908806-003
1/4-inch VCR (E.P.)	908864-002
3/8-inch VCR (E.P.)	908947-002
1/4-inch Weld tube (E.P.)	908806-002
Metal seals, fittings (2 required)	908928-004
Sensor Assembly	908858-001
Metal seals only (2 required)	908928-006
Mass Flow Controller Only	
Valve Assembly	
Normally Closed	
10-499 SCCM	908778-004
500-9999 SCCM	908778-001
10-30 SLM	908778-003
(Normally Open)	
10-499 SCCM	908778-008
500-9999 SCCM	908778-005
10-30 SLM	908778-007
Metal seal	908928-005
Valve Orifices (including teflon seal)	
10 sccm - 99 sccm	908812-001
100 sccm - 499 sccm	908812-002
0.5 slm - 2.9 slm	908812-003
3 slm - 9.9 slm	908812-004
10 slm - 30 slm	908812-006
Teflon seal only	200635-002

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APPENDIX A

MASS FLOW CONVERSION FACTORS

Gas Conversion Factors Equations (PS-1064I)
Gas Conversion Factor Table (PS-1064T)
(relative to nitrogen N_2 and test gas)

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Tylan General

PROCEDURE

REV	DESCRIPTION	DATE	DCR/EO	INITL
P	ADD FORMULA IN SIMPLIFIED FORM FOR MFG PRODUCTION		16433	
N	ADD EQUATION FOR FLOW LIMITER OPTION	8/27/96	16064	BG
M	GAS CONVERSION FACTOR EQUATIONS	2/9/96	15031	AFG
L	(SHT 4 & 5) CHG CALIBRATION TEST GAS FOR CF_4 WAS C_2H_6 IS CF_4 CHG CONV FACTOR REL TO TEST GAS FOR CF_4 WAS .874 IS 1.000 AND TO N_2 FOR CF_4 WAS .420 IS .428	08/11/95	14612	KDM
K	(SHT 5) CHG DYNAMIC RESPONSE TEST GAS FOR HCI FROM Ar TO N20	04/17/95	14068	KDM
J	REVISED C2F6 CONVERSION FACTOR RELATIVE TO SF6 (WAS.911 IS .942), ADDED NOTE "**DERIVED EMPIRICALLY", ADDED EQUATION 11, SEE E.O.	04/05/94	13684	KDM
H	ELIMINATE USE OF FREON GASES/UPDATE TABLES, REVISED CONVERSION TABLES & EQUATIONS PER E.O. (SEE HISTORY FILE)	02/01/94	13602	KDM
G	BCI3 C.F. FOR N2 WAS .408 IS.430, BCI3 C.F. FOR TEST GAS WAS .815 IS .88 WATER VAPOR C.F. WAS .815 IS .88	12/20/93	12212	KDM
F	REVISED PER E.O.	10/16/90	11404	KDM
E	REVISED SHT'S 5 & 7 PER E.O.	07/15/88	9843	KDM
D	EXTENSIVE CHANGES, SEE HISTORY FILE	11/11/87	9477	KM
C	REVISED & RETYPED	11/07/84	5403	RB
B	REVISED FORMAT	01/10/84	3695	NN
A	INITIAL RELEASE	09/22/83	3126	RB

1.0 PURPOSE

DOCUMENT TITLE: GAS CONVERSION FACTOR EQUATIONS		
ORIGINATOR	DOCUMENT NUMBER	REVISION
MICHAEL RAYCRAFT	PS-1064I	P

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THIS DOCUMENT IS INTENDED TO HELP SELECT A CONVERSION FACTOR RELATIVE TO THE TEST GAS THAT CLOSELY MATCHES THE SAMPLE GAS.

EXAMPLE:

SAMPLE GAS	TEST GAS	CONVERSION FACTOR RELATING TO TEST GAS
AMMONIA (NH3) C.F. = 0.719	NITROUS OXIDE (N20) C.F. = 0.709	.719/.709 = 101.4%

ONCE A TEST GAS CONVERSION FACTOR IS SELECTED (101.4%), DIVIDE THE DESIRED RATE OF FLOW BY THAT PERCENTAGE TO OBTAIN THE ADJUSTED FLOW RATE FOR THE STANDARD UNIT (UNIT CALIBRATED FOR THE TEST GAS).
DECENT

2.0 EXAMPLES

EXAMPLE 1: (USING C.F. RELATIVE TO TEST GAS)
REQUIRED CALB: 1000 SCCM BROMINE (BR2)(c.f. = 0.808)

A) USING N2 STANDARD AND N2 TEST GAS (C.F. = 100)

$\frac{0.808}{1.00}$ = 0.808 FOR 1000 SCCM BR2 NEED $\frac{1000}{0.808}$ = 1238 N2

USE A 1500 SCCM STANDARD, Vo OF STANDARD $\frac{1238}{1500}$ X 5V = 4.13V OR 82.5% F.S.

B) USING N2O STANDARD AND N20 TEST GAS (C.F. = 0.709)

$\frac{0.808}{0.709}$ = 1.140 FOR 1000 SCCM BR2 NEED $\frac{1000}{1.140}$ = 877 N2O

USE A 1000 SCCM STANDARD, Vo OF STANDARD $\frac{877}{1000}$ X 5V = 4.38V OR 87.6% F.S.

EXAMPLE 2:
REQUIRED CALB: 1000 SCCM 10% ARGON 90% CO2
(C.F. RELATIVE TO N2 = .777) SEE PAGE 7 EXAMPLE

A) USING N2O STANDARD AND N2O TEST GAS (C.F. = 0.709)

$\frac{0.777}{0.709}$ = 1.096 FOR 1000 SCCM 10% Ar 90% CO2 NEED $\frac{1000}{1.096}$ = 912.4 N2O

USE A 1000 SCCM STANDARD Vo OF STANDARD $\frac{912.4}{1000}$ X 5V =

= 91.2% FULL SCALE TEST STANDARD
= 91.2%/2 = 45.6% LINEARITY TEST STANDARD



1 SLPM 10% ARGON 90% CO2 @ 1 SLPM N20

100% (F.S.) = 109.6% 91.2%
LINEARITY = 50% 45.6%

B) ALTERNATIVE METHOD USING N20 STD & TEST GAS (C.F. = 0.709)

$\frac{0.777}{0.709}$ = 1.096 X 100%(F.S.) = 109.6%

= 1.096 X 50%(LIN) = 54.8%



1 SLPM 10% ARGON 90% CO2 @ 1 SLPM N20

100% (F.S.) = 100% 100%
50% (LIN) = 54.8% 50%

NOTE: ASSUMING ACTUAL FLOW OF TEST STANDARD PER CALIBRATION STICKER IS 100% @ F.S. AND 50% @ LINEARITY.

3.0 DETERMINING MASS FLOW CONVERSION FACTORS

THE RATIO OF A FLOW W_1 FOR A PARTICULAR GAS, TO THE FLOW W_2 OF A DIFFERENT GAS WHICH WILL PRODUCE THE SAME OUTPUT VOLTAGE IN A FLOW METER OR CONTROLLER IS EQUAL TO THE RATIO OF THEIR RESPECTIVE CORRECTION FACTORS. THIS RELATIONSHIP IS SHOWN IN EQUATION 1 BELOW:

$$\frac{C_1}{C_2} = \frac{W_1}{W_2} \qquad \text{(EQUATION 1)}$$

EQUATION 2 SHOWS THE RELATIONSHIP BETWEEN THE PROPERTIES OF A GAS AND ITS RATE OF FLOW IN STANDARD CUBIC CENTIMETERS PER MINUTE FOR EQUAL OUTPUT VOLTAGES:

$$W = \frac{KN}{\rho Cp} \qquad \text{(EQUATION 2)}$$

Cp IS THE SPECIFIC HEAT IN CAL/g-C IS ρ IS THE DENSITY OF THE GAS AT OC IN g/LITER, AND N IS A CORRECTION FACTOR FOR THE MOLECULAR STRUCTURE OF THE GAS. THE VALUES OF N ARE LISTED IN TABLE 1. FOR A PARTICULAR FLOW METER OR CONTROLLER, K IS A CONSTANT.

SUBSTITUTING EQUATION 2 INTO EQUATION 1 YIELDS THE FOLLOWING RELATIONSHIP:

$$\frac{C_1}{C_2} = \frac{\dfrac{KN_1}{\rho_1 Cp_1}}{\dfrac{KN_2}{\rho_2 Cp_2}} \qquad \text{(EQUATION 3)}$$

IF WE DEFINE BOTH THE CONVERSION FACTOR AND THE CORRECTION FACTOR N, FOR THE GAS NITROGEN TO BE 1.000, EQUATION 3 CAN BE REDUCED TO THE FOLLOWING:

$$C = \frac{(\rho N_2 Cp_{N2})N_1}{\rho_1 Cp_1} \qquad \text{(EQUATION 4)}$$

BY DROPPING THE SUBSCRIPT 2 AND REPLACING ρN_2 AND CpN_2, THE DENSITY AND SPECIFIC HEAT RESPECTIVELY OF NITROGEN, WITH THE ACTUAL VALUES, E HAVE AN EQUATION FOR CALCULATING THE FLOW CONVERSION FACTORS:

$$C = \frac{.3108}{\rho Cp} \times N \qquad \text{(EQUATION 5)}$$

DOCUMENT PS-1064T LISTS THE CONVERSION FACTORS FOR COMMONLY USED GASES AS DERIVED FROM EQUATION 5.

SUBSTITUTING THE FLOW RANGE AND CONVERSION FACTOR OF THE FACTORY-CALIBRATED GAS AND THE CONVERSION FACTOR OF A DIFFERENT GAS INTO EQUATION 1 YIELDS THE FLOW RANGE FOR THE DIFFERENT GAS. EXAMPLE:

A FLOW METER IS CALIBRATED FOR H2 AND THE FLOW RATE IS 5000 SCCM FOR 5.00 VDC OUTPUT. THE FLOW RATE FOR NITROUS OXIDE AT 5.00 VDC OUTPUT IS:

$$\frac{1.01}{.709} = \frac{5000}{W}$$

$$W = \frac{.709 \times 5000}{1.01} = 3510 \text{ SCCM}$$

DOCUMENT PS-1064T LISTS THE CONVERSION FACTORS FOR INSERT, LIMITER FOR COMMONLY USED GASES.

WHEN CALCULATING THE CONVERSION FACTOR FOR FLOW LIMITER OPTION COLUMN, THE FOLLOWING EQUATION SHOULD BE USED:

$$\text{FLOW}_{.0145} \text{ @45 PSIG} = 4 \text{ SLM} \times \sqrt{\frac{\rho N_2}{\rho Gas}}$$

EXAMPLE:

$$\text{FLOW } \varnothing_{ORIFICE} = \text{FLOW } \varnothing_{.0145} \times \left(\frac{\varnothing_{ORIFICE}}{.0145}\right)^2$$

WHEN CALCULATING THE CONVERSION FACTOR FOR A MIXTURE OF TWO OR MORE GASES, THE FOLLOWING EQUATIONS SHOULD BE USED:

THE EQUIVALENT DENSITY IS GIVEN BY THE RELATIONSHIP:

$$\rho = \frac{W_1}{W_T}\rho_1 + \frac{W_2}{W_T}\rho_2 + \ldots + \frac{W_n}{W_T}\rho_n \qquad \text{(EQUATION 6)}$$

WHERE:

W_1 IS THE FLOW OF GAS #1 — N_1 IS THE MOLECULAR C.F. OF GAS #1
W_2 IS THE FLOW OF GAS #2 — N_2 IS THE MOLECULAR C.F. OF GAS #2
W_n IS THE FLOW OF GAS #n — N_n IS THE MOLECULAR C.F. OF GAS #n
W_T IS THE TOTAL FLOW

ρ_1 IS THE DENSITY OF GAS #1
ρ_2 IS THE DENSITY OF GAS #2
ρ_n IS THE DENSITY OF GAS #n

CP_1 IS THE SPECIFIC HEAT GAS #1
CP_2 IS THE SPECIFIC HEAT GAS #2
CP_n IS THE SPECIFIC HEAT GAS #n
THE EQUIVALENT SPECIFIC HEAT IS GIVEN BY EQUATION:

$$Cp = F_1Cp_1 + F_2Cp_2 + \ldots + F_nCp_n \qquad \text{(EQUATION 7)}$$

WHERE:

$$F1 = \frac{W_1\rho_1}{W_T\rho}, \quad F2 = \frac{W_2\rho_2}{W_T\rho} + \ldots Fn = \frac{W_n\rho_n}{W_T\rho} \qquad \text{(EQUATION 8)}$$

THE EQUIVALENT VALUE OF N IS:

$$N = \frac{W_1}{W_T}N1 + \frac{W_2}{W_T} + \ldots + \frac{W_n}{W_T}N \qquad \text{(EQUATION 9)}$$

FOR GAS MIXTURES, THE VALUES ρ, Cp, AND N FROM EQUATIONS 6, 7, & 9 ARE USED IN EQUATION 5.

COMBINING NUMBERS 5 THROUGH 9 RESULTS IN THE FOLLOWING EQUATION:

$$C = \frac{.3106\ \left(\frac{W_1}{W_T}N1 + \frac{W_2}{W_T}N_2 + \ldots + \frac{W_n}{W_T}N_n\right)}{\frac{W_1}{W_T}\rho_1 Cp_1 + \frac{W_2}{W_T}\rho_2 Cp_2 + \ldots + \frac{W_n}{W_T}\rho_n Cp_n} \quad \text{(EQUATION 10)}$$

OR

$$C = \frac{\frac{W_1}{W_T}N1 + \frac{W_2}{W_T}N_2 + \ldots + \frac{W_n}{W_T}N_n\right)}{\frac{W_1}{W_T} X \frac{N_1}{C_T} + \frac{W_2}{W_T} X \frac{N_2}{C_T} + \ldots + \frac{W_N}{W_T} X \frac{N_n}{C_T}} \quad \text{(EQUATION 11)}$$

EXAMPLE: 10% ARGON 90% CO2
LET ARGON BE GAS #1 & CO2 GAS #2

$$C.F. = \frac{.3106\ \left(\frac{10}{100} x\ 1.01 + \frac{90}{100} x\ 0.94\right)}{\left(\frac{10}{100}\ 1.782\ x\ .1245\right) + \left(\frac{90}{100}\ x\ 1.964\ x\ .2017\right)}$$

$$= \frac{.3106\left(0.101 + 0.846\right)}{0.022186 + 0.356525} = \frac{.3106\ \left(0.947\right)}{0.378711} = \frac{0.2941382}{0.378711}$$

C.F. = 0.777 RELATIVE TO N2

MIX GAS CONVERSION FACTOR to N2 Equivalent

The following equation represents the calculation necessary to determine the Conversion Factor to N_2 equivalent where :

Constant= 0.3106
MF_1 = Molecular Factor of Gas 1
DEN_1 = Density of Gas 1
SH_1 = Specific Heat @ 25° C of Gas 1

$$\frac{.3106 \times ((\%\ \text{of}\ GAS_1 \times MF_1) + (\%\ \text{of}\ GAS_2 \times MF_2) + (\%\ \text{of}\ GAS_3 \times MF_3)}{(\%\ \text{of}\ GAS_1 \times DEN_1 \times SH_1) + (\%\ \text{of}\ GAS_2 \times DEN_2 \times SH_2) + (\%\ \text{of}\ GAS_3 \times DEN_3 \times SH_3)}$$

NOTE:M
1.) The percentage of Gases must add up to 100 % . When using formula % is stated in decimals. e.g. 15 % = 0.15
2.) If there are more than three gases, insert additional data of GAS_4.
3.) If there are less than three, do not put any data in for three.

Instructions:

1.) FIND MF for each GAS from Table 1 of PS-1064 (PS 1064 T might have factor installed later)
Monotomic Gases (gases with only one molecule) e.g. AR, HE, NE etc. = 1.01
Diatomic Gases (Gases with two molecules) e.g. CO, N_2, O_2, NO etc. = 1.00
Triatomic Gases (Gases with three molecules) e.g. CO_2, N_2O, SO_2 etc. = 0.94
Polyatomic Gases (gases with four or more molecules) e.g. NH_3, PH_3, CH_4 etc. =0.88

2.) FIND DEN for each GAS from PS-1064T column headed "DENSITY"
3.) FIND SH for each GAS from PS-1064T column headed "Specific Heat @ 25° C
4.) Use formula above:

Step 1.) Multiply % of Gas times MF for each gas
Step 2.) Add the results for each MF Gas
Step 3.) Multiply that result by the Constant 0.3106

Step 4.) Multiply the % of Gas times it's DENSITY times it's SPECIFIC HEAT for each gas.
Step 5.) Add those results together
Step 6.) Divide the result from Step 3 by the result from Step 5.

FLOW LIMITER SELECTION INSTRUCTIONS
for use with PS-1064T Limiter Options

The flow limiter option table in PS-1064T is to be used to determine the appropriate size limiter to be installed for specific gases and ranges. The PS-1064T table refers to (-001, -002,-003,-005,-006,-011) dash numbers to be installed. Use Part Number 909804-xxx for FC-2979 and FM-3979 series. Use Part Number 909805-xxx for FC-2936M , FM-3936, FC-2900M, and FM-3900M series.

The flow rates listed in the tabular section of PS-1064T under Limiter Options are the maximum purge flow rates (in SLPM) for the different flow limiter dash numbers.

NOTE: Do NOT use conversion factors to N_2 equivalent when using this table. Take the range and gas as stated.

Convert all ranges to SLPM equivalent. If SCCM divide by 1,000.
e.g. 500 SCCM becomes 0.5 SLPM, 10,000 SCCM becomes 10.0 SLPM

There are three multiplication factors to size the purge. SEE table below:

Flow in SCCM	0 to 1,000	1,001 to 1,700	1,701 to up
Flow in SLPM	0 to 1.0	1.001 to 1.700	1.701 to up
Multiply SLPM by	10	1	6

Find the appropriate gas.

If the Range is 0 to 1.0 SLPM:
Multiply the range (in SLPM) by 10 and find the column where the flow rate first exceeds the calculated number. The Dash number in that column is the correct size.

If the Range is 1.001 to 1.700 SLPM:
Multiply the range (in SLPM) by 1 and find the column where the flow rate first exceeds 10 SLPM. The Dash number in that column is the correct size.

If the Range is 1.701 to up:
Multiply the range (in SLPM) by 6 and find the column where the flow rate first exceeds the calculated number. The Dash number in that column is the correct size.

Examples:

FC-2979M 40 SCCM of HBR
1.) 40 SCCM = 0.04 SLPM of Hbr
2.) 0.04 x 10 = 0.4 SLPM
3.) 0.4 SLPM is smaller than 1.12 SLPM in column 1
4.) use part number 909804-001

FC-2979M 1.5 SLPM of HBR
1.) 1.5 x 1 =1.5
2.) 11.464 in column 5 is the first value greater than 10 SLPM
3.) use part number 909804-005

FC-2900M 5.0 SLPM of Ar
1.) 5.0 x 6 = 30 SLPM
2.) 101.978 in column 7 is the largest greater than 30.0 SLPM
3.) use part number 909805-011

4.0 CONVERSION TABLE

PSI (LBS/IN^2) X 51.7149 mmHg (O·C)

mmHg @ O·C X 0.0193368 = INCHES OF Hg

PSI (LBS/IN^2) X 2.03602 = INCHES OF Hg

Hg X (0.491154) = LBS/SQ INCHES (PSI)

STD PRESSURE IN ABSOLUTE UNIT = 14.696 PSI

(ATMOSPHERE) = 760 mmHg = 29.92 IN. OF MERCURY

ATMOSPHERIC PRESSURE (SEA LEVEL, O·C) = ONE ATMOSPHERE

INCHES OF H2O (4·C) x 0.03612628 = LBS/SQ INCH (PSI)

CUBIC FEET PER HOUR x 0.4719342 = LITERS PER MINUTE

CUBIC FEET PER MINUTE x 28.316052 = LITERS PER MINUTE (lpm)

SLPM (LITERS/MIN.) x 0.0353157 = cfm

CUBIC CENTIMETER x 0.001 - LITERS

LITERS x 1000 = CUBIC CENTIMETERS

CUBIC FEET/MINUTE x 60 = CUBIC FEET/HOUR

C° = 5/9 (F-32°)

F° = 9/5C° + 32

1 TORR = 1 mmHg @ O°C = 0.0013157895 ATMOSPHERE

1 TORR x 0.0193368 = PSI

PSI x 51.7149 = TORR OF mmHg

LB/HOUR x 6.0533 = L/MINUTE

TABLE 1

VALUES FOR THE MOLECULAR CORRECTION FACTOR

	N
MONOTAMIC GASES (e.g. ARGON, HELIUM, XENON)	1.01
DIATOMIC GASES (e.g. CARBON MONOXIDE, NITROGEN, OXYGEN, NITRIC OXIDE)	1.00
TRIAMTOMIC GASES (e.g. CARBON DIOXIDE, NITROUS OXIDE, SULFUR DIOXIDE)	0.94
POLYATOMIC GASES (e.g. AMMONIA, ARSINE, DIBORANE, ETHANE, METHANE, PHOSPHINE)	0.88

NOTE: STANDARD PRESSURE IS DEFINED AS 760mmHg (14.7 PSIG).
STANDARD TEMPERATURE IS DEFINED AS 0°C.

5.0 APPLICABLE DOCUMENTS

5.1 PS-1033, CALIBRATION PROCEDURE FOR MFC'S & MFM
PS-1183, BYPASS SELECTION CHARTS

5.2 SOURCES OF DATA

1. MATHESON GAS DATA BOOK; BRAKER, WILLIAM & MOSSMAN, ALLENL.; 6th EDITION 1980
2. JANAF THERMOCHEMICAL TABLES; CHASE, M.W., et.at.; 3rd EDITION, 1985
3. LANGE'S HANDBOOK OF CHEMISTRY; 13TH EDITION; DEAN, JOHN A., McGRAW HILL, 1985
4. TOULOUKIAN, Y.S. et.al.; VISCOSITY, NONMETALLIC LIQUIDS & GASES, IFI/PLENUM DATA CORP., 1975
5. HANDBOOK OF CHEMISTRY & PHYSIS; 70TH EDITION; CRC PRESS, 1989-90
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14. CALCULATED FROM EXPERIMENTAL DATA
15. SCHUMACHER TRANS-LC DATA SHEET5

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UPDATES TO PS1064T (TABLE) WILL NOT REQUIRE UPDATE TO PS1064I (INSTRUCTIONS).

COLOR SEQUENCE:

REVISION	COLOR	REVISION	COLOR
L	WHITE/BLUE	AF	GREEN
M	PINK	AG	BLUE
N	YELLOW	AH	PINK
O	NOT USED	AJ	YELLOW
P	GREEN	AK	GREEN
Q	BLUE	AL	BLUE
R	PINK	AM	PINK
S	YELLOW	AN	YELLOW
T	GREEN	AO	GREEN
U	BLUE	AP	BLUE
V	PINK	AQ	PINK
W	YELLOW	AR	YELLOW
X	GREEN	AS	GREEN
Y	BLUE	AT	BLUE
Z	PINK	AU	PINK
AA	YELLOW	AW	YELLOW
AB	GREEN	AX	GREEN
AC	BLUE	AY	BLUE
AD	PINK	AZ	PINK
AE	YELLOW		

7.0 *PRESSURE OPERATIONS*

(ALL GAS RANGES ARE RELATIVE TO N2)

Product	Range	Operating Pressure	
FC-2900	10 SCCM - 9.99 SLPM 10 SLPM - 30 SLPM	10 - 40 PSI 15 - 40 PSI	
FC-2979	10 SCCM - 9.99 SLPM 10 SLPM - 20 SLPM	10 - 40 PSI 15 - 40 PSI	
FC-2930	10 SCCM - 9.99 SLPM 10 SLPM - 30 SLPM	10 - 40 PSI 15 - 40 PSI	
VC-4900	10 SCCM - 9.99 SLPM 10 SLPM - 30 SLPM	10 - 40 PSI* 15 - 40 PSI*	*Unless Specified by the customer.
VC-4979	10 SCCM - 9.99 SLPM 10 SLPM - 30 SLPM	10 - 40 PSI* 15 - 40 PSI*	*Unless Specified by the customer.
FC-2950	10 SCCM - 9.99 SLPM 10 SLPM- 20 SLPM 20.1 SLPM - 30 SLPM	10 - 40 PSI* 15 - 40 PSI* 30 - 60 PSI*	*Unless Specified by the customer.
FC-2960	50 SCCM - 9.99 SLPM 10 SLPM - 20 SLPM	10 - 40 PSI* 15 - 40 PSI*	*Unless Specified by the customer

TEL OPERATING PRESSURE

Product	Range	Operating Pressure
FC-2936 MO	5 SCCM - 5.9 SLPM 6 SLPM - 20 SLPM 21 SLPM - 30 SLPM	7 - 40 PSID 14 - 40 PSID 21 - 30 PSID
FC-2936 MC	5 SCCM - 5.9 SLPM 6 SLPM - 20 SLPM 21 SLPM - 30 SLPM	7 - 40 PSID 14 - 40 PSID 21 - 30 PSID
FM-3936	5 SCCM - 5.9 SLPM 6 SLPM - 20 SLPM 21 SLPM - 30 SLPM	7 - 40 PSID 14 - 40 PSID 21 - 30 PSID*

**Tylan General guideline for 21-30 SLPM (14 - 40 PSID)*

SPECIAL GASES
(MOST COMMONLY USED)

Name plate Gas	Range	Operating Pressure
CLF3	N/A	3 - 12 PSI
C4F8	N/A	3 - 12 PSI
BCL3	N/A	3 - 12 PSI
SICL4	N/A	3 - 12 PSI
WF6	N/A	3 - 12 PSI
HF	N/A	3 - 12 PSI
C4H6	N/A	3 - 12 PSI

NOTE: ***IF OTHER GASES ARE USED, OTHER THAN THE LIST ABOVE, PLEASE REFER TO PS-1064T.***

Tylan General

PROCEDURE

REV	DESCRIPTION	DATE	DCR/EO	INITL
R	DELETED NOTE 8,MODIFIED NOTE 7,MODIFIED CODE 174.		16910	
P	CORRECTED: DENSITY, STD CAL PRESS/ADD:HEAT CP/DELETED: DUPLICATE GAS DATA/INCORP. R1 CHG'S	5/21/97	16433	BG
N	ADD COLUMN: FLOW LIMITER OPTION SEE E.O.	8/27/96	16064	BG
M	INITIAL RELEASE	2/9/96	15031	BG

DOCUMENT TITLE:

GAS CONVERSION FACTOR TABLE

ORIGINATOR	*DOCUMENT NUMBER*	*REVISION*
MICHAEL RAYCRAFT	PS-1064T	R

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COD	GAS	SYMBOL	CAL. TEST GAS	CONV FACTOR REL. TO CAL TEST GAS	CONV FACTOR REL. TO CAL N2	DYNAMIC RESP. TEST GAS	HEAT Cp x 10³ (J/K·mol)	SPECIFIC HEAT TEMP (°C)	SPECIFIC Heat @ 25°C Cp(cal/g°C)	DENSITY g/l @ 0°C	SUG. SEAL MAT.	STD CAL PRESS (PSID)	MOLECULAR FACTOR
25	Acetylene	C2H2	C2H4	0.973	0.581	N2	44.0950	25.00	0.4049	1.1620	V	10 to 40	0.88
1	Air	-	N2	1.006	1.006	N2	29.1300	26.85	0.2389	1.2930	V	10 to 40	1.00
32	Allene (Propadiene)	C3H4	C2H6	0.877	0.421	Ar	60.8400	25.00	0.3633	1.7870	V	10 to 40	0.88
125	Ammonia	NH3	N2O	1.014	0.719	N2	35.6520	25.00	0.5005	0.7600	N	10 to 40	0.88
2	Argon	Ar	Ar	1.000	1.415	Ar	20.7860	25.00	0.1245	1.7820	V	10 to 40	1.01
4	Arsine	AsH3	N2O	0.950	0.673	N2O	38.0744	25.00	0.1168	3.4780	V	10 to 40	0.88
201	Acetaldehyde	CH3CH0	C2H4	0.944	0.634	N20	40.5500	25.00	0.2200	1.9648	KZ	3 to 12	0.88
200	Benzene	C6H6	C2H4	0.682	0.482	N2	137.3400	25.00	0.4202	0.9550	V	3 to 12	0.88
7	Boron Tribromide	BBr3	C2H6	0.787	0.231	SF6	67.7710	25.00	0.0647	11.1800	KZ	3 to 12	0.88
8	Boron Trichloride	BCl3	C2H6	0.895	0.430	SF6	59.6600	25.00	0.1217	5.2270	KZ	3 to 12	0.88
9	Boron Trifluoride	BF3	C2H6	1.057	0.508	N2O	50.4460	25.00	0.1779	3.0250	KZ	10 to 40	0.88
10	Bromine	Br2	N2O	1.140	0.808	SF6	36.0480	25.00	0.0539	7.1300	V	10 to 40	1.00
12	Bromine Pentafluoride	BrF5	SF6	0.956	0.253	SF6	101.4040	25.00	0.1387	7.8030	KZ	3 to 12	0.88
11	Bromine Trifluoride	BrF3	C2H6	0.802	0.385	SF6	66.4970	25.00	0.1162	6.1080	KZ	10 to 40	0.88
63	1,3-Butadiene	C4H6	SF6	1.181	0.312	N2O	82.1320	25.00	0.3632	2.4130	V	6 to 24	0.88
61	n-Butane	n-C4H10	SF6	0.966	0.255	N2O	100.3650	25.00	0.4130	2.5930	V	6 to 24	0.88
67	1-Butene	C4H8	SF6	1.110	0.294	N2O	87.3290	25.00	0.3787	2.4603	V	6 to 24	0.88
161	2-Butene CIS	C4H8	SF6	1.070	0.283	N2O	90.6160	25.00	0.3930	2.5030	V	6 to 24	0.88
162	2-Butene TRANS	C4H8	SF6	1.187	0.314	N2O	81.6950	25.00	0.3483	2.5030	V	6 to 24	0.88
93	Carbon Dioxide	CO2	N2O	1.040	0.737	N2O	37.1290	25.00	0.2017	1.9640	V	10 to 40	0.94
97	Carbon Disulfide	CS2	C2H4	1.003	0.600	N2O	45.6640	25.00	0.1434	3.3970	V	10 to 40	0.94
92	Carbon Monoxide	CO	N2	1.000	1.000	N2	29.1420	25.00	0.2488	1.2500	V	10 to 40	1.00
77	Carbon Tetrachloride	CCl4	SF6	1.163	0.307	SF6	83.4010	25.00	0.1297	6.8600	V	3 to 12	0.88
95	Carbonyl Fluoride	COF2	C2H4	0.908	0.542	N2O	47.2550	25.00	0.1712	2.9450	KZ	10 to 40	0.88
96	Carbonyl Sulfide	COS	N2O	0.931	0.660	N2O	41.4970	25.00	0.1652	2.6800	KZ	10 to 40	0.94
13	Chlorine	Cl2	N2O	1.210	0.858	N2O	33.9490	25.00	0.1145	3.1630	V	10 to 40	1.00
91	Chlorine Trifluoride	ClF3	C2H6	0.836	0.401	N2O	63.8450	25.00	0.1651	4.1250	KZ	3 to 12	0.88
88	Chloroform	CHCl3	C2H6	0.816	0.392	SF6	65.3830	25.00	0.1310	5.3260	V	3 to 12	0.88
42	Cyanogen	C2N2	C2H6	0.940	0.452	N2O	56.7460	25.00	0.2608	2.3220	V	10 to 40	0.88
16	Cyanogen Chloride	ClCN	C2H4	1.012	0.605	N2O	45.2710	25.00	0.1762	2.7420	V	10 to 40	0.94
202	Cyclopentane	C5H10	C2H6	0.883	0.424	N20	60.4000	25.00	0.2060	3.1290	KZ	3 to 12	0.88
48	Cyclopropane	C3H6	C2H6	0.927	0.445	N2O	57.5590	25.00	0.3272	1.8770	V	10 to 40	0.88
98	Deuterium*	D2	N2	0.998	0.998	He	29.1940	25.00	1.7325	0.1798	V	10 to 40	1.00
5	Diborane	B2H6	C2H6	0.918	0.441	N2	58.1000	25.00	0.5020	1.2350	V	10 to 40	0.88
163	Dibromodiflouromethane	CBr2F2	SF6	1.210	0.320	SF6	80.1238	25.00	0.0913	9.3620	V	3 to 12	0.88
164	Dibromomethane	CH2Br2	C2H6	0.978	0.470	SF6	54.5400	25.00	0.0750	7.7600	V	10 to 40	0.88
165	Dichloromethylsilane	(CH3)2SiCl2	SF6	0.955	0.252	SF6	101.6100	25.00	0.1882	5.7580	KZ	3 to 12	0.88
143	Dichlorosilane	SiH2Cl2	C2H6	0.858	0.412	SF6	62.1740	25.00	0.1472	4.5060	KZ	3 to 12	0.88
166	Trans 1,2-Dichloroethylene	C2H2Cl2	SF6	0.885	0.234	SF6	109.5200	25.00	0.2700	4.3280	V	10 to 40	0.88

COD	GAS	SYMBOL	CAL. TEST GAS	CONV FACTOR REL. TO CAL TEST GAS	CONV FACTOR REL. TO CAL N2	DYNAMIC RESP. TEST GAS	HEAT Cp x 10^3 (J/K-mol)	SPECIFIC HEAT TEMP (°C)	SPECIFIC Heat @ 25°C Cp(cal/g°C)	DENSITY g/l @ 0°C	SUG. SEAL MAT.	STD CAL PRESS (PSID)	MOLECULAR FACTOR
167	Difluoromethane	CH2F2	C2H4	1.000	0.598	N20	43.0900	25.00	0.1980	2.3210	V	10 to 40	0.88
26	1,1-Difluoroethylene	C2H2F2	C2H6	0.887	0.426	N2O	60.1240	25.00	0.2245	2.8570	V	10 to 40	0.88
144	Dimethylamine	(CH3)2NH	C2H6	1.346	0.356	N2O	72.0400	25.00	0.3822	2.0110	KZ	3 to 12	0.88
168	Dimethyl Cadmium	(CH3)2Cd	SF6	0.736	0.194	SF6	131.8000	25.00	0.2211	6.3607	KZ	to	0.88
38	Dimethyl Ether	(CH3)2O	C2H6	0.812	0.390	N2O	65.6900	25.00	0.3411	2.0550	KZ	10 to 40	0.88
70	2,2-Dimethylpropane	C5H12	SF6	0.821	0.217	N2O	118.1560	25.00	0.3916	3.2190	V	3 to 12	0.88
140	Disilane	Si2H6	SF6	1.200	0.317	N2O	80.7930	25.00	0.3105	2.7760	V	10 to 40	0.88
37	Ethane	C2H6	C2H6	1.000	0.481	C2H6	53.3460	25.00	0.4241	1.3420	V	10 to 40	0.88
36	Ethanol	C2H6O	C2H6	0.815	0.392	N2O	65.43776	25.00	0.3398	2.0550	V	3 to 12	0.88
64	Ethyl Acetylene (1-Butyne)	C4H6	SF6	1.220	0.322	N2O	79.5000	25.00	0.3515	2.4130	V	3 to 12	0.88
169	Ethylamine	C2H5NH2	SF6	1.335	0.353	N2O	72.6342	25.00	0.3854	2.0110	V	10 to 40	0.88
35	Ethyl Chloride	C2H5Cl	SF6	1.475	0.390	N2O	65.7300	25.00	0.2436	2.8790	V	3 to 12	0.88
31	Ethylene	C2H4	C2H4	1.000	0.598	C2H4	42.8860	25.00	0.3658	1.2510	V	10 to 40	0.88
34	Ethylene Oxide	C2H4O	C2H6	1.114	0.535	N2O	47.9010	25.00	0.2601	1.9650	KZ	3 to 12	0.88
99	Fluorine	F2	N2	0.931	0.931	Ar	31.3020	25.00	0.1970	1.6950	V	10 to 40	1.00
90	Fluroform (Freon-23)	CHF3	C2H6	1.045	0.502	N2O	51.0400	25.00	0.1742	3.1270	V	10 to 40	0.88
170	Trichlorofluoromethane Freon-11	CCl3F	SF6	1.163	0.307	SF6	83.4010	25.00	0.1452	6.1290	V	3 to 12	0.88
171	Dichlorodiflouromethane Freon-12	CCl2F2	SF6	1.339	0.354	SF6	72.4040	25.00	0.1432	5.3950	V	10 to 40	0.88
172	Cholrotriflouromethane Freon-13	CClF3	SF6	1.243	0.329	SF6	78.0180	25.00	0.1786	4.6600	V	10 to 40	0.88
173	Bromotriflouromethane Freon-13B1	CBrF3	SF6	1.044	0.370	SF6	69.3300	25.00	0.1113	6.6440	V	10 to 40	0.88
174	Carbon Tetraflouride Freon-14	(7) CF4	C2H6	**0.874	**0.428	N20	61.0540	25.00	0.1659	3.9260	V	10 to 40	0.88
175	Dichloroflouromethane Freon-21	CHCl2F	C2H6	0.875	0.420	SF6	60.9950	25.00	0.1417	4.5920	KZ	3 to 12	0.88
176	Chlorodiflouromethane Freon-22	CHClF2	C2H6	0.934	0.449	N2O	57.1260	25.00	0.1580	3.8580	KZ	10 to 40	0.88
177	Trichloro-triflouroethane Freon-113	CCl2FCClF2	SF6	0.858	0.277	SF6	126.2210	60.00	0.1442	8.3600	V	3 to 12	0.88
178	1,2-Dichlorotetraflouroethane Freon-114	C2Cl2F4	SF6	0.858	0.227	SF6	112.9917	25.00	0.1581	7.6260	V	3 to 12	0.88
179	Chloropentaflouroethane Freon-115	C2ClF5	SF6	0.915	0.242	SF6	105.9700	25.00	0.1640	6.8959	V	10 to 40	0.88
24	Hexafluoroethane (Freon-116)	C2F6	SF6	**0.942	0.241	SF6	106.4070	25.00	0.1844	6.1570	KZ	10 to 40	0.88
180	Octaflourocyclobutane Freon-C318	C4F8	SF6	0.610	0.161	SF6	159.0170	25.00	0.1899	8.9370	V	6 to 24	0.88
181	Freon-152A	C2H4F2	C2H6	0.788	0.378	SF6	15.7500	25.00	0.0570	12.6800	N	10 to 40	0.88
197	Pentaflouroethane Freon-125	C2HF5	SF6	1.000	0.264	SF6	97.1400	25.00	0.1935	5.3547	V	10 to 40	0.88
198	Tetraflourethane Freon-134A	C2H2F4	C2H4	1.088	0.288	SF6	89.9800	25.00	0.2088	4.5540	V	10 to 40	0.88
104	Germane	GeH4	C2H4	0.953	0.569	N2O	45.0198	25.00	0.1405	3.4180	V	10 to 40	0.88
102	Germanium Tetrachloride	GeCl4	SF6	1.009	0.267	SF6	96.1065	25.00	0.1072	9.5650	V	3 to 12	0.88
182	Hexamethyl Disiloxane	(CH3)3Si-O-Si(CH3)3	SF6	0.310	0.082	SF6	0.0000	25.00	0.4600	7.2500	KZ	1 to 5	0.88
112	Helium*	He	He	1.000	1.415	He	20.7860	25.00	1.2418	0.1786	V	10 to 40	1.01
203	Hexafloropropolylene	C3F6	SF6	0.539	0.142	SF6	180.0900	25.00	0.2870	6.6933	V	10 to 40	0.88
71	n-Hexane	C6H14	SF6	0.678	0.179	N2O	143.0928	25.00	0.3971	3.8450	V	3 to 12	0.88
105	Hydrogen*	H2	H2	1.000	1.010	H2	28.8360	25.00	3.4224	0.0899	V	10 to 40	1.00

*INDEX ID FOR AUTOMATIC TEST PROGRAMS USED IN MANUFACTURING
**DERIVED EMPIRICALLY

COD	GAS	SYMBOL	CAL. TEST GAS	CONV FACTOR REL. TO CAL TEST GAS	CONV FACTOR REL. TO CAL N2	DYNAMIC RESP. TEST GAS	HEAT Cp x 10 ^{3} (J/K·mol)	SPECIFIC HEAT TEMP (°C)	SPECIFIC Heat @ 25°C Cp(cal/g°C)	DENSITY g/l @ 0°C	SUG. SEAL MAT.	STD CAL PRESS (PSID)	MOLECULAR FACTOR
109	Hydrogen Bromide	HBr	N2	1.000	1.000	N2O	29.1410	25.00	0.0861	3.6100	KZ	10 to 40	1.00
110	Hydrogen Chloride	HCl	N2	1.000	1.000	N2O	29.1360	25.00	0.1911	1.6270	KZ	10 to 40	1.00
111	Hydrogen Cyanide	HCN	N2O	1.077	0.764	N2	35.8570	25.00	0.3172	1.2060	KZ	10 to 40	0.94
113	Hydrogen Fluoride	HF	N2	1.000	1.000	N2	29.1380	25.00	0.3482	0.8930	KZ	3 to 12	1.00
114	Hydrogen Iodide	HI	N2	0.999	0.999	SF6	29.1560	25.00	0.0545	5.7070	KZ	10 to 40	1.00
108	Hydrogen Selenide	H2Se	N2O	1.112	0.788	N2O	34.7272	25.00	0.1026	3.6130	KZ	10 to 40	0.94
107	Hydrogen Sulfide	H2S	N2O	1.190	0.844	C2H6	32.4470	25.00	0.2278	1.5200	KZ	10 to 40	0.94
115	Iodine Pentafluoride	IF5	SF6	0.942	0.249	SF6	102.8780	25.00	0.1109	9.9000	KZ	3 to 12	0.88
66	Isobutane	CH(CH3)3	SF6	0.997	0.264	N2O	94.1630	15.60	0.3998	2.5950	V	6 to 24	0.88
183	Isobutylene	C4H6	SF6	1.081	0.286	N2O	86.8830	15.60	0.3824	2.5030	V	6 to 24	0.88
184	Isopropanol	C3H7OH	SF6	1.093	0.289	N2O	88.7426	25.00	0.3529	2.6830	V	3 to 12	0.88
116	Krypton	Kr	Ar	1.000	1.415	N20	20.7860	25.00	0.0593	3.7390	V	10 to 40	1.01
84	Methane	CH4	N2O	1.014	0.719	N2	35.6390	25.00	0.5318	0.7150	V	10 to 40	0.88
185	Methanol	CH3OH	C2H4	0.977	0.584	C2H6	43.8902	25.00	0.3277	1.4290	V	10 to 40	0.88
46	Methylacetylene (Propyne)	C3H4	C2H6	0.867	0.417	Ar	61.5050	25.00	0.3672	1.7870	V	10 to 40	0.88
186	Methylamine	CH3NH2	C2H6	1.065	0.512	C2H6	50.0825	25.00	0.3856	1.3860	KZ	10 to 40	0.88
81	Methyl Bromide	CH3Br	C2H4	0.953	0.569	N2O	45.0200	25.00	0.1134	4.2360	KZ	6 to 24	0.88
80	Methyl Chloride	CH3Cl	C2H4	1.053	0.629	N2O	40.7310	25.00	0.1929	2.2530	V	10 to 40	0.88
82	Methyl Fluoride	CH3F	N2O	0.964	0.684	C2H6	37.5000	25.00	0.2636	1.5180	V	10 to 40	0.88
83	Methyl Mercaptan	CH3SH	C2H6	1.044	0.502	N2O	49.4910	15.60	0.2541	2.1460	V	6 to 24	0.88
187	Methyl Trichlorosilane	(CH3)SiCL3	SF6	0.947	0.250	SF6	102.3820	25.00	0.1638	6.6690	KZ	3 to 12	0.88
117	Molybdenum Hexafluoride	MoF6	SF6	0.571	0.151	SF6	169.7950	25.00	0.1933	9.3719	KZ	3 to 12	0.88
123	Neon	Ne	Ar	1.000	1.415	N2	20.7860	25.00	0.2464	0.9000	V	10 to 40	1.01
127	Nitric Oxide	NO	N2	0.976	0.976	C2H6	29.8450	25.00	0.2378	1.3390	V	10 to 40	1.00
118	Nitrogen	N2	N2	1.000	1.000	N2	29.1240	25.00	0.2486	1.2500	V	10 to 40	1.00
128	Nitrogen Dioxide	NO2	N2O	1.044	0.741	N2O	36.9740	25.00	0.1923	2.0520	KZ	10 to 40	0.94
124	Nitrogen Trifluoride	NF3	C2H6	1.000	0.480	N2O	53.3700	25.00	0.1798	3.1680	KZ	10 to 40	0.88
119	Nitrogen Trioxide	N2O3	C2H6	0.813	0.391	N2O	65.6150	25.00	0.2063	3.3930	V	10 to 40	0.88
129	Nitrosyl Chloride	NOCl	C2H4	1.027	0.614	N2O	44.5890	25.00	0.1629	2.9200	V	10 to 40	0.94
122	Nitrous Oxide	N2O	N2O	1.000	0.709	N2O	38.6170	25.00	0.2098	1.9640	V	10 to 40	0.94
130	Oxygen	O2	N2	0.992	0.992	N2	29.3760	25.00	0.2196	1.4270	V	10 to 40	1.00
132	Oxygen Difluoride	OF2	C2H4	1.058	0.632	N2O	43.3010	25.00	0.1918	2.4090	KZ	10 to 40	0.94
131	Ozone	O3	N2O	0.984	0.698	N2O	39.2380	25.00	0.1954	2.1428	V	10 to 40	0.94
6	Pentaborane	B5H9	SF6	1.033	0.273	N2O	93.8180	25.00	0.3555	2.8160	V	3 to 12	0.88
189	n-Pentane	n-C5H12	SF6	0.807	0.213	N2O	120.2063	25.00	0.3984	3.2190	V	3 to 12	0.88
17	Perchloryl Fluoride	ClO3F	C2H6	0.822	0.395	SF6	64.924	25.00	0.1515	4.5710	V	10 to 40	0.88
44	Perfluoropropane	C3F8	SF6	0.635	0.168	SF6	145.1010	10.00	0.1944	8.3880	V	10 to 40	0.88
94	Phosgene	COCl2	C2H6	0.925	0.444	SF6	57.6950	25.00	0.1393	4.4180	V	3 to 12	0.88
135	Phosphine	PH3	N2O	0.974	0.691	C2H6	37.1020	25.00	0.2610	1.5170	V	10 to 40	0.88

COD	GAS	SYMBOL	CAL. TEST GAS	CONV FACTOR REL. TO CAL TEST GAS	CONV FACTOR REL. TO CAL N2	DYNAMIC RESP. TEST GAS	HEAT Cp x 10 ^{3} (J/K-mol)	SPECIFIC HEAT TEMP (°C)	SPECIFIC Heat @ 25°C Cp(cal/g°C)	DENSITY g/l @ 0°C	SUG. SEAL MAT.	STD CAL PRESS (PSID)	MOLECULAR FACTOR
136	Phosphorous Oxychloride	POCl3	SF6	1.142	0.302	SF6	84.9170	25.00	0.1324	6.8452	V	3 to 12	0.88
134	Phosphorous Pentafluoride	PF5	SF6	1.143	0.302	SF6	84.8360	25.00	0.1611	5.6200	KZ	10 to 40	0.88
133	Phosphorous Tribromide	PBr3	SF6	1.275	0.337	SF6	76.0330	25.00	0.0671	12.0842	V	3 to 12	0.88
190	Phosphorous Trichloride	PCl3	SF6	1.355	0.358	SF6	71.5810	25.00	0.1247	6.1270	V	3 to 12	0.88
50	Propane	C3H8	C2H6	0.725	0.348	N20	74.01	26.80	0.3991	1.9670	V	10 to 40	0.88
47	Propylene	CH3CH:CH2	C2H6	0.829	0.398	N2O	62.3450	15.60	0.3659	1.8770	V	10 to 40	0.88
145	Silane	SiH4	C2H4	1.001	0.599	C2H6	42.8270	25.00	0.3189	1.4330	V	10 to 40	0.88
141	Silicon Tetrachloride	SiCl4	SF6	1.074	0.284	SF6	90.2610	25.00	0.1270	7.5847	KZ	3 to 12	0.88
142	Silicon Tetrafluoride	SiF4	SF6	1.317	0.348	SF6	73.6210	25.00	0.1692	4.6430	KZ	10 to 40	0.88
148	Sulfur Dioxide	SO2	N2O	0.968	0.687	N2O	39.8780	25.00	0.1489	2.8580	V	10 to 40	0.94
139	Sulfur Hexafluoride	SF6	SF6	1.000	0.264	SF6	96.9600	25.00	0.1588	6.5160	V	10 to 40	0.88
138	Sulfur Tetrafluoride	SF4	SF6	1.250	0.330	SF6	77.5980	25.00	0.1717	4.8210	KZ	10 to 40	0.88
149	Sulfuryl Fluoride	SO2F2	C2H6	0.808	0.388	SF6	66.0540	25.00	0.1545	4.5620	V	3 to 12	0.88
73	Tetraethoxysilane (TEOS)	Si(OC2H5)4	SF6	0.295	0.078	SF6	328.6200	25.00	0.3770	9.3004	KZ	1 to 5	0.88
191	Tetrafluorahydrazine	N2F4	SF6	1.225	0.324	SF6	79.1690	25.00	0.1821	4.6400	V	3 to 12	0.88
159	1,3,5,7-Tetramethylcyclotetrasiloxane	[-HSi(CH3)O-]4	SF6	0.000	0.000	SF6	0.0000	25.00	0.0000	10.7366	KZ	1 to 5	0.88
150	Titanium Tetrachloride	TiCL4	SF6	0.777	0.206	SF6	124.7100	25.00	0.1572	8.4650	V	10 to 40	0.88
199	Trans-LC	C2H2Cl2	SF6	1.468	0.388	SF6	66.1200	25.00	0.1630	4.3250	KZ	3 to 12	0.88
28	Trichlorethane (TCA)	C2H3Cl3	SF6	1.051	0.278	SF6	90.2300	25.00	0.1654	5.9500	V	10 to 40	0.88
193	Trichloroethylene (TCE)	C2HCl3	SF6	1.210	0.320	SF6	80.2100	25.00	0.1459	5.8620	V	10 to 40	0.88
146	Trichlorosilane	SiHCl3	SF6	1.285	0.340	SF6	75.4550	25.00	0.1332	6.0430	KZ	3 to 12	0.88
194	Triisobutyl Aluminum	(C4H9)3Al	SF6	0.230	0.061	SF6	421.5500	25.00	0.5080	8.8480	V	3 to 12	0.88
56	Trimethylamine	(CH3)3N	SF6	1.055	0.279	N2O	91.9310	25.00	0.3717	2.6390	V	6 to 24	0.88
195	Trimethylborate (TMB)	B(OCH3)3	SF6	0.769	0.203	SF6	126.0700	25.00	0.2900	4.6388	V	1 to 5	0.88
58	Trimethylphosphite (TMP)	P(OCH3)3	SF6	0.599	0.158	SF6	161.7700	25.00	0.3116	5.5393	V	1 to 5	0.88
196	Tritium*	T2	N2	0.986	0.986	He	29.5520	25.00	1.1709	0.2693	(6)	10 to 40	1.00
152	Tungsten Hexafluoride	WF6	SF6	0.815	0.215	SF6	119.0290	25.00	0.0956	13.2900	KZ	3 to 12	0.88
151	Uranium Hexafluoride	UF6	SF6	0.748	0.198	SF6	129.6203	25.00	0.0881	15.7000	KZ	3 to 12	0.88
27	Vinyl Bromide	CH2CHBr	C2H6	0.960	0.462	SF6	55.5350	25.00	0.1242	4.7720	V	10 to 40	0.88
30	Vinyl Chloride	CH2CHCl	C2H6	0.995	0.478	N2O	53.6070	25.00	0.2052	2.7880	V	10 to 40	0.88
106	Water Vapor	H2O	N2	**0.876	0.876	N2	33.5900	25.00	0.4458	0.8040	V	3 to 12	0.94
153	Xenon	Xe	Ar	1.000	1.415	SF6	20.7860	25.00	0.0379	5.8580	V	10 to 40	1.01

*INDEX ID FOR AUTOMATIC TEST PROGRAMS USED IN MANUFACTURING
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				CONV FACTOR REL. TO		DYNAMIC	HEAT	SPECIFIC	SPECIFIC		SUG.	STD CAL	MOLECULAR
			CAL.	CAL		RESP.	Cp	HEAT	Heat @ 25°C	DENSITY	SEAL	PRESS	FACTOR
							x 10 ^{3}						
COD	GAS	SYMBOL	TEST GAS	TEST GAS	N2	TEST GAS	(J/K·mol)	TEMP (°C)	Cp(cal/g°C)	g/l @ 0°C	MAT.	(PSID)	
	NOTE: Conversion of controller to or from these gases may alter dynamic response or stability.												
	In accordance with SEMI Standard E12-86, Standard Pressure is defined as 760 mm Hg (14.7 psia), Standard Temperature is defined as 0 degrees C.												
	1. Matheson Gas Data Book; Braker, William and Mossman, Allen L.; Sixth Edition, 1980												
	2. JANAF Thermochemical Tables; Chase, M.W., et. al.; Third Edition, 1985												
	3. Lange's handbook of Chemistry, Thirteenth Edition; Dean, John A., McGraw Hill, 1985												
	4. Touloukian, Y. S. et. al.; Viscosity, Nonmetallic Liquids and Gases, IFI/Plenum Data Corp., 1975												
	5. Handbook of Chemistry and Physics, 70th Edition, CRC Press, 1989-90												
	(6) Suggested seal is KZ with a composite of 2035.												
	(7) Use CF4 test gas on customer request. (Conversion factor rel to gas = 1.000, dynamic response test gas = CF4) .												
	(8) -Deleted-												

					STD CAL	Limiter Dash Number						
			CAL.	DENSITY	PRESS	Purge Rate, sim (+/- 10% Est.) @45 PSIG						
CODE	GAS	SYMBOL	TEST GAS	g/l @ 0°C	(PSID)	-001	-002	-003	-004	-005	-006	-011
25	Acetylene	C2H2	C2H4	1.1620	10 to 40	1.973	4.149	7.123	13.339	20.206	28.493	126.286
1	Air	-	N2	1.2930	10 to 40	1.871	3.933	6.753	12.645	19.155	27.011	119.718
32	Allene (Propadiene)	C3H4	C2H6	1.7870	10 to 40	1.591	3.345	5.744	10.756	16.294	22.977	101.835
125	Ammonia	NH3	N2O	0.7600	10 to 40	2.440	5.130	8.808	16.494	24.985	35.232	156.154
2	Argon	Ar	Ar	1.7820	10 to 40	1.593	3.350	5.752	10.771	16.316	23.009	101.978
4	Arsine	AsH3	N2O	3.4780	10 to 40	1.141	2.398	4.117	7.710	11.679	16.470	72.995
201	Acetaldehyde	CH3CHO	C2H4	1.9648	3 to 12	1.900	3.190	6.850	12.827	19.430	27.400	121.440
200	Benzene	C6H6	C2H4	0.9550	3 to 12	2.720	4.576	9.820	18.389	27.855	39.280	174.094
7	Boron Tribromide	BBr3	C2H6	11.1800	3 to 12	0.636	1.338	2.296	4.300	6.514	9.186	40.713
8	Boron Trichloride	BCl3	C2H6	5.2270	3 to 12	0.930	1.956	3.359	6.289	9.527	13.434	59.543
9	Boron Trifluoride	BF3	C2H6	3.0250	10 to 40	1.223	2.571	4.415	8.267	12.523	17.660	78.270
10	Bromine	Br2	N2O	7.1300	10 to 40	0.797	1.675	2.876	5.385	8.157	11.503	50.982
12	Bromine Pentafluoride	BrF5	SF6	7.8030	3 to 12	0.761	1.601	2.749	5.147	7.797	10.996	48.734
11	Bromine Trifluoride	BrF3	C2H6	6.1080	10 to 40	0.861	1.810	3.107	5.818	8.813	12.428	55.082
63	1,3-Butadiene	C4H6	SF6	2.4130	6 to 24	1.369	2.879	4.943	9.257	14.022	19.773	87.636
61	n-Butane	n-C4H10	SF6	2.5930	6 to 24	1.321	2.777	4.769	8.929	13.526	19.074	84.539
67	1-Butene	C4H8	SF6	2.4603	6 to 24	1.356	2.851	4.895	9.167	13.886	19.582	86.789
161	2-Butene CIS	C4H8	SF6	2.4603	6 to 24	1.356	2.851	4.895	9.167	13.886	19.582	86.789
162	2-Butene TRANS	C4H8	SF6	2.5030	6 to 24	1.344	2.827	4.854	9.089	13.767	19.414	86.046
93	Carbon Dioxide	CO2	N2O	1.9640	10 to 40	1.518	3.191	5.479	10.260	15.542	21.917	97.138
97	Carbon Disulfide	CS2	C2H4	3.3970	10 to 40	1.154	2.426	4.166	7.801	11.818	16.665	73.860
92	Carbon Monoxide	CO	N2	1.2500	10 to 40	1.902	4.000	6.868	12.861	19.482	27.472	121.760
77	Carbon Tetrachloride	CCl4	SF6	6.8600	3 to 12	0.812	1.707	2.932	5.490	8.316	11.727	51.975
95	Carbonyl Fluoride	COF2	C2H4	2.9450	10 to 40	1.239	2.606	4.474	8.379	12.692	17.898	79.326
96	Carbonyl Sulfide	COS	N2O	2.6800	10 to 40	1.299	2.732	4.690	8.783	13.305	18.762	83.156
13	Chlorine	Cl2	N2O	3.1630	10 to 40	1.196	2.515	4.318	8.085	12.247	17.270	76.544

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					STD CAL	Limiter Dash Number						
			CAL.	DENSITY	PRESS	Purge Rate, slm (+/- 10% Est.) @45 PSIG						
CODE	GAS	SYMBOL	TEST GAS	g/l @ 0°C	(PSID)	-001	-002	-003	-004	-005	-006	-011
91	Chlorine Trifluoride	ClF3	C2H6	4.1250	3 to 12	1.047	2.202	3.781	7.080	10.724	15.123	67.027
88	Chloroform	CHCl3	C2H6	5.3260	3 to 12	0.922	1.938	3.327	6.231	9.438	13.309	58.987
42	Cyanogen	C2N2	C2H6	2.3220	10 to 40	1.396	2.935	5.039	9.436	14.294	20.156	89.336
16	Cyanogen Chloride	ClCN	C2H4	2.7420	10 to 40	1.285	2.701	4.637	8.683	13.154	18.549	82.210
202	Cyclopentane	C5H10	C2H6	3.1290	3 to 12	1.500	2.528	5.430	10.168	15.403	21.720	96.266
48	Cyclopropane	C3H6	C2H6	1.8770	10 to 40	1.553	3.264	5.605	10.495	15.898	22.419	99.363
98	Deuterium*	D2	N2	0.1798	10 to 40	5.016	10.547	18.109	33.910	51.367	72.435	321.044
5	Diborane	B2H6	C2H6	1.2350	10 to 40	1.914	4.024	6.910	12.939	19.600	27.638	122.497
163	Dibromodiflouromethane	CBr2F2	SF6	9.3620	3 to 12	0.695	1.462	2.510	4.699	7.119	10.038	44.491
164	Dibromomethane	CH2Br2	C2H6	7.7600	10 to 40	0.764	1.605	2.756	5.162	7.819	11.026	48.868
165	Dichloromethylsilane	(CH3)2SiCl2	SF6	5.7580	3 to 12	0.886	1.864	3.200	5.992	9.077	12.800	56.731
143	Dichlorosilane	SiH2Cl2	C2H6	4.5060	3 to 12	1.002	2.107	3.617	6.774	10.261	14.469	64.130
166	Trans 1,2-Dichloroethylene	C2H2Cl2	SF6	4.3280	10 to 40	1.022	2.150	3.691	6.912	10.470	14.764	65.436
167	Difluoromethane	CH2F2	C2H4	2.3210	10 to 40	1.396	2.935	5.040	9.438	14.297	20.161	89.356
26	1,1-Difluoroethylene	C2H2F2	C2H6	2.8570	10 to 40	1.258	2.646	4.543	8.507	12.886	18.172	80.539
144	Dimethylamine	(CH3)2NH	C2H6	2.0110	3 to 12	1.500	3.154	5.415	10.140	15.359	21.659	95.996
168	Dimethyl Cadmium	(CH3)2Cd	SF6	6.3607	0	0.843	1.773	3.045	5.701	8.636	12.178	53.977
38	Dimethyl Ether	(CH3)2O	C2H6	2.0550	10 to 40	1.484	3.120	5.356	10.030	15.194	21.426	94.963
70	2,2-Dimethylpropane	C5H12	SF6	3.2190	3 to 12	1.186	2.493	4.280	8.014	12.140	17.119	75.875
140	Disilane	Si2H6	SF6	2.7760	10 to 40	1.277	2.684	4.609	8.630	13.073	18.435	81.705
37	Ethane	C2H6	C2H6	1.3420	10 to 40	1.836	3.860	6.628	12.412	18.802	26.514	117.512
36	Ethanol	C2H6O	C2H6	2.0550	3 to 12	1.484	3.120	5.356	10.030	15.194	21.426	94.963
64	Ethyl Acetylene (1-Butyne)	C4H6	SF6	2.4130	3 to 12	1.369	2.879	4.943	9.257	14.022	19.773	87.636

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CODE	GAS	SYMBOL	CAL. TEST GAS	DENSITY g/l @ 0°C	STD CAL PRESS (PSID)	Limiter Dash Number Purge Rate, slm (+/- 10% Est.) @45 PSIG -001	-002	-003	-004	-005	-006	-011
169	Ethylamine	C2H5NH2	SF6	2.0110	10 to 40	1.500	3.154	5.415	10.140	15.359	21.659	95.996
35	Ethyl Chloride	C2H5Cl	SF6	2.8790	3 to 12	1.254	2.636	4.525	8.474	12.837	18.102	80.230
31	Ethylene	C2H4	C2H4	1.2510	10 to 40	1.902	3.998	6.865	12.856	19.474	27.461	121.711
34	Ethylene Oxide	C2H4O	C2H6	1.9650	3 to 12	1.517	3.190	5.478	10.258	15.538	21.911	97.113
99	Fluorine	F2	N2	1.6950	10 to 40	1.634	3.435	5.898	11.044	16.730	23.592	104.562
90	Fluroform (Freon-23)	CHF3	C2H6	3.1270	10 to 40	1.203	2.529	4.342	8.131	12.317	17.369	76.983
170	Trichlorofluoromethane Freon-11	CCl3F	SF6	6.1290	3 to 12	0.859	1.806	3.102	5.808	8.798	12.407	54.988
171	Dichlorodiflouromethane Freon-12	CCl2F2	SF6	5.3950	10 to 40	0.916	1.925	3.306	6.191	9.377	13.224	58.609
172	Cholrotriflouromethane Freon-13	CClF3	SF6	4.6600	10 to 40	0.985	2.072	3.557	6.661	10.090	14.228	63.062
173	Bromotriflouromethane Freon-13B1	CBrF3	SF6	6.6440	10 to 40	0.825	1.735	2.979	5.578	8.450	11.916	52.813
174	Carbon Tetraflouride Freon-14	(7) CF4	(8) CF4	3.9260	10 to 40	1.074	2.257	3.875	7.257	10.993	15.501	68.704
175	Dichloroflouromethane Freon-21	CHCl2F	C2H6	4.5920	3 to 12	0.993	2.087	3.583	6.710	10.164	14.333	63.527
176	Chlorodiflouromethane Freon-22	CHClF2	C2H6	3.8580	10 to 40	1.083	2.277	3.909	7.321	11.089	15.637	69.307
177	Trichloro-triflouroethane Freon-113	CCl2FCClF2	SF6	8.3600	3 to 12	0.736	1.547	2.656	4.973	7.533	10.623	47.082
178	1,2-Dichlorotetraflouroethane Freon-114	C2Cl2F4	SF6	7.6260	3 to 12	0.770	1.619	2.781	5.207	7.887	11.122	49.296
179	Chloropentaflouroethane Freon-115	C2ClF5	SF6	6.8959	10 to 40	0.810	1.703	2.924	5.476	8.294	11.696	51.840
24	Hexafluoroethane (Freon-116)	C2F6	SF6	6.1570	10 to 40	0.857	1.802	3.095	5.795	8.778	12.378	54.862
180	Octaflourocyclobutane Freon-C318	C4F8	SF6	8.9370	6 to 24	0.712	1.496	2.569	4.810	7.286	10.274	45.537
181	Freon-152A	C2H4F2	C2H6	12.6800	10 to 40	0.597	1.256	2.156	4.038	6.117	8.626	38.230
197	Pentaflouroethane Freon-125	C2HF5	SF6	5.3547	10 to 40	1.150	1.933	4.150	7.771	11.772	16.600	73.573
198	Tetraflourethane Freon-134A	C2H2F4	C2H4	4.5540	10 to 40	1.250	2.096	4.500	8.427	12.765	18.000	79.778
104	Germane	GeH4	C2H4	3.4180	10 to 40	1.151	2.419	4.153	7.777	11.781	16.613	73.633
102	Germanium Tetrachloride	GeCl4	SF6	9.5650	3 to 12	0.688	1.446	2.483	4.649	7.043	9.931	44.017
182	Hexamethyl Disiloxane	H3)3Si-O-Si(CH3)	SF6	7.2500	1 to 5	0.790	1.661	2.852	5.340	8.089	11.407	50.558
112	Helium*	He	He	0.1786	10 to 40	5.033	10.582	18.170	34.024	51.539	72.678	322.120
203	Hexafloropropolylene	C3F6	SF6	6.6933	10 to 40	1.030	1.729	3.710	6.947	10.524	14.840	65.773

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**DERIVED EMPIRICALLY

					STD CAL	Limiter Dash Number						
			CAL.	DENSITY	PRESS	Purge Rate, slm (+/- 10% Est.) @45 PSIG						
CODE	GAS	SYMBOL	TEST GAS	g/l @ 0°C	(PSID)	-001	-002	-003	-004	-005	-006	-011
71	n-Hexane	C6H14	SF6	3.8450	3 to 12	1.085	2.281	3.916	7.333	11.108	15.664	69.424
105	Hydrogen*	H2	H2	0.0899	10 to 40	7.094	14.915	25.610	47.956	72.644	102.439	454.024
109	Hydrogen Bromide	HBr	N2	3.6100	10 to 40	1.120	2.354	4.041	7.568	11.464	16.166	71.648
110	Hydrogen Chloride	HCl	N2	1.6270	10 to 40	1.668	3.506	6.020	11.273	17.076	24.080	106.725
111	Hydrogen Cyanide	HCN	N2O	1.2060	10 to 40	1.937	4.072	6.992	13.093	19.834	27.969	123.961
113	Hydrogen Fluoride	HF	N2	0.8930	3 to 12	2.251	4.732	8.126	15.216	23.049	32.503	144.057
114	Hydrogen Iodide	HI	N2	5.7070	10 to 40	0.890	1.872	3.214	6.019	9.117	12.857	56.984
108	Hydrogen Selenide	H2Se	N2O	3.6130	10 to 40	1.119	2.353	4.040	7.565	11.459	16.159	71.618
107	Hydrogen Sulfide	H2S	N2O	1.5200	10 to 40	1.725	3.627	6.228	11.663	17.667	24.913	110.417
115	Iodine Pentafluoride	IF5	SF6	9.9000	3 to 12	0.676	1.421	2.440	4.570	6.922	9.762	43.265
66	Isobutane	CH(CH3)3	SF6	2.5950	6 to 24	1.320	2.776	4.767	8.926	13.521	19.067	84.507
183	Isobutylene	C4H6	SF6	2.5030	6 to 24	1.344	2.827	4.854	9.089	13.767	19.414	86.046
184	Isopropanol	C3H7OH	SF6	2.6830	3 to 12	1.299	2.730	4.688	8.778	13.297	18.751	83.109
116	Krypton	Kr	Ar	3.7390	10 to 40	1.100	2.313	3.971	7.436	11.264	15.884	70.401
84	Methane	CH4	N2O	0.7150	10 to 40	2.516	5.289	9.081	17.005	25.759	36.324	160.993
185	Methanol	CH3OH	C2H4	1.4290	10 to 40	1.779	3.741	6.423	12.028	18.221	25.694	113.879
46	Methylacetylene (Propyne)	C3H4	C2H6	1.7870	10 to 40	1.591	3.345	5.744	10.756	16.294	22.977	101.835
186	Methylamine	CH3NH2	C2H6	1.3860	10 to 40	1.807	3.799	6.522	12.214	18.501	26.089	115.632
81	Methyl Bromide	CH3Br	C2H4	4.2360	6 to 24	1.033	2.173	3.731	6.986	10.583	14.923	66.143
80	Methyl Chloride	CH3Cl	C2H4	2.2530	10 to 40	1.417	2.979	5.116	9.580	14.511	20.463	90.694
82	Methyl Fluoride	CH3F	N2O	1.5180	10 to 40	1.726	3.630	6.232	11.671	17.678	24.929	110.490
83	Methyl Mercaptan	CH3SH	C2H6	2.1460	6 to 24	1.452	3.053	5.242	9.815	14.868	20.967	92.927
187	Methyl Trichlorosilane	(CH3)SiCL3	SF6	6.6690	3 to 12	0.824	1.732	2.973	5.568	8.434	11.894	52.714
117	Molybdenum Hexafluoride	MoF6	SF6	9.3719	3 to 12	0.695	1.461	2.508	4.697	7.115	10.033	44.468
123	Neon	Ne	Ar	0.9000	10 to 40	2.242	4.714	8.094	15.157	22.959	32.376	143.495
127	Nitric Oxide	NO	N2	1.3390	10 to 40	1.838	3.865	6.636	12.426	18.823	26.543	117.644

*INDEX ID FOR AUTOMATIC TEST PROGRAMS USED IN MANUFACTURING.
**DERIVED EMPIRICALLY

CODE	GAS	SYMBOL	CAL. TEST GAS	DENSITY g/l @ 0°C	STD CAL PRESS (PSID)	Limiter Dash Number: Purge Rate, slm (+/- 10% Est.) @45 PSIG						
						-001	-002	-003	-004	-005	-006	-011
118	Nitrogen	N2	N2	1.2500	10 to 40	1.902	4.000	6.868	12.861	19.482	27.472	121.760
128	Nitrogen Dioxide	NO2	N2O	2.0520	10 to 40	1.485	3.122	5.360	10.038	15.205	21.442	95.032
124	Nitrogen Trifluoride	NF3	C2H6	3.1680	10 to 40	1.195	2.513	4.314	8.079	12.237	17.257	76.483
119	Nitrogen Trioxide	N2O3	C2H6	3.3930	10 to 40	1.155	2.428	4.169	7.806	11.825	16.675	73.904
129	Nitrosyl Chloride	NOCl	C2H4	2.9200	10 to 40	1.245	2.617	4.494	8.415	12.746	17.974	79.665
122	Nitrous Oxide	N2O	N2O	1.9640	10 to 40	1.518	3.191	5.479	10.260	15.542	21.917	97.138
130	Oxygen	O2	N2	1.4270	10 to 40	1.781	3.744	6.428	12.037	18.233	25.712	113.959
132	Oxygen Difluoride	OF2	C2H4	2.4090	10 to 40	1.370	2.881	4.947	9.264	14.033	19.789	87.708
131	Ozone	O3	N2O	2.1428	10 to 40	1.453	3.055	5.246	9.823	14.880	20.983	92.997
6	Pentaborane	B5H9	SF6	2.8160	3 to 12	1.268	2.665	4.576	8.569	12.980	18.303	81.123
189	n-Pentane	n-C5H12	SF6	3.2190	3 to 12	1.186	2.493	4.280	8.014	12.140	17.119	75.875
17	Perchloryl Fluoride	ClO3F	C2H6	4.5710	10 to 40	0.995	2.092	3.592	6.725	10.188	14.366	63.673
44	Perfluoropropane	C3F8	SF6	8.3880	10 to 40	0.734	1.544	2.651	4.965	7.521	10.605	47.003
94	Phosgene	COCl2	C2H6	4.4180	3 to 12	1.012	2.128	3.653	6.841	10.363	14.613	64.766
135	Phosphine	PH3	N2O	1.5170	10 to 40	1.727	3.631	6.234	11.674	17.684	24.938	110.526
136	Phosphorous Oxychloride	POCl3	SF6	6.8452	3 to 12	0.813	1.709	2.935	5.496	8.325	11.740	52.031
134	Phosphorous Pentafluoride	PF5	SF6	5.6200	10 to 40	0.897	1.886	3.239	6.065	9.188	12.956	57.424
133	Phosphorous Tribromide	PBr3	SF6	12.0842	3 to 12	0.612	1.286	2.209	4.136	6.266	8.836	39.161
190	Phosphorous Trichloride	PCl3	SF6	6.1270	3 to 12	0.859	1.807	3.102	5.809	8.799	12.409	54.996
50	Propane	C3H8	C2H6	1.9670	10 to 40	1.517	3.189	5.475	10.252	15.530	21.900	97.064
47	Propylene	CH3CH:CH2	C2H6	1.8770	10 to 40	1.553	3.264	5.605	10.495	15.898	22.419	99.363
145	Silane	SiH4	C2H4	1.4330	10 to 40	1.777	3.736	6.415	12.012	18.195	25.658	113.720
141	Silicon Tetrachloride	SiCl4	SF6	7.5847	3 to 12	0.772	1.624	2.788	5.221	7.909	11.153	49.430
142	Silicon Tetrafluoride	SiF4	SF6	4.6430	10 to 40	0.987	2.075	3.564	6.673	10.108	14.254	63.177
148	Sulfur Dioxide	SO2	N2O	2.8580	10 to 40	1.258	2.645	4.542	8.505	12.884	18.168	80.524
139	Sulfur Hexafluoride	SF6	SF6	6.5160	10 to 40	0.833	1.752	3.008	5.633	8.533	12.032	53.330

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					STD CAL	Limiter Dash Number						
			CAL.	DENSITY	PRESS	Purge Rate, slm (+/- 10% Est.) @45 PSIG						
CODE	GAS	SYMBOL	TEST GAS	g/l @ 0°C	(PSID)	-001	-002	-003	-004	-005	-006	-011
138	Sulfur Tetrafluoride	SF4	SF6	4.8210	10 to 40	0.969	2.037	3.497	6.549	9.920	13.989	62.000
149	Sulfuryl Fluoride	SO2F2	C2H6	4.5620	3 to 12	0.996	2.094	3.595	6.732	10.198	14.380	63.735
73	Tetraethoxysilane (TEOS)	Si(OC2H5)4	SF6	9.3004	1 to 5	0.697	1.466	2.518	4.715	7.142	10.072	44.638
191	Tetrafluorahydrazine	N2F4	SF6	4.6400	3 to 12	0.987	2.076	3.565	6.675	10.112	14.259	63.198
159	1,3,5,7-Tetramethylcyclotetrasiloxane	[-HSi(CH3)O-]4	SF6	10.7366	1 to 5	0.649	1.365	2.343	4.388	6.647	9.374	41.546
150	Titanium Tetrachloride	TiCL4	SF6	8.4650	10 to 40	0.731	1.537	2.639	4.942	7.486	10.557	46.789
199	Trans-LC	C2H2Cl2	SF6	4.3250	3 to 12	1.280	2.150	4.620	8.651	13.105	18.480	81.906
28	Trichlorethane (TCA)	C2H3Cl3	SF6	5.9500	10 to 40	0.872	1.833	3.148	5.895	8.929	12.592	55.809
193	Trichloroethylene (TCE)	C2HCl3	SF6	5.8620	10 to 40	0.879	1.847	3.171	5.939	8.996	12.686	56.226
146	Trichlorosilane	SiHCl3	SF6	6.0430	3 to 12	0.865	1.819	3.124	5.849	8.860	12.495	55.377
194	Triisobutyl Aluminum	(C4H9)3Al	SF6	8.8480	3 to 12	0.715	1.503	2.581	4.834	7.322	10.326	45.765
56	Trimethylamine	(CH3)3N	SF6	2.6390	6 to 24	1.309	2.753	4.727	8.851	13.408	18.907	83.799
195	Trimethylborate (TMB)	B(OCH3)3	SF6	4.6388	1 to 5	0.988	2.076	3.565	6.676	10.113	14.261	63.205
58	Trimethylphosphite (TMP)	P(OCH3)3	SF6	5.5393	1 to 5	0.904	1.900	3.263	6.109	9.254	13.050	57.841
196	Tritium*	T2	N2	0.2693	10 to 40	4.099	8.618	14.797	27.708	41.972	59.187	262.326
152	Tungsten Hexafluoride	WF6	SF6	13.2900	3 to 12	0.583	1.227	2.106	3.944	5.975	8.425	37.342
151	Uranium Hexafluoride	UF6	SF6	15.7000	3 to 12	0.537	1.129	1.938	3.629	5.497	7.752	34.357
27	Vinyl Bromide	CH2CHBr	C2H6	4.7720	10 to 40	0.974	2.047	3.515	6.582	9.971	14.060	62.317
30	Vinyl Chloride	CH2CHCl	C2H6	2.7880	10 to 40	1.274	2.678	4.599	8.612	13.045	18.395	81.529
106	Water Vapor	H2O	N2	0.8040	3 to 12	2.372	4.988	8.564	16.036	24.291	34.255	151.821
153	Xenon	Xe	Ar	5.8580	10 to 40	0.879	1.848	3.173	5.941	8.999	12.690	56.245

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**DERIVED EMPIRICALLY

APPENDIX B

RECOMMENDED PROCEDURES TO AVOID CONTAMINATION OF HIGHLY REACTIVE GAS SYSTEMS

(Technical Bulletin 8901)

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Tylan General

Technical Bulletin No. 8901

Recommended Procedures to Avoid Contamination of Highly Reactive Gas Systems

Gas systems for silane and other highly reactive gases are extremely vulnerable to contamination. When oxygen, water vapor or other gases combine with highly reactive molecules, solid contaminants, such as silicon dioxide, form in the system and can cause numerous problems with measurement and control equipment. These particles can be transported in sub-micron sizes that pass through filters, or, can be formed by reactions in the plumbing downstream of the filter. Although proper use of filters reduces contamination, this potential problem exists throughout the entire system.

The three primary causes of contamination are leaks, impure purge gases, and improper purging procedures. These problems and their solutions are discussed below.

Leaks

Plumbing leaks cannot be tolerated in a pure gas system. It is commonly believed that if a gas line pressure is above that of the ambient, the surrounding atmosphere will be prevented from entering the system. However, for any flow through an orifice or leak path, back diffusion from the surrounding gas (air) takes place into the gas stream. The system can then become contaminated with oxygen or moisture. THEREFORE, AVOID ALL LEAKS. Check the system when new, after replacing any equipment in the plumbing circuit, and on a periodic schedule. A leak detector capable of detecting leaks in the range of less than 1×10^{-10} atm cc/sec of helium, is recommended.

Impure Purging Gases

Before and after a highly reactive gas system has been exposed to air, it is necessary to purge the system with an inert gas, i.e. N_2 or Ar with less than 10 ppm O_2 and water vapor.

PURGING IS NOT NECESSARY, NOR IS IT RECOMMENDED, AT ANY OTHER TIME

Each time a transition is made from purge gas to process gas or process gas to purge gas, there is mixing of the two gases throughout the plumbing; any oxygen or moisture in the purge gas reacts and can be left behind as contamination. The practice of purging process lines between runs gives two potential mixing events per run, and can cause a cumulative buildup or contamination. With leak-tight plumbing it is not necessary to remove process gas from the system. Buildup resulting from repetitive purging can be avoided by not purging between runs.

In some installations process gas is fed into a reaction chamber that can be exposed to air, thereby causing oxidation of residual gas in the gas supply line. As a preventative measure, installation of a shutoff valve in the process line, close to the reaction chamber, is recommended.

Proper Purge Procedure

The most effective method for removing process gases from process tubing is cycle purging. Cycle purging alternates evacuating the gas system and pressurizing the gas system with an inert purge gas. This procedure causes absorbed molecules and trapped gases to be removed much more quickly than straight purging. Contaminants trapped in plumbing dead space and microcavities can communicate with the process stream to cause virtual leaks. Cycle purging helps to purge these cavities. Although the application of vacuum to the gas system is very effective, a similar procedure can be adapted for atmospheric systems.

Technical Bulletin No. 8901

Vacuum Systems

Figure 1 is a picture of a typical reactive gas flow controller installation. To initiate purge procedure, close Valves 2 and 3, and open Valve 1. Evacuate the line to base pressure. Close Valve 1 and open Valve 3 for one or two minutes to pressurize the controller with purge gas. Close Valve 3 and open Valve 1 to evacuate the controller back down to base pressure. Repeat the alternating application of evacuation and pressurization at least 20 times. When removing the flow controller, pressurize it with the purge gas and then close all 3 valves.

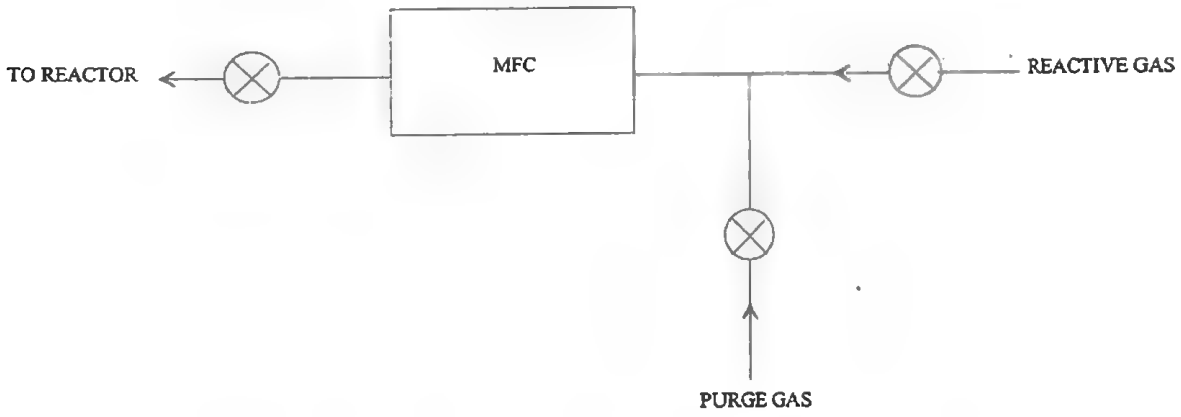
Atmospheric Systems

Although a vacuum pump is highly desirable, a similar procedure can be performed without a pump. An aspirator powered by a high rate of purge gas flow to exhaust can be used to create a subatmospheric pressure in the plumbing system. If an aspirator is not available, cycling between atmospheric pressure and purge gas line pressure is better than simply flowing purge gas. Pressurize and depressurize the controller for about 5 minutes, and repeat the procedure at least 40 times, rather than the 20 used for vacuum systems.

CAUTION

When highly reactive gases are being evacuated by pumping, the pump should contain a nitrogen blanket above the oil level to dilute the gases and prevent air from being absorbed into the pump. Take extreme precautions to dispose of any unreacted process gas.

CONSULT YOUR PLANT SAFETY ENGINEER.



Highly Reactive Gas System

921214

APPENDIX C

FC-2900 Series Autozero Feature

(Technical Bulletin 8906)

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Tylan General

Technical Bulletin No. 8906

FC-2900 Series Autozero Feature

What it is
How it works
How to use it

OVERVIEW

An automatic re-zeroing circuit is a standard feature on all FC-2900 series mass flow controllers. FC-2900 autozero helps compensate for small changes in electronic components and the environment which result in calibration changes commonly referred to as "drift."

In most applications, FC-2900 autozero will operate automatically with no change in the plumbing or control system of the process tool. The only requirement is that there is no gas flowing through the MFC during the autozero procedure. Because this action also ensures that the MFC will start without overshoot when gas flow begins, it does not represent an additional control requirement.

WHY AUTOZERO IS USEFUL

Tylan General has developed the most stable flow sensor technology currently available in the marketplace. Our products routinely come out on top in competitive evaluations of MFC stability. Nevertheless, no sensor is absolutely stable. In critical applications, even a few tenths of a percent drift between MFC recalibrations may be objectionable.

FC-2900 autozero adds one more level of stability enhancement to answer this requirement. In less demanding applications, using a FC-2900 series MFC may still improve process control due to the more consistent, repeatable flows they produce. Users may also find that equipment stays in spec longer so that downtime for recalibration is reduced.

FC-2900 autozeroing also cancels out several small environmental effects on the MFC. Like all Tylan General MFCs, the FC-2900s can be considered essentially insensitive to mounting attitude, exhibiting a zero shift of less than 0.1 percent with any change in mounting position. Even these small changes can be cancelled out with FC-2900 auto-zero, which will resolve errors as small as 0.05%.

FC-2900 MFCs also have enhanced temperature stability which, by itself, reduces the effects of ambient temperature change. A 10 degree Celsius change in ambient temperature will affect MFC zero less than 0.5%. FC-2900 autozero can cancel this residual effect as well.

WHAT AUTOZERO CANNOT DO

MFCs can be affected in ways which change their full-scale sensitivity, or span calibration, independently from zero calibration. An example of such a phenomenon is contamination or corrosion of the internal components of the MFC. FC-2900 autozero cannot completely compensate for these effects because they do not necessarily involve any zero error. Fortunately, in the FC-2900 series, most span effects have been stabilized or compensated. FC-2900 autozero provides significant improvements in the remaining zero effects which by their nature are larger in magnitude.

HOW IT WORKS

In normal operation the electrical output of the mass flowmeter which is a part of every MFC will slowly drift away from its calibrated setting. After some time, this output will no longer be zero volts under no-flow conditions. When this happens, the entire calibration of the MFC is shifted by a constant inversely equivalent to the shift in the zero point. For example, if the flowmeter zero were to drift high by 1% (50 mV) a setpoint of 50% would produce a 49% flow. Adjusting the flowmeter back to a zero output in the absence of flow shifts the entire range back into calibration. FC-2900 autozero performs these corrections automaticallycally, without the need for removing the cover and manually adjusting the zero potentiometer.

FC-2900 autozeroing is an intermittent process which is triggered by the equipment process control system under

conditions where there should be no gas flow through the MFC. The process starts after the setpoint to the MFC has been commanded to less than 1% for 80 seconds or more. The actual zeroing process takes about 2-1/2 seconds and repeats itself every 80 seconds as long as the triggering condition continues to be met. This continuous re-zeroing prevents drift in the zero point over long idle periods. In short-cycle processes where a gas loop may not be inactive for the 80 or 90 seconds which autozero requires, autozeroing may only occur at some interruption in the process cycle such as during the changing of cassettes.

The FC-2900 autozero circuit looks at any net error which remains in the zero signal after previous corrections and creates a correction voltage which, when fed back to the flowmeter, cancels the error. The magnitude of this correction voltage is stored as a digital number. As the correction is stored digitally, it does not drift or change with time as would an analog value stored in a capacitor.

FC-2900 autozero has a total correction range of 5% or ±2.5% with a resolution of ±0.05%. This correction is non-cumulative. It cannot offset the real or "natural" zero of the MFC by more than 5%. Successive zero corrections cannot accumulate.

This correction range is intentionally restricted. The flow sensor in the FC-2900 is very stable by design and zero errors which fall outside the correction range can indicate a damaged MFC. Apparent zero errors caused by shutoff valve leaks will often be greater in magnitude than the correction range and are thereby also more readily detectable by the process control system.

FC-2900 AUTOZERO MEMORY

Because the autozero correction voltage is stored digitally in active memory, conventional autozero circuits can be vulnerable to power interruptions. During a brief power interruption, after which processing is expected to resume, the correction value is lost. The MFC will recover to some nominal zero value which could be several percent different from what was in effect immediately before the interruption. This new zero value translates into a virtual calibration shift upon resumption of flow. This shift is invisible to the process control system. The result can be needlessly scrapped product.

FC-2900 autozero solves this problem by storing the memory value in non-volatile EEROM. Upon power-up, the value in EEROM is transferred to the active autozero circuit.

Momentarily connecting Pin K to common stores the current autozero offset in non-volatile memory. For maximum benefit EEROM memory should by updated in this way on a periodic basis.

CAUTION: PIN K IS DESIGNATED PIN I ON UNIT INSTRUMENT PRODUCTS. SEE NOTE BELOW.

ENSURING NO FLOW WHEN AUTOZEROING

When adjusting the no-flow or zero indication of a mass flow controller one should be quite certain that there is no gas flowing through the device. The FC-2900 is very sensitive and may respond to room air currents if operated on a service bench without end caps. In a gas system, shutoff valves in series with the MFC should be used to ensure that gas does not flow when the gas loop is off.

Even in the presence of both up and downstream shutoff valves, it is possible for gas to flow through the MFC for some time after the shutoff valves have been closed. Gas can flow as long as there is a pressure difference across the MFC. This is an especially common problem in vacuum systems where flow rates are low. Some variation of the following control sequence can help in these cases:

1. Close the upstream gas valve and set the MFC for its maximum flow rate. Pump the MFC and its plumbing to vacuum.

2. When evacuation of the system has been confirmed by measuring a very low flow value for some period of time, close the downstream shutoff valve. Wait again to ensure that any remaining pressure differential across the MFC has had the opportunity to discharge. Any non-zero reading from the MFC flow output at this point should be due to zero error or shutoff valve leakage.

3. Initiate the autozero process by taking the MFC setpoint to zero or by connecting pin L to common. The control valve will close. If no shutoff valve leakage exists, the flow output reading should not change. The FC-2900 will recompute its zero setting after 80 seconds.

The normal rate of zero drift is low enough that run-to-run zeroing is not required in most cases. Consequently, the above sequence can be run as part of a regular weekly maintenance routine. Autozero memory can be updated as part of the same procedure. As noted below, autozero

operation which would normally take place at other times can be inhibited by using Pin J.

No mass flow controller, including the FC-2900, should be absolutely relied upon to be a positive shut off device. The FC-2900 control valve with elastomeric poppet has been tested to shut off positively after over a million operations. Nevertheless, there are fundamental differences in the design of proportional control valves and shutoff valves such as those manufactured by Nupro, Parker or Fujikin.

The forces available to hold the valve poppet closed against the seat of a proportional control valve are much lower than those in a shutoff valve. The possibility that small leaks will develop as the valve materials age cannot be discounted. The direct effect of such a leak in the process may be undetectable. It may be their effect on the autozeroing process that brings them to attention. Consequently, autozeroing conditions should not rely on the seal of the MFC control valve for maximum system robustness and reliability.

CONFIRMING AUTOZERO OPERATION

When the MFC actually autozeros, connector Pin J goes high. This state lasts approximately 2.5 seconds. This signal can be used to alert a process controller that the MFC is autozeroing or it can be tied into an LED, giving a visual display of MFC autozeroing. It should be noted that Pin J functions as both an input and an output. Connecting the pin to common will inhibit autozero updating as discussed below.

Figure 1 shows the output configuration on Pin J and an example of how to drive an LED which will illuminate while the controller is autozeroing. Note that the pin will not source enough current in the high state to drive an LED directly. Within the FC-2900, Pin J is pulled up to +5V through 10k. It is normally driven to common by a transistor. During the autozeroing process the transistor turns off. The transistor will sink 500 mA. Pin J must reach at least 2.8V for the autozeroing process to proceed.

CAUTION: PIN J IS DESIGNATED PIN H ON UNIT INSTRUMENT PRODUCTS. SEE NOTE BELOW.

DISABLING FC-2900 AUTOZERO

There may be circumstances in which the user does not want autozero operation. Two different methods of disabling the autozero are available in the FC-2900. The zero correction output of the autozero circuit may be disconnected from the flowmeter circuit by removing PC board jumper J2. Removing this jumper disables the effect of the autozero circuit and permits the uncorrected "natural zero" of the flowmeter to be observed and adjusted if desired.

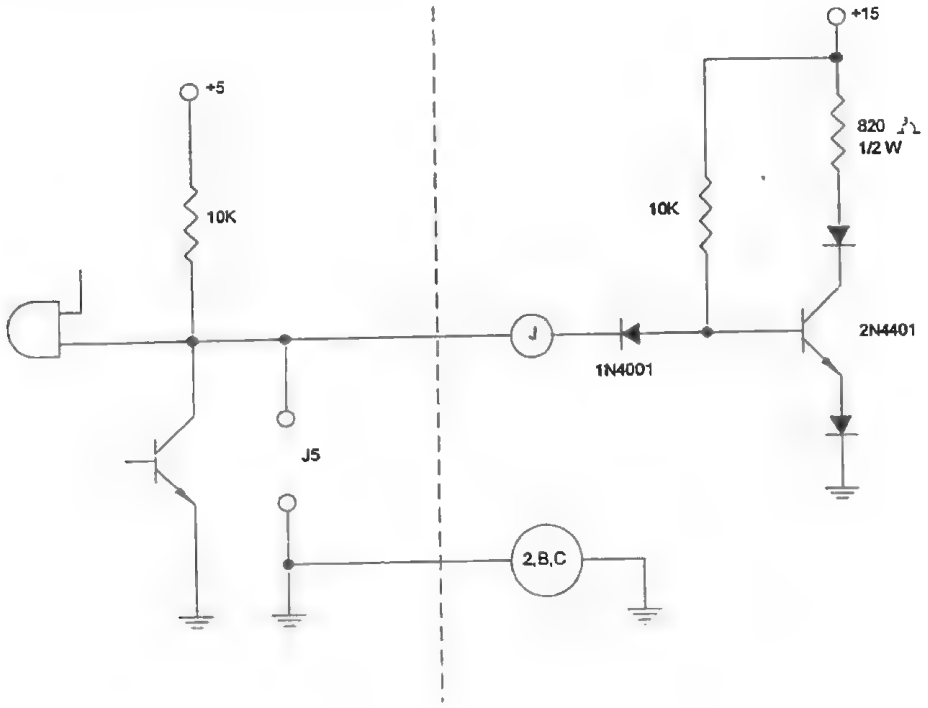


Figure 1

Connecting Pin J to common disables autozero updating. The current correction value stored will still be used to zero the MFC, but will not be recomputed under ordinary triggering conditions. This feature can be used to bring autozeroing under additional user control. For example, it may be desirable to re-zero only at routine maintenance intervals where some special condition can be set up. Such a condition might involve the closing of manual valves or the complete evacuation of the gas system to ensure that a no-flow condition exists when zero is recomputed. By linking this pin to a switch on an air actuated valve, the user can also prevent the MFC from autozeroing if the valve sticks open.

CAUTION: PIN J IS DESIGNATED PIN H ON UNIT INSTRUMENT PRODUCTS. SEE NOTE BELOW.

CAUTION
THE SAME MFC PINS ARE NOT DESIGNATED THE SAME WAY
BY ALL MANUFACTURERS

The 10 position bifurcated edge connector originally introduced by Tylan General and currently used on many mass flow products references pins by numbers along one edge and letters along the other. Tylan General, following the common practice of connector manufacturers, does not use the letters "G" and "I" in its pinout designations because they are too easily confused with "6" and "1" when molded almost illegibly into a plastic connector. Another manufacturer, Unit Instruments, does not follow this procedure. This can lead to confusion. The system is as follows:

Numbered side of connector:	1	2	3	4	5	6	7	8	9	10
Lettered side of connector:										
Industry Standard (Tylan General):	A	B	C	D	E	F	H	J	K	L
Unit Instruments:	A	B	C	D	E	F	G	H	I	J

The difference is in the last four positions on the lettered side. The functions commonly found on these pins are as follows:

	Tylan	Unit	
Function	Pin	Pin	Function
Keyway	H	G	Keyway
Autozero output/disable	J	H	none
Autozero memory store	K	I	none
Valve off/Autozero Trigger	L	J	Valve off

As can be seen, the pin letters may be different but the same functions appear in the same position. The Tylan General FC-2900 can therefore be used with confidence to upgrade a Unit flow controller, although the advanced autozero feature pins found on the FC-2900 will not necessarily be available in the Unit cable as-wired.

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CERTIFICATE OF NON-HAZARDOUS CONTAMINATION

Return Acceptance Policy

In accordance with California Hazard Communication Regulation (General Industry Safety Orders Sec. 5194 Title 8, California Code CCR Title 22, California Hazardous Substances Information and Training Law and Department of Transportation requirements) Tylan General requires this Certificate of Non-Hazardous Contamination to be completed for all products received for service or repair. Complete and affix to the outside of the shipping container.

This form must be completed by a person with knowledge of all types of processes in which the equipment has been used. Non-completion of this form will cause delays and possible rejection of your equipment.

SEC. 1

Model _____ Serial Number _____

Has the equipment been used? Y N If Yes, go to section 2
If No, go to section 6

Sec. 2

List all substances, gases or by-products which have come in contact with the equipment.

Chemical/Substance:

_____

_____

_____

Has the equipment been purged on-site? Y N If Yes, go to section 3
If no, go to section 4.

Sec. 3

Substance used in purge/flush process:

_____

Sec. 4

Please attach all required Material Safety Data Sheets (MSDS) for all substances listed in Sec.2. Exposed equipment cannot be transported without the above safety information. Please contact your carrier for more specific instructions.

Sec. 5

The undersigned certifies that the composition of residue on the equipment listed above is of a non-hazardous nature and/or that the equipment has been purged or flushed on-site. All exposed equipment will be cleaned. Tylan General reserves the right to reject any equipment based on contamination resulting from exposure to severely toxic substances.

Sec. 6

Signed: _____
Print Name: _____
Title: _____
Company: _____
Telephone: _____
Date of Signature: _____

FORM 5238, REV. C

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